

# Zolotarev problem case 1a

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## Abstract

Zolotarev

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# **1 Description of the report and overview of the results**

## **1.1 Methodology**

## **1.2 Overview of the results**

## **1.3 Preliminary conclusions**

## 2 Approximation objective $r = 3$

### 2.1 Numerator and denominator coefficients

The numerator and denominator coefficients (both real and complex parts) are given in the following tables. Each table gives the results obtained on different platform.

| Basis | Opt., $\sigma_3 = 0.00037$ |       | LF, $\sigma_3 = 0.0021$ |                       | AAA-sign-L200, $\sigma_3 = 0.00044$ |                      | AAA-sign-L1000, $\sigma_3 = 0.0005$ |       |
|-------|----------------------------|-------|-------------------------|-----------------------|-------------------------------------|----------------------|-------------------------------------|-------|
|       | real                       | imag. | real                    | imag.                 | real                                | imag.                | real                                | imag. |
| $z^3$ | 0                          | 0     | 0                       | 0                     | -0.011                              | $6.5 \cdot 10^{-4}$  | -70.0                               | 12.0  |
| $z^2$ | 2.6                        | 0     | 3.7                     | $1.8 \cdot 10^{-16}$  | 2.6                                 | $-2.2 \cdot 10^{-3}$ | 0.95                                | 0.064 |
| $z$   | 0                          | 0     | $-6.8 \cdot 10^{-15}$   | $-1.2 \cdot 10^{-15}$ | $9.9 \cdot 10^{-3}$                 | $1.0 \cdot 10^{-3}$  | -150.0                              | 27.0  |
| 1.0   | 0.65                       | 0     | 1.5                     | $1.9 \cdot 10^{-16}$  | 0.65                                | $-6.1 \cdot 10^{-4}$ | -0.36                               | 0.082 |
| $z^3$ | 1.0                        | 0     | 1.0                     | 0                     | 1.0                                 | 0                    | 1.0                                 | 0     |
| $z^2$ | 0                          | 0     | $-4.2 \cdot 10^{-15}$   | $-1.3 \cdot 10^{-15}$ | $-6.6 \cdot 10^{-3}$                | $2.1 \cdot 10^{-3}$  | -180.0                              | 31.0  |
| $z$   | 2.2                        | 0     | 4.2                     | $4.4 \cdot 10^{-16}$  | 2.3                                 | $-1.9 \cdot 10^{-3}$ | -0.38                               | 0.084 |
| 1.0   | 0                          | 0     | $-2.8 \cdot 10^{-15}$   | $-2.0 \cdot 10^{-16}$ | $7.3 \cdot 10^{-3}$                 | $2.0 \cdot 10^{-4}$  | -44.0                               | 7.6   |

Table 1: Case 1a (cpv MACA64 MR2023b),  $r = 3$ , Z4: numerator (first lines) and denominator (last lines) coefficients.

| Basis | Opt., $\sigma_3 = 0.00037$ |       | LF, $\sigma_3 = 0.0021$ |                       | AAA-sign-L200, $\sigma_3 = 0.00044$ |                      | AAA-sign-L1000, $\sigma_3 = 0.0005$ |       |
|-------|----------------------------|-------|-------------------------|-----------------------|-------------------------------------|----------------------|-------------------------------------|-------|
|       | real                       | imag. | real                    | imag.                 | real                                | imag.                | real                                | imag. |
| $z^3$ | 0                          | 0     | 0                       | 0                     | -0.011                              | $6.5 \cdot 10^{-4}$  | -70.0                               | 12.0  |
| $z^2$ | 2.6                        | 0     | 3.7                     | $6.2 \cdot 10^{-16}$  | 2.6                                 | $-2.2 \cdot 10^{-3}$ | 0.95                                | 0.064 |
| $z$   | 0                          | 0     | $-7.6 \cdot 10^{-15}$   | $1.4 \cdot 10^{-16}$  | $9.9 \cdot 10^{-3}$                 | $1.0 \cdot 10^{-3}$  | -150.0                              | 27.0  |
| 1.0   | 0.65                       | 0     | 1.5                     | $3.3 \cdot 10^{-17}$  | 0.65                                | $-6.1 \cdot 10^{-4}$ | -0.36                               | 0.082 |
| $z^3$ | 1.0                        | 0     | 1.0                     | 0                     | 1.0                                 | 0                    | 1.0                                 | 0     |
| $z^2$ | 0                          | 0     | $-3.8 \cdot 10^{-15}$   | $2.7 \cdot 10^{-16}$  | $-6.6 \cdot 10^{-3}$                | $2.1 \cdot 10^{-3}$  | -180.0                              | 31.0  |
| $z$   | 2.2                        | 0     | 4.2                     | $6.2 \cdot 10^{-16}$  | 2.3                                 | $-1.9 \cdot 10^{-3}$ | -0.38                               | 0.084 |
| 1.0   | 0                          | 0     | $-3.4 \cdot 10^{-15}$   | $-9.1 \cdot 10^{-17}$ | $7.3 \cdot 10^{-3}$                 | $2.0 \cdot 10^{-4}$  | -44.0                               | 7.6   |

Table 2: Case 1a (cpv MACI64 MR2022b),  $r = 3$ , Z4: numerator (first lines) and denominator (last lines) coefficients.

| Basis | Opt., $\sigma_3 = 0.00037$ |       | LF, $\sigma_3 = 0.0021$ |                       | AAA-sign-L200, $\sigma_3 = 0.00037$ |                       | AAA-sign-L1000, $\sigma_3 = 0.00037$ |                       |
|-------|----------------------------|-------|-------------------------|-----------------------|-------------------------------------|-----------------------|--------------------------------------|-----------------------|
|       | real                       | imag. | real                    | imag.                 | real                                | imag.                 | real                                 | imag.                 |
| $z^3$ | 0                          | 0     | 0                       | 0                     | $-2.4 \cdot 10^{-11}$               | $2.7 \cdot 10^{-13}$  | $-9.5 \cdot 10^{-15}$                | $5.6 \cdot 10^{-13}$  |
| $z^2$ | 2.6                        | 0     | 3.7                     | $-1.8 \cdot 10^{-16}$ | 2.6                                 | $-5.8 \cdot 10^{-15}$ | 2.6                                  | $-4.2 \cdot 10^{-15}$ |
| $z$   | 0                          | 0     | $-5.1 \cdot 10^{-15}$   | $6.8 \cdot 10^{-16}$  | $-5.4 \cdot 10^{-11}$               | $6.3 \cdot 10^{-13}$  | $-2.1 \cdot 10^{-14}$                | $1.2 \cdot 10^{-12}$  |
| 1.0   | 0.65                       | 0     | 1.5                     | $-2.4 \cdot 10^{-16}$ | 0.65                                | $-9.7 \cdot 10^{-15}$ | 0.65                                 | $-6.4 \cdot 10^{-15}$ |
| $z^3$ | 1.0                        | 0     | 1.0                     | 0                     | 1.0                                 | 0                     | 1.0                                  | 0                     |
| $z^2$ | 0                          | 0     | $-3.2 \cdot 10^{-15}$   | $3.0 \cdot 10^{-18}$  | $-6.2 \cdot 10^{-11}$               | $7.1 \cdot 10^{-13}$  | $-2.3 \cdot 10^{-14}$                | $1.4 \cdot 10^{-12}$  |
| $z$   | 2.2                        | 0     | 4.2                     | $-4.0 \cdot 10^{-16}$ | 2.2                                 | $-9.9 \cdot 10^{-15}$ | 2.2                                  | $-9.8 \cdot 10^{-15}$ |
| 1.0   | 0                          | 0     | $-1.8 \cdot 10^{-15}$   | $4.3 \cdot 10^{-16}$  | $-1.6 \cdot 10^{-11}$               | $1.9 \cdot 10^{-13}$  | $-7.1 \cdot 10^{-15}$                | $3.6 \cdot 10^{-13}$  |

Table 3: Case 1a (cpv PCWIN64 MR2025b),  $r = 3$ , Z4: numerator (first lines) and denominator (last lines) coefficients.

| Basis | Opt., $\sigma_3 = 0.00037$ |       | LF, $\sigma_3 = 0.0021$ |                       | AAA-sign-L200, $\sigma_3 = 0.00037$ |                       | AAA-sign-L1000, $\sigma_3 = 0.00037$ |                       |
|-------|----------------------------|-------|-------------------------|-----------------------|-------------------------------------|-----------------------|--------------------------------------|-----------------------|
|       | real                       | imag. | real                    | imag.                 | real                                | imag.                 | real                                 | imag.                 |
| $z^3$ | 0                          | 0     | 0                       | 0                     | $-2.4 \cdot 10^{-11}$               | $2.7 \cdot 10^{-13}$  | $-9.5 \cdot 10^{-15}$                | $5.6 \cdot 10^{-13}$  |
| $z^2$ | 2.6                        | 0     | 3.7                     | $-1.8 \cdot 10^{-16}$ | 2.6                                 | $-5.8 \cdot 10^{-15}$ | 2.6                                  | $-4.2 \cdot 10^{-15}$ |
| $z$   | 0                          | 0     | $-5.1 \cdot 10^{-15}$   | $6.8 \cdot 10^{-16}$  | $-5.4 \cdot 10^{-11}$               | $6.3 \cdot 10^{-13}$  | $-2.1 \cdot 10^{-14}$                | $1.2 \cdot 10^{-12}$  |
| 1.0   | 0.65                       | 0     | 1.5                     | $-2.4 \cdot 10^{-16}$ | 0.65                                | $-9.7 \cdot 10^{-15}$ | 0.65                                 | $-6.4 \cdot 10^{-15}$ |
| $z^3$ | 1.0                        | 0     | 1.0                     | 0                     | 1.0                                 | 0                     | 1.0                                  | 0                     |
| $z^2$ | 0                          | 0     | $-3.2 \cdot 10^{-15}$   | $3.0 \cdot 10^{-18}$  | $-6.2 \cdot 10^{-11}$               | $7.1 \cdot 10^{-13}$  | $-2.3 \cdot 10^{-14}$                | $1.4 \cdot 10^{-12}$  |
| $z$   | 2.2                        | 0     | 4.2                     | $-4.0 \cdot 10^{-16}$ | 2.2                                 | $-9.9 \cdot 10^{-15}$ | 2.2                                  | $-9.8 \cdot 10^{-15}$ |
| 1.0   | 0                          | 0     | $-1.8 \cdot 10^{-15}$   | $4.3 \cdot 10^{-16}$  | $-1.6 \cdot 10^{-11}$               | $1.9 \cdot 10^{-13}$  | $-7.1 \cdot 10^{-15}$                | $3.6 \cdot 10^{-13}$  |

Table 4: Case 1a (AlexReis PCWIN64 MR2025b),  $r = 3$ , Z4: numerator (first lines) and denominator (last lines) coefficients.

## 2.2 Poles

The poles are given in the following tables. Each table gives the results obtained on different platform.

|     | Opt., $\sigma_3 = 0.00037$ | LF, $\sigma_3 = 0.0021$                      | AAA-sign-L200, $\sigma_3 = 0.00044$ | AAA-sign-L1000, $\sigma_3 = 0.0005$ |
|-----|----------------------------|----------------------------------------------|-------------------------------------|-------------------------------------|
| 1.0 | 0                          | $6.58 \cdot 10^{-16} + 4.81 \cdot 10^{-17}i$ | $-0.00323 - 9.0 \cdot 10^{-5}i$     | $-0.00181 + 0.493i$                 |
| 2.0 | $1.5i$                     | $2.14 \cdot 10^{-15} - 2.06i$                | $0.00557 + 1.5i$                    | $-0.00165 - 0.494i$                 |
| 3.0 | $-1.5i$                    | $1.49 \cdot 10^{-15} + 2.06i$                | $0.00428 - 1.5i$                    | $180.0 - 31.3i$                     |

Table 5: Case 1a (cpv MACA64 MR2023b),  $r = 3$ , Z4: poles.

|     | Opt., $\sigma_3 = 0.00037$ | LF, $\sigma_3 = 0.0021$                      | AAA-sign-L200, $\sigma_3 = 0.00044$ | AAA-sign-L1000, $\sigma_3 = 0.0005$ |
|-----|----------------------------|----------------------------------------------|-------------------------------------|-------------------------------------|
| 1.0 | 0                          | $8.09 \cdot 10^{-16} + 2.14 \cdot 10^{-17}i$ | $-0.00323 - 9.0 \cdot 10^{-5}i$     | $-0.00181 + 0.493i$                 |
| 2.0 | $1.5i$                     | $1.75 \cdot 10^{-15} - 2.06i$                | $0.00557 + 1.5i$                    | $-0.00165 - 0.494i$                 |
| 3.0 | $-1.5i$                    | $1.29 \cdot 10^{-15} + 2.06i$                | $0.00428 - 1.5i$                    | $180.0 - 31.3i$                     |

Table 6: Case 1a (cpv MACI64 MR2022b),  $r = 3$ , Z4: poles.

|     | Opt., $\sigma_3 = 0.00037$ | LF, $\sigma_3 = 0.0021$                      | AAA-sign-L200, $\sigma_3 = 0.00037$         | AAA-sign-L1000, $\sigma_3 = 0.00037$         |
|-----|----------------------------|----------------------------------------------|---------------------------------------------|----------------------------------------------|
| 1.0 | 0                          | $4.35 \cdot 10^{-16} - 1.02 \cdot 10^{-16}i$ | $6.9 \cdot 10^{-12} - 8.47 \cdot 10^{-14}i$ | $3.42 \cdot 10^{-15} - 1.59 \cdot 10^{-13}i$ |
| 2.0 | $1.5i$                     | $1.25 \cdot 10^{-15} - 2.06i$                | $2.76 \cdot 10^{-11} + 1.5i$                | $1.22 \cdot 10^{-14} + 1.5i$                 |
| 3.0 | $-1.5i$                    | $1.47 \cdot 10^{-15} + 2.06i$                | $2.76 \cdot 10^{-11} - 1.5i$                | $7.73 \cdot 10^{-15} - 1.5i$                 |

Table 7: Case 1a (cpv PCWIN64 MR2025b),  $r = 3$ , Z4: poles.

|     | Opt., $\sigma_3 = 0.00037$ | LF, $\sigma_3 = 0.0021$                      | AAA-sign-L200, $\sigma_3 = 0.00037$         | AAA-sign-L1000, $\sigma_3 = 0.00037$         |
|-----|----------------------------|----------------------------------------------|---------------------------------------------|----------------------------------------------|
| 1.0 | 0                          | $4.35 \cdot 10^{-16} - 1.02 \cdot 10^{-16}i$ | $6.9 \cdot 10^{-12} - 8.47 \cdot 10^{-14}i$ | $3.42 \cdot 10^{-15} - 1.59 \cdot 10^{-13}i$ |
| 2.0 | $1.5i$                     | $1.25 \cdot 10^{-15} - 2.06i$                | $2.76 \cdot 10^{-11} + 1.5i$                | $1.22 \cdot 10^{-14} + 1.5i$                 |
| 3.0 | $-1.5i$                    | $1.47 \cdot 10^{-15} + 2.06i$                | $2.76 \cdot 10^{-11} - 1.5i$                | $7.73 \cdot 10^{-15} - 1.5i$                 |

Table 8: Case 1a (AlexReis PCWIN64 MR2025b),  $r = 3$ , Z4: poles.

## 2.3 Zeros

The zeros are given in the following tables. Each table gives the results obtained on different platform.

|     | Opt., $\sigma_3 = 0.00037$ | LF, $\sigma_3 = 0.0021$        | AAA-sign-L200, $\sigma_3 = 0.00044$ | AAA-sign-L1000, $\sigma_3 = 0.0005$ |
|-----|----------------------------|--------------------------------|-------------------------------------|-------------------------------------|
| 1.0 | NaN                        | NaN                            | $-0.00241 + 0.5i$                   | $-0.00236 + 1.2 \cdot 10^{-4}i$     |
| 2.0 | $0.5i$                     | $9.48 \cdot 10^{-16} - 0.637i$ | $-0.00245 - 0.501i$                 | $0.00667 - 1.48i$                   |
| 3.0 | $-0.5i$                    | $8.89 \cdot 10^{-16} + 0.637i$ | $237.0 + 13.9i$                     | $0.00871 + 1.48i$                   |

Table 9: Case 1a (cpv MACA64 MR2023b),  $r = 3$ , Z4: zeros.

|     | Opt., $\sigma_3 = 0.00037$ | LF, $\sigma_3 = 0.0021$        | AAA-sign-L200, $\sigma_3 = 0.00044$ | AAA-sign-L1000, $\sigma_3 = 0.0005$ |
|-----|----------------------------|--------------------------------|-------------------------------------|-------------------------------------|
| 1.0 | NaN                        | NaN                            | $-0.00241 + 0.5i$                   | $-0.00236 + 1.2 \cdot 10^{-4}i$     |
| 2.0 | $0.5i$                     | $9.48 \cdot 10^{-16} - 0.637i$ | $-0.00245 - 0.501i$                 | $0.00667 - 1.48i$                   |
| 3.0 | $-0.5i$                    | $1.1 \cdot 10^{-15} + 0.637i$  | $237.0 + 13.9i$                     | $0.00871 + 1.48i$                   |

Table 10: Case 1a (cpv MACI64 MR2022b),  $r = 3$ , Z4: zeros.

|     | Opt., $\sigma_3 = 0.00037$ | LF, $\sigma_3 = 0.0021$        | AAA-sign-L200, $\sigma_3 = 0.00037$     | AAA-sign-L1000, $\sigma_3 = 0.00037$       |
|-----|----------------------------|--------------------------------|-----------------------------------------|--------------------------------------------|
| 1.0 | NaN                        | NaN                            | $9.21 \cdot 10^{-12} + 0.5i$            | $4.98 \cdot 10^{-15} + 0.5i$               |
| 2.0 | $0.5i$                     | $7.16 \cdot 10^{-16} + 0.637i$ | $9.2 \cdot 10^{-12} - 0.5i$             | $2.0 \cdot 10^{-15} - 0.5i$                |
| 3.0 | $-0.5i$                    | $6.45 \cdot 10^{-16} - 0.637i$ | $1.09 \cdot 10^{11} + 1.22 \cdot 10^9i$ | $7.82 \cdot 10^{10} + 4.66 \cdot 10^{12}i$ |

Table 11: Case 1a (cpv PCWIN64 MR2025b),  $r = 3$ , Z4: zeros.

|     | Opt., $\sigma_3 = 0.00037$ | LF, $\sigma_3 = 0.0021$        | AAA-sign-L200, $\sigma_3 = 0.00037$     | AAA-sign-L1000, $\sigma_3 = 0.00037$       |
|-----|----------------------------|--------------------------------|-----------------------------------------|--------------------------------------------|
| 1.0 | NaN                        | NaN                            | $9.21 \cdot 10^{-12} + 0.5i$            | $4.98 \cdot 10^{-15} + 0.5i$               |
| 2.0 | $0.5i$                     | $7.16 \cdot 10^{-16} + 0.637i$ | $9.2 \cdot 10^{-12} - 0.5i$             | $2.0 \cdot 10^{-15} - 0.5i$                |
| 3.0 | $-0.5i$                    | $6.45 \cdot 10^{-16} - 0.637i$ | $1.09 \cdot 10^{11} + 1.22 \cdot 10^9i$ | $7.82 \cdot 10^{10} + 4.66 \cdot 10^{12}i$ |

Table 12: Case 1a (AlexReis PCWIN64 MR2025b),  $r = 3$ , Z4: zeros.

## 2.4 Poles and zeros visualisation

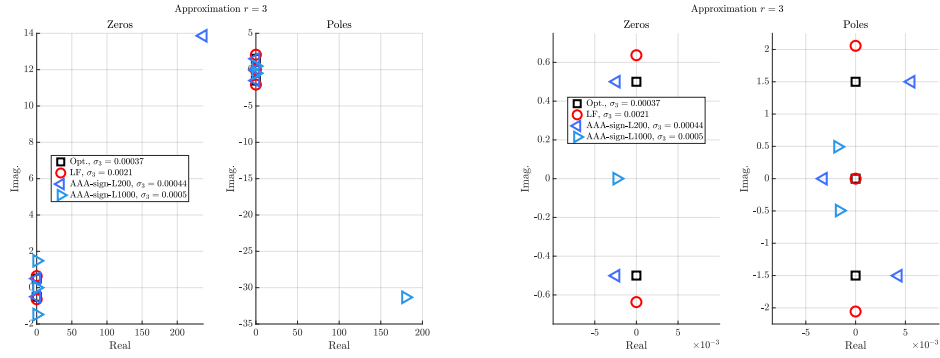


Figure 1: Case 1a (cpv MACA64 MR2023b),  $r = 3$ , Z4. Left frame: zeros (left) and poles (right). Right frame: same but discarding the largest real entry.

## 2.5 Approximation visualization

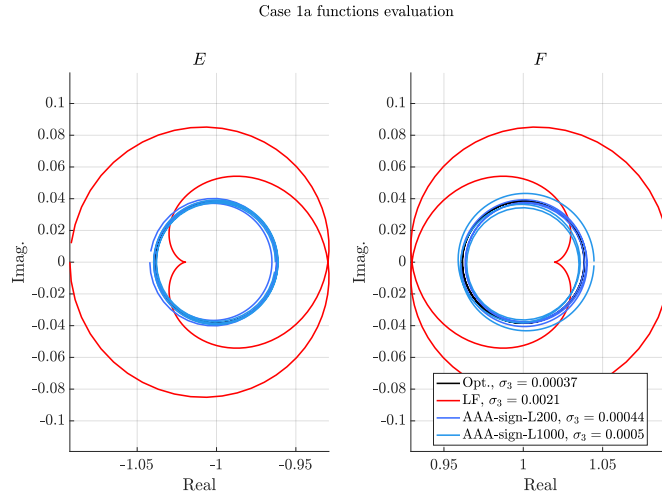


Figure 2: Case 1a (cpv MACA64 MR2023b),  $r = 3$ , Z4.  $E$  and  $F$  evaluations.

### 3 Approximation objective $r = 4$

#### 3.1 Numerator and denominator coefficients

The numerator and denominator coefficients (both real and complex parts) are given in the following tables. Each table gives the results obtained on different platform.

| Basis | Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$ |       | LF, $\sigma_4 = 6 \cdot 10^{-05}$ |                       | AAA-sign-L200, $\sigma_4 = 2.8 \cdot 10^{-05}$ |       | AAA-sign-L1000, $\sigma_4 = 2.8 \cdot 10^{-05}$ |        |
|-------|---------------------------------------|-------|-----------------------------------|-----------------------|------------------------------------------------|-------|-------------------------------------------------|--------|
|       | real                                  | imag. | real                              | imag.                 | real                                           | imag. | real                                            | imag.  |
| $z^4$ | 0                                     | 0     | 0                                 | 0                     | -140.0                                         | 16.0  | -150.0                                          | 7.3    |
| $z^3$ | 3.5                                   | 0     | 3.7                               | $4.1 \cdot 10^{-16}$  | 4.7                                            | -0.29 | 4.8                                             | -0.22  |
| $z^2$ | 0                                     | 0     | $-4.6 \cdot 10^{-15}$             | $3.8 \cdot 10^{-16}$  | -650.0                                         | 70.0  | -660.0                                          | 33.0   |
| $z$   | 2.6                                   | 0     | 3.2                               | $2.0 \cdot 10^{-16}$  | 5.5                                            | -0.52 | 5.6                                             | -0.34  |
| 1.0   | 0                                     | 0     | $-1.3 \cdot 10^{-15}$             | $-1.6 \cdot 10^{-15}$ | -80.0                                          | 8.8   | -82.0                                           | 4.1    |
| $z^4$ | 1.0                                   | 0     | 1.0                               | 0                     | 1.0                                            | 0     | 1.0                                             | 0      |
| $z^3$ | 0                                     | 0     | $-2.7 \cdot 10^{-15}$             | $-8.4 \cdot 10^{-16}$ | -500.0                                         | 54.0  | -510.0                                          | 25.0   |
| $z^2$ | 4.5                                   | 0     | 5.2                               | $8.2 \cdot 10^{-16}$  | 7.9                                            | -0.7  | 7.9                                             | -0.49  |
| $z$   | 0                                     | 0     | $-3.5 \cdot 10^{-15}$             | $-1.1 \cdot 10^{-15}$ | -370.0                                         | 41.0  | -380.0                                          | 19.0   |
| 1.0   | 0.56                                  | 0     | 0.75                              | $1.0 \cdot 10^{-16}$  | 1.4                                            | -0.13 | 1.4                                             | -0.078 |

Table 13: Case 1a (cpv MACA64 MR2023b),  $r = 4$ , Z4: numerator (first lines) and denominator (last lines) coefficients.

| Basis | Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$ |       | LF, $\sigma_4 = 6 \cdot 10^{-05}$ |                      | AAA-sign-L200, $\sigma_4 = 2.8 \cdot 10^{-05}$ |       | AAA-sign-L1000, $\sigma_4 = 2.8 \cdot 10^{-05}$ |        |
|-------|---------------------------------------|-------|-----------------------------------|----------------------|------------------------------------------------|-------|-------------------------------------------------|--------|
|       | real                                  | imag. | real                              | imag.                | real                                           | imag. | real                                            | imag.  |
| $z^4$ | 0                                     | 0     | 0                                 | 0                    | -140.0                                         | 16.0  | -150.0                                          | 7.3    |
| $z^3$ | 3.5                                   | 0     | 3.7                               | $6.4 \cdot 10^{-16}$ | 4.7                                            | -0.29 | 4.8                                             | -0.22  |
| $z^2$ | 0                                     | 0     | $-4.8 \cdot 10^{-15}$             | $3.7 \cdot 10^{-16}$ | -650.0                                         | 70.0  | -660.0                                          | 33.0   |
| $z$   | 2.6                                   | 0     | 3.2                               | $1.8 \cdot 10^{-15}$ | 5.5                                            | -0.52 | 5.6                                             | -0.34  |
| 1.0   | 0                                     | 0     | $-1.0 \cdot 10^{-15}$             | $2.6 \cdot 10^{-15}$ | -80.0                                          | 8.8   | -82.0                                           | 4.1    |
| $z^4$ | 1.0                                   | 0     | 1.0                               | 0                    | 1.0                                            | 0     | 1.0                                             | 0      |
| $z^3$ | 0                                     | 0     | $-9.8 \cdot 10^{-16}$             | $8.7 \cdot 10^{-16}$ | -500.0                                         | 54.0  | -510.0                                          | 25.0   |
| $z^2$ | 4.5                                   | 0     | 5.2                               | $1.9 \cdot 10^{-15}$ | 7.9                                            | -0.7  | 7.9                                             | -0.49  |
| $z$   | 0                                     | 0     | $-4.0 \cdot 10^{-15}$             | $2.8 \cdot 10^{-15}$ | -370.0                                         | 41.0  | -380.0                                          | 19.0   |
| 1.0   | 0.56                                  | 0     | 0.75                              | $5.9 \cdot 10^{-16}$ | 1.4                                            | -0.13 | 1.4                                             | -0.078 |

Table 14: Case 1a (cpv MACI64 MR2022b),  $r = 4$ , Z4: numerator (first lines) and denominator (last lines) coefficients.

| Basis | Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$ |       | LF, $\sigma_4 = 6 \cdot 10^{-05}$ |                       | AAA-sign-L200, $\sigma_4 = 3.3 \cdot 10^{-05}$ |                      | AAA-sign-L1000, $\sigma_4 = 3.4 \cdot 10^{-05}$ |                      |
|-------|---------------------------------------|-------|-----------------------------------|-----------------------|------------------------------------------------|----------------------|-------------------------------------------------|----------------------|
|       | real                                  | imag. | real                              | imag.                 | real                                           | imag.                | real                                            | imag.                |
| $z^4$ | 0                                     | 0     | 0                                 | 0                     | $-5.4 \cdot 10^{-3}$                           | $-3.1 \cdot 10^{-3}$ | $-4.8 \cdot 10^{-3}$                            | $-2.8 \cdot 10^{-3}$ |
| $z^3$ | 3.5                                   | 0     | 3.7                               | $1.8 \cdot 10^{-16}$  | 3.5                                            | $-3.7 \cdot 10^{-3}$ | 3.5                                             | $-3.8 \cdot 10^{-3}$ |
| $z^2$ | 0                                     | 0     | $-2.8 \cdot 10^{-15}$             | $-9.9 \cdot 10^{-16}$ | -0.07                                          | $-2.6 \cdot 10^{-3}$ | -0.078                                          | $-2.5 \cdot 10^{-3}$ |
| $z$   | 2.6                                   | 0     | 3.2                               | $-5.5 \cdot 10^{-16}$ | 2.6                                            | -0.011               | 2.6                                             | -0.011               |
| 1.0   | 0                                     | 0     | $-1.8 \cdot 10^{-15}$             | $-7.7 \cdot 10^{-16}$ | -0.015                                         | $1.3 \cdot 10^{-3}$  | -0.017                                          | $1.4 \cdot 10^{-3}$  |
| $z^4$ | 1.0                                   | 0     | 1.0                               | 0                     | 1.0                                            | 0                    | 1.0                                             | 0                    |
| $z^3$ | 0                                     | 0     | $-4.6 \cdot 10^{-16}$             | $-1.2 \cdot 10^{-15}$ | -0.036                                         | $-6.9 \cdot 10^{-3}$ | -0.038                                          | $-6.5 \cdot 10^{-3}$ |
| $z^2$ | 4.5                                   | 0     | 5.2                               | $2.7 \cdot 10^{-16}$  | 4.5                                            | -0.011               | 4.5                                             | -0.011               |
| $z$   | 0                                     | 0     | $-4.2 \cdot 10^{-15}$             | $-4.2 \cdot 10^{-16}$ | -0.054                                         | $2.2 \cdot 10^{-3}$  | -0.062                                          | $2.5 \cdot 10^{-3}$  |
| 1.0   | 0.56                                  | 0     | 0.75                              | $-5.2 \cdot 10^{-16}$ | 0.57                                           | $-3.3 \cdot 10^{-3}$ | 0.57                                            | $-3.3 \cdot 10^{-3}$ |

Table 15: Case 1a (cpv PCWIN64 MR2025b),  $r = 4$ , Z4: numerator (first lines) and denominator (last lines) coefficients.

| Basis | Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$ |       | LF, $\sigma_4 = 6 \cdot 10^{-05}$ |                       | AAA-sign-L200, $\sigma_4 = 3.3 \cdot 10^{-05}$ |                      | AAA-sign-L1000, $\sigma_4 = 3.4 \cdot 10^{-05}$ |                      |
|-------|---------------------------------------|-------|-----------------------------------|-----------------------|------------------------------------------------|----------------------|-------------------------------------------------|----------------------|
|       | real                                  | imag. | real                              | imag.                 | real                                           | imag.                | real                                            | imag.                |
| $z^4$ | 0                                     | 0     | 0                                 | 0                     | $-5.4 \cdot 10^{-3}$                           | $-3.1 \cdot 10^{-3}$ | $-4.8 \cdot 10^{-3}$                            | $-2.8 \cdot 10^{-3}$ |
| $z^3$ | 3.5                                   | 0     | 3.7                               | $1.8 \cdot 10^{-16}$  | 3.5                                            | $-3.7 \cdot 10^{-3}$ | 3.5                                             | $-3.8 \cdot 10^{-3}$ |
| $z^2$ | 0                                     | 0     | $-2.8 \cdot 10^{-15}$             | $-9.9 \cdot 10^{-16}$ | -0.07                                          | $-2.6 \cdot 10^{-3}$ | -0.078                                          | $-2.5 \cdot 10^{-3}$ |
| $z$   | 2.6                                   | 0     | 3.2                               | $-5.5 \cdot 10^{-16}$ | 2.6                                            | -0.011               | 2.6                                             | -0.011               |
| 1.0   | 0                                     | 0     | $-1.8 \cdot 10^{-15}$             | $-7.7 \cdot 10^{-16}$ | -0.015                                         | $1.3 \cdot 10^{-3}$  | -0.017                                          | $1.4 \cdot 10^{-3}$  |
| $z^4$ | 1.0                                   | 0     | 1.0                               | 0                     | 1.0                                            | 0                    | 1.0                                             | 0                    |
| $z^3$ | 0                                     | 0     | $-4.6 \cdot 10^{-16}$             | $-1.2 \cdot 10^{-15}$ | -0.036                                         | $-6.9 \cdot 10^{-3}$ | -0.038                                          | $-6.5 \cdot 10^{-3}$ |
| $z^2$ | 4.5                                   | 0     | 5.2                               | $2.7 \cdot 10^{-16}$  | 4.5                                            | -0.011               | 4.5                                             | -0.011               |
| $z$   | 0                                     | 0     | $-4.2 \cdot 10^{-15}$             | $-4.2 \cdot 10^{-16}$ | -0.054                                         | $2.2 \cdot 10^{-3}$  | -0.062                                          | $2.5 \cdot 10^{-3}$  |
| 1.0   | 0.56                                  | 0     | 0.75                              | $-5.2 \cdot 10^{-16}$ | 0.57                                           | $-3.3 \cdot 10^{-3}$ | 0.57                                            | $-3.3 \cdot 10^{-3}$ |

Table 16: Case 1a (AlexReis PCWIN64 MR2025b),  $r = 4$ , Z4: numerator (first lines) and denominator (last lines) coefficients.

### 3.2 Poles

The poles are given in the following tables. Each table gives the results obtained on different platform.

|     | Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$ | LF, $\sigma_4 = 6 \cdot 10^{-05}$ | AAA-sign-L200, $\sigma_4 = 2.8 \cdot 10^{-05}$ | AAA-sign-L1000, $\sigma_4 = 2.8 \cdot 10^{-05}$ |
|-----|---------------------------------------|-----------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | $-1.94 \cdot 10^{-16} + 2.09i$        | $3.21 \cdot 10^{-16} - 0.384i$    | $0.00377 + 6.65 \cdot 10^{-5}i$                | $0.00375 - 2.07 \cdot 10^{-5}i$                 |
| 2.0 | $-1.94 \cdot 10^{-16} - 2.09i$        | $3.09 \cdot 10^{-16} + 0.384i$    | $0.00535 - 0.864i$                             | $0.00513 + 0.864i$                              |
| 3.0 | $-1.39 \cdot 10^{-17} + 0.359i$       | $1.12 \cdot 10^{-15} - 2.25i$     | $0.00516 + 0.864i$                             | $0.00532 - 0.864i$                              |
| 4.0 | $-1.39 \cdot 10^{-17} - 0.359i$       | $9.09 \cdot 10^{-16} + 2.25i$     | $499.0 - 54.4i$                                | $507.0 - 25.2i$                                 |

Table 17: Case 1a (cpv MACA64 MR2023b),  $r = 4$ , Z4: poles.

|     | Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$ | LF, $\sigma_4 = 6 \cdot 10^{-05}$ | AAA-sign-L200, $\sigma_4 = 2.8 \cdot 10^{-05}$ | AAA-sign-L1000, $\sigma_4 = 2.8 \cdot 10^{-05}$ |
|-----|---------------------------------------|-----------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | $-1.94 \cdot 10^{-16} + 2.09i$        | $3.19 \cdot 10^{-16} + 0.384i$    | $0.00377 + 6.65 \cdot 10^{-5}i$                | $0.00375 - 2.07 \cdot 10^{-5}i$                 |
| 2.0 | $-1.94 \cdot 10^{-16} - 2.09i$        | $4.66 \cdot 10^{-16} - 0.384i$    | $0.00535 - 0.864i$                             | $0.00513 + 0.864i$                              |
| 3.0 | $-1.39 \cdot 10^{-17} + 0.359i$       | $4.91 \cdot 10^{-16} - 2.25i$     | $0.00516 + 0.864i$                             | $0.00532 - 0.864i$                              |
| 4.0 | $-1.39 \cdot 10^{-17} - 0.359i$       | $-3.02 \cdot 10^{-16} + 2.25i$    | $499.0 - 54.4i$                                | $507.0 - 25.2i$                                 |

Table 18: Case 1a (cpv MACI64 MR2022b),  $r = 4$ , Z4: poles.

|     | Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$ | LF, $\sigma_4 = 6 \cdot 10^{-05}$ | AAA-sign-L200, $\sigma_4 = 3.3 \cdot 10^{-05}$ | AAA-sign-L1000, $\sigma_4 = 3.4 \cdot 10^{-05}$ |
|-----|---------------------------------------|-----------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | $-1.94 \cdot 10^{-16} + 2.09i$        | $5.75 \cdot 10^{-16} + 0.384i$    | $0.00636 + 0.36i$                              | $0.00731 + 0.36i$                               |
| 2.0 | $-1.94 \cdot 10^{-16} - 2.09i$        | $2.75 \cdot 10^{-16} - 0.384i$    | $0.0052 - 0.361i$                              | $0.00612 - 0.361i$                              |
| 3.0 | $-1.39 \cdot 10^{-17} + 0.359i$       | $-1.15 \cdot 10^{-16} - 2.25i$    | $0.00935 - 2.1i$                               | $0.00961 - 2.09i$                               |
| 4.0 | $-1.39 \cdot 10^{-17} - 0.359i$       | $-2.71 \cdot 10^{-16} + 2.25i$    | $0.0147 + 2.1i$                                | $0.0149 + 2.1i$                                 |

Table 19: Case 1a (cpv PCWIN64 MR2025b),  $r = 4$ , Z4: poles.

|     | Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$ | LF, $\sigma_4 = 6 \cdot 10^{-05}$ | AAA-sign-L200, $\sigma_4 = 3.3 \cdot 10^{-05}$ | AAA-sign-L1000, $\sigma_4 = 3.4 \cdot 10^{-05}$ |
|-----|---------------------------------------|-----------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | $-1.94 \cdot 10^{-16} + 2.09i$        | $5.75 \cdot 10^{-16} + 0.384i$    | $0.00636 + 0.36i$                              | $0.00731 + 0.36i$                               |
| 2.0 | $-1.94 \cdot 10^{-16} - 2.09i$        | $2.75 \cdot 10^{-16} - 0.384i$    | $0.0052 - 0.361i$                              | $0.00612 - 0.361i$                              |
| 3.0 | $-1.39 \cdot 10^{-17} + 0.359i$       | $-1.15 \cdot 10^{-16} - 2.25i$    | $0.00935 - 2.1i$                               | $0.00961 - 2.09i$                               |
| 4.0 | $-1.39 \cdot 10^{-17} - 0.359i$       | $-2.71 \cdot 10^{-16} + 2.25i$    | $0.0147 + 2.1i$                                | $0.0149 + 2.1i$                                 |

Table 20: Case 1a (AlexReis PCWIN64 MR2025b),  $r = 4$ , Z4: poles.



### 3.3 Zeros

The zeros are given in the following tables. Each table gives the results obtained on different platform.

|     | Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$ | LF, $\sigma_4 = 6 \cdot 10^{-05}$           | AAA-sign-L200, $\sigma_4 = 2.8 \cdot 10^{-05}$ | AAA-sign-L1000, $\sigma_4 = 2.8 \cdot 10^{-05}$ |
|-----|---------------------------------------|---------------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | NaN                                   | NaN                                         | $0.00403 - 0.358i$                             | $0.00404 + 0.358i$                              |
| 2.0 | 0                                     | $4.1 \cdot 10^{-16} + 5.92 \cdot 10^{-16}i$ | $0.00406 + 0.358i$                             | $0.00402 - 0.358i$                              |
| 3.0 | $0.866i$                              | $3.69 \cdot 10^{-16} + 0.929i$              | $0.0128 - 2.08i$                               | $0.0127 - 2.09i$                                |
| 4.0 | $-0.866i$                             | $4.46 \cdot 10^{-16} - 0.929i$              | $0.0117 + 2.09i$                               | $0.0116 + 2.09i$                                |

Table 21: Case 1a (cpv MACA64 MR2023b),  $r = 4$ , Z4: zeros.

|     | Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$ | LF, $\sigma_4 = 6 \cdot 10^{-05}$            | AAA-sign-L200, $\sigma_4 = 2.8 \cdot 10^{-05}$ | AAA-sign-L1000, $\sigma_4 = 2.8 \cdot 10^{-05}$ |
|-----|---------------------------------------|----------------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | NaN                                   | NaN                                          | $0.00403 - 0.358i$                             | $0.00404 + 0.358i$                              |
| 2.0 | 0                                     | $3.12 \cdot 10^{-16} - 7.29 \cdot 10^{-16}i$ | $0.00406 + 0.358i$                             | $0.00402 - 0.358i$                              |
| 3.0 | $0.866i$                              | $6.57 \cdot 10^{-16} - 0.929i$               | $0.0128 - 2.08i$                               | $0.0127 - 2.09i$                                |
| 4.0 | $-0.866i$                             | $3.16 \cdot 10^{-16} + 0.929i$               | $0.0117 + 2.09i$                               | $0.0116 + 2.09i$                                |

Table 22: Case 1a (cpv MACI64 MR2022b),  $r = 4$ , Z4: zeros.

|     | Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$ | LF, $\sigma_4 = 6 \cdot 10^{-05}$            | AAA-sign-L200, $\sigma_4 = 3.3 \cdot 10^{-05}$ | AAA-sign-L1000, $\sigma_4 = 3.4 \cdot 10^{-05}$ |
|-----|---------------------------------------|----------------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | NaN                                   | NaN                                          | $0.00562 - 4.73 \cdot 10^{-4}i$                | $0.00656 - 5.2 \cdot 10^{-4}i$                  |
| 2.0 | 0                                     | $5.68 \cdot 10^{-16} + 2.27 \cdot 10^{-16}i$ | $0.00528 - 0.87i$                              | $0.00611 - 0.869i$                              |
| 3.0 | $0.866i$                              | $3.38 \cdot 10^{-18} - 0.929i$               | $0.00792 + 0.871i$                             | $0.00875 + 0.87i$                               |
| 4.0 | $-0.866i$                             | $1.73 \cdot 10^{-16} + 0.929i$               | $486.0 - 282.0i$                               | $539.0 - 319.0i$                                |

Table 23: Case 1a (cpv PCWIN64 MR2025b),  $r = 4$ , Z4: zeros.

|     | Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$ | LF, $\sigma_4 = 6 \cdot 10^{-05}$            | AAA-sign-L200, $\sigma_4 = 3.3 \cdot 10^{-05}$ | AAA-sign-L1000, $\sigma_4 = 3.4 \cdot 10^{-05}$ |
|-----|---------------------------------------|----------------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | NaN                                   | NaN                                          | $0.00562 - 4.73 \cdot 10^{-4}i$                | $0.00656 - 5.2 \cdot 10^{-4}i$                  |
| 2.0 | 0                                     | $5.68 \cdot 10^{-16} + 2.27 \cdot 10^{-16}i$ | $0.00528 - 0.87i$                              | $0.00611 - 0.869i$                              |
| 3.0 | $0.866i$                              | $3.38 \cdot 10^{-18} - 0.929i$               | $0.00792 + 0.871i$                             | $0.00875 + 0.87i$                               |
| 4.0 | $-0.866i$                             | $1.73 \cdot 10^{-16} + 0.929i$               | $486.0 - 282.0i$                               | $539.0 - 319.0i$                                |

Table 24: Case 1a (AlexReis PCWIN64 MR2025b),  $r = 4$ , Z4: zeros.

### 3.4 Poles and zeros visualisation

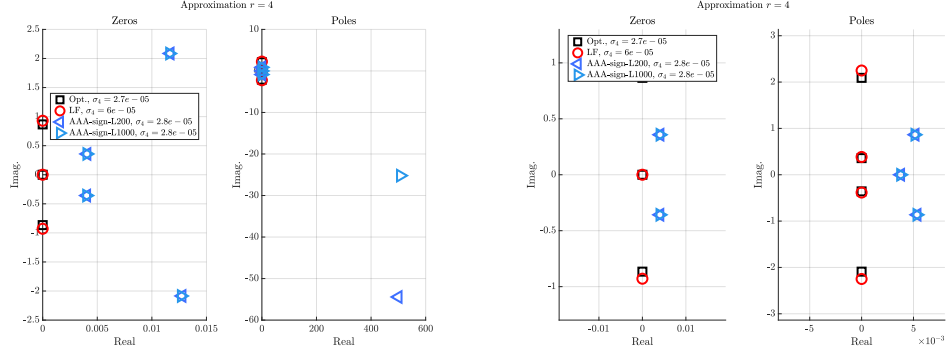


Figure 3: Case 1a (cpv MACA64 MR2023b),  $r = 4$ , Z4. Left frame: zeros (left) and poles (right). Right frame: same but discarding the largest real entry.

### 3.5 Approximation visualization

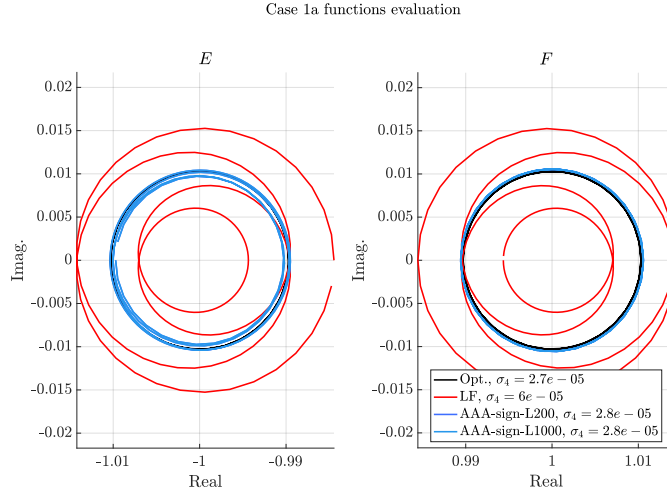


Figure 4: Case 1a (cpv MACA64 MR2023b),  $r = 4$ , Z4.  $E$  and  $F$  evaluations.

## 4 Approximation objective $r = 5$

### 4.1 Numerator and denominator coefficients

The numerator and denominator coefficients (both real and complex parts) are given in the following tables. Each table gives the results obtained on different platform.

| Basis | Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$ |       | LF, $\sigma_5 = 1 \cdot 10^{-05}$ |                       | AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$ |                       | AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$ |                      |
|-------|---------------------------------------|-------|-----------------------------------|-----------------------|------------------------------------------------|-----------------------|-------------------------------------------------|----------------------|
|       | real                                  | imag. | real                              | imag.                 | real                                           | imag.                 | real                                            | imag.                |
| $z^5$ | 0                                     | 0     | 0                                 | 0                     | $-1.1 \cdot 10^{-11}$                          | $-4.8 \cdot 10^{-11}$ | $2.1 \cdot 10^{-13}$                            | $4.5 \cdot 10^{-11}$ |
| $z^4$ | 4.3                                   | 0     | 5.5                               | $-5.9 \cdot 10^{-15}$ | 4.3                                            | $-6.4 \cdot 10^{-13}$ | 4.3                                             | $1.2 \cdot 10^{-12}$ |
| $z^3$ | 0                                     | 0     | $-4.1 \cdot 10^{-15}$             | $-3.4 \cdot 10^{-15}$ | $-8.3 \cdot 10^{-11}$                          | $-3.6 \cdot 10^{-10}$ | $1.5 \cdot 10^{-12}$                            | $3.4 \cdot 10^{-10}$ |
| $z^2$ | 6.5                                   | 0     | 12.0                              | $-1.7 \cdot 10^{-14}$ | 6.5                                            | $-2.9 \cdot 10^{-12}$ | 6.5                                             | $5.2 \cdot 10^{-12}$ |
| $z$   | 0                                     | 0     | $-4.0 \cdot 10^{-15}$             | $-9.0 \cdot 10^{-15}$ | $-3.1 \cdot 10^{-11}$                          | $-1.4 \cdot 10^{-10}$ | $5.5 \cdot 10^{-13}$                            | $1.3 \cdot 10^{-10}$ |
| 1.0   | 0.49                                  | 0     | 1.1                               | $-1.3 \cdot 10^{-15}$ | 0.49                                           | $-3.6 \cdot 10^{-13}$ | 0.49                                            | $6.3 \cdot 10^{-13}$ |
| $z^5$ | 1.0                                   | 0     | 1.0                               | 0                     | 1.0                                            | 0                     | 1.0                                             | 0                    |
| $z^4$ | 0                                     | 0     | $-3.6 \cdot 10^{-15}$             | $-3.0 \cdot 10^{-15}$ | $-4.8 \cdot 10^{-11}$                          | $-2.1 \cdot 10^{-10}$ | $8.8 \cdot 10^{-13}$                            | $2.0 \cdot 10^{-10}$ |
| $z^3$ | 7.5                                   | 0     | 11.0                              | $-1.6 \cdot 10^{-14}$ | 7.5                                            | $-2.2 \cdot 10^{-12}$ | 7.5                                             | $4.0 \cdot 10^{-12}$ |
| $z^2$ | 0                                     | 0     | $-4.4 \cdot 10^{-15}$             | $-1.0 \cdot 10^{-14}$ | $-7.2 \cdot 10^{-11}$                          | $-3.1 \cdot 10^{-10}$ | $1.3 \cdot 10^{-12}$                            | $3.0 \cdot 10^{-10}$ |
| $z$   | 2.8                                   | 0     | 5.8                               | $-7.1 \cdot 10^{-15}$ | 2.8                                            | $-1.7 \cdot 10^{-12}$ | 2.8                                             | $3.0 \cdot 10^{-12}$ |
| 1.0   | 0                                     | 0     | $-9.8 \cdot 10^{-16}$             | $-1.1 \cdot 10^{-15}$ | $-5.4 \cdot 10^{-12}$                          | $-2.4 \cdot 10^{-11}$ | $9.5 \cdot 10^{-14}$                            | $2.3 \cdot 10^{-11}$ |

Table 25: Case 1a (cpv MACA64 MR2023b),  $r = 5$ , Z4: numerator (first lines) and denominator (last lines) coefficients.

| Basis | Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$ |       | LF, $\sigma_5 = 1 \cdot 10^{-05}$ |                       | AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$ |                       | AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$ |                      |
|-------|---------------------------------------|-------|-----------------------------------|-----------------------|------------------------------------------------|-----------------------|-------------------------------------------------|----------------------|
|       | real                                  | imag. | real                              | imag.                 | real                                           | imag.                 | real                                            | imag.                |
| $z^5$ | 0                                     | 0     | 0                                 | 0                     | $-4.1 \cdot 10^{-11}$                          | $3.9 \cdot 10^{-11}$  | $-2.9 \cdot 10^{-14}$                           | $1.2 \cdot 10^{-11}$ |
| $z^4$ | 4.3                                   | 0     | 5.5                               | $-6.5 \cdot 10^{-16}$ | 4.3                                            | $-5.4 \cdot 10^{-13}$ | 4.3                                             | $1.3 \cdot 10^{-13}$ |
| $z^3$ | 0                                     | 0     | $-2.1 \cdot 10^{-14}$             | $4.3 \cdot 10^{-15}$  | $-3.1 \cdot 10^{-10}$                          | $2.9 \cdot 10^{-10}$  | $-2.9 \cdot 10^{-13}$                           | $9.0 \cdot 10^{-11}$ |
| $z^2$ | 6.5                                   | 0     | 12.0                              | $7.5 \cdot 10^{-15}$  | 6.5                                            | $-2.4 \cdot 10^{-12}$ | 6.5                                             | $5.9 \cdot 10^{-13}$ |
| $z$   | 0                                     | 0     | $-1.3 \cdot 10^{-14}$             | $1.6 \cdot 10^{-14}$  | $-1.2 \cdot 10^{-10}$                          | $1.1 \cdot 10^{-10}$  | $-6.8 \cdot 10^{-14}$                           | $3.4 \cdot 10^{-11}$ |
| 1.0   | 0.49                                  | 0     | 1.1                               | $2.0 \cdot 10^{-16}$  | 0.49                                           | $-3.0 \cdot 10^{-13}$ | 0.49                                            | $8.0 \cdot 10^{-14}$ |
| $z^5$ | 1.0                                   | 0     | 1.0                               | 0                     | 1.0                                            | 0                     | 1.0                                             | 0                    |
| $z^4$ | 0                                     | 0     | $-5.8 \cdot 10^{-15}$             | $1.8 \cdot 10^{-15}$  | $-1.8 \cdot 10^{-10}$                          | $1.7 \cdot 10^{-10}$  | $-1.7 \cdot 10^{-13}$                           | $5.2 \cdot 10^{-11}$ |
| $z^3$ | 7.5                                   | 0     | 11.0                              | $3.2 \cdot 10^{-15}$  | 7.5                                            | $-1.8 \cdot 10^{-12}$ | 7.5                                             | $4.5 \cdot 10^{-13}$ |
| $z^2$ | 0                                     | 0     | $-2.3 \cdot 10^{-14}$             | $1.8 \cdot 10^{-14}$  | $-2.7 \cdot 10^{-10}$                          | $2.5 \cdot 10^{-10}$  | $-2.2 \cdot 10^{-13}$                           | $7.8 \cdot 10^{-11}$ |
| $z$   | 2.8                                   | 0     | 5.8                               | $4.1 \cdot 10^{-15}$  | 2.8                                            | $-1.4 \cdot 10^{-12}$ | 2.8                                             | $3.5 \cdot 10^{-13}$ |
| 1.0   | 0                                     | 0     | $-3.4 \cdot 10^{-15}$             | $2.4 \cdot 10^{-15}$  | $-2.0 \cdot 10^{-11}$                          | $1.9 \cdot 10^{-11}$  | $-2.3 \cdot 10^{-15}$                           | $5.9 \cdot 10^{-12}$ |

Table 26: Case 1a (cpv MACI64 MR2022b),  $r = 5$ , Z4: numerator (first lines) and denominator (last lines) coefficients.

| Basis | Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$ |       | LF, $\sigma_5 = 1 \cdot 10^{-05}$ |                       | AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$ |                     | AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$ |                       |
|-------|---------------------------------------|-------|-----------------------------------|-----------------------|------------------------------------------------|---------------------|-------------------------------------------------|-----------------------|
|       | real                                  | imag. | real                              | imag.                 | real                                           | imag.               | real                                            | imag.                 |
| $z^5$ | 0                                     | 0     | 0                                 | 0                     | $1.1 \cdot 10^3$                               | -31.0               | $1.1 \cdot 10^3$                                | -31.0                 |
| $z^4$ | 4.3                                   | 0     | 5.5                               | $-3.7 \cdot 10^{-15}$ | 3.8                                            | 0.021               | 3.8                                             | 0.021                 |
| $z^3$ | 0                                     | 0     | $-6.7 \cdot 10^{-15}$             | $2.5 \cdot 10^{-15}$  | $8.1 \cdot 10^3$                               | -240.0              | $8.1 \cdot 10^3$                                | -230.0                |
| $z^2$ | 6.5                                   | 0     | 12.0                              | $-1.4 \cdot 10^{-14}$ | 4.1                                            | 0.086               | 4.1                                             | 0.085                 |
| $z$   | 0                                     | 0     | $-1.4 \cdot 10^{-14}$             | $6.9 \cdot 10^{-17}$  | $3.1 \cdot 10^3$                               | -89.0               | $3.1 \cdot 10^3$                                | -87.0                 |
| 1.0   | 0.49                                  | 0     | 1.1                               | $-1.9 \cdot 10^{-15}$ | 0.19                                           | $9.6 \cdot 10^{-3}$ | 0.19                                            | $9.5 \cdot 10^{-3}$   |
| $z^5$ | 1.0                                   | 0     | 1.0                               | 0                     | 1.0                                            | 0                   | 1.0                                             | $-5.8 \cdot 10^{-17}$ |
| $z^4$ | 0                                     | 0     | $1.9 \cdot 10^{-16}$              | $-1.4 \cdot 10^{-15}$ | $4.7 \cdot 10^3$                               | -140.0              | $4.7 \cdot 10^3$                                | -130.0                |
| $z^3$ | 7.5                                   | 0     | 11.0                              | $-1.1 \cdot 10^{-14}$ | 5.7                                            | 0.071               | 5.7                                             | 0.07                  |
| $z^2$ | 0                                     | 0     | $-1.7 \cdot 10^{-14}$             | $3.9 \cdot 10^{-15}$  | $7.1 \cdot 10^3$                               | -200.0              | $7.1 \cdot 10^3$                                | -200.0                |
| $z$   | 2.8                                   | 0     | 5.8                               | $-8.6 \cdot 10^{-15}$ | 1.4                                            | 0.047               | 1.4                                             | 0.046                 |
| 1.0   | 0                                     | 0     | $-3.3 \cdot 10^{-15}$             | $5.1 \cdot 10^{-16}$  | 530.0                                          | -15.0               | 530.0                                           | -15.0                 |

Table 27: Case 1a (cpv PCWIN64 MR2025b),  $r = 5$ , Z4: numerator (first lines) and denominator (last lines) coefficients.

| Basis | Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$ |       | LF, $\sigma_5 = 1 \cdot 10^{-05}$ |                       | AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$ |                     | AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$ |                       |
|-------|---------------------------------------|-------|-----------------------------------|-----------------------|------------------------------------------------|---------------------|-------------------------------------------------|-----------------------|
|       | real                                  | imag. | real                              | imag.                 | real                                           | imag.               | real                                            | imag.                 |
| $z^5$ | 0                                     | 0     | 0                                 | 0                     | $1.1 \cdot 10^3$                               | -31.0               | $1.1 \cdot 10^3$                                | -31.0                 |
| $z^4$ | 4.3                                   | 0     | 5.5                               | $-3.7 \cdot 10^{-15}$ | 3.8                                            | 0.021               | 3.8                                             | 0.021                 |
| $z^3$ | 0                                     | 0     | $-6.7 \cdot 10^{-15}$             | $2.5 \cdot 10^{-15}$  | $8.1 \cdot 10^3$                               | -240.0              | $8.1 \cdot 10^3$                                | -230.0                |
| $z^2$ | 6.5                                   | 0     | 12.0                              | $-1.4 \cdot 10^{-14}$ | 4.1                                            | 0.086               | 4.1                                             | 0.085                 |
| $z$   | 0                                     | 0     | $-1.4 \cdot 10^{-14}$             | $6.9 \cdot 10^{-17}$  | $3.1 \cdot 10^3$                               | -89.0               | $3.1 \cdot 10^3$                                | -87.0                 |
| 1.0   | 0.49                                  | 0     | 1.1                               | $-1.9 \cdot 10^{-15}$ | 0.19                                           | $9.6 \cdot 10^{-3}$ | 0.19                                            | $9.5 \cdot 10^{-3}$   |
| $z^5$ | 1.0                                   | 0     | 1.0                               | 0                     | 1.0                                            | 0                   | 1.0                                             | $-5.8 \cdot 10^{-17}$ |
| $z^4$ | 0                                     | 0     | $1.9 \cdot 10^{-16}$              | $-1.4 \cdot 10^{-15}$ | $4.7 \cdot 10^3$                               | -140.0              | $4.7 \cdot 10^3$                                | -130.0                |
| $z^3$ | 7.5                                   | 0     | 11.0                              | $-1.1 \cdot 10^{-14}$ | 5.7                                            | 0.071               | 5.7                                             | 0.07                  |
| $z^2$ | 0                                     | 0     | $-1.7 \cdot 10^{-14}$             | $3.9 \cdot 10^{-15}$  | $7.1 \cdot 10^3$                               | -200.0              | $7.1 \cdot 10^3$                                | -200.0                |
| $z$   | 2.8                                   | 0     | 5.8                               | $-8.6 \cdot 10^{-15}$ | 1.4                                            | 0.047               | 1.4                                             | 0.046                 |
| 1.0   | 0                                     | 0     | $-3.3 \cdot 10^{-15}$             | $5.1 \cdot 10^{-16}$  | 530.0                                          | -15.0               | 530.0                                           | -15.0                 |

Table 28: Case 1a (AlexReis PCWIN64 MR2025b),  $r = 5$ , Z4: numerator (first lines) and denominator (last lines) coefficients.

## 4.2 Poles

The poles are given in the following tables. Each table gives the results obtained on different platform.

|     | Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$ | LF, $\sigma_5 = 1 \cdot 10^{-05}$            | AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$ | AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$ |
|-----|---------------------------------------|----------------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | 0                                     | $1.69 \cdot 10^{-16} + 1.84 \cdot 10^{-16}i$ | $1.92 \cdot 10^{-12} + 8.44 \cdot 10^{-12}i$   | $-3.41 \cdot 10^{-14} - 8.04 \cdot 10^{-12}i$   |
| 2.0 | $3.89 \cdot 10^{-16} + 2.67i$         | $2.45 \cdot 10^{-16} - 0.729i$               | $2.84 \cdot 10^{-12} - 0.629i$                 | $-2.12 \cdot 10^{-13} + 0.629i$                 |
| 3.0 | $3.89 \cdot 10^{-16} - 2.67i$         | $-1.57 \cdot 10^{-16} + 0.729i$              | $3.03 \cdot 10^{-12} + 0.629i$                 | $1.08 \cdot 10^{-13} - 0.629i$                  |
| 4.0 | $-5.55 \cdot 10^{-17} + 0.629i$       | $-1.18 \cdot 10^{-15} - 3.3i$                | $1.98 \cdot 10^{-11} - 2.67i$                  | $-1.09 \cdot 10^{-12} + 2.67i$                  |
| 5.0 | $-5.55 \cdot 10^{-17} - 0.629i$       | $4.72 \cdot 10^{-15} + 3.3i$                 | $2.05 \cdot 10^{-11} + 2.67i$                  | $3.47 \cdot 10^{-13} - 2.67i$                   |

Table 29: Case 1a (cpv MACA64 MR2023b),  $r = 5$ , Z4: poles.

|     | Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$ | LF, $\sigma_5 = 1 \cdot 10^{-05}$            | AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$ | AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$ |
|-----|---------------------------------------|----------------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | 0                                     | $5.89 \cdot 10^{-16} - 4.22 \cdot 10^{-16}i$ | $7.1 \cdot 10^{-12} - 6.72 \cdot 10^{-12}i$    | $-8.34 \cdot 10^{-16} - 2.1 \cdot 10^{-12}i$    |
| 2.0 | $3.89 \cdot 10^{-16} + 2.67i$         | $4.48 \cdot 10^{-16} + 0.729i$               | $1.09 \cdot 10^{-11} + 0.629i$                 | $-8.47 \cdot 10^{-15} + 0.629i$                 |
| 3.0 | $3.89 \cdot 10^{-16} - 2.67i$         | $8.9 \cdot 10^{-16} - 0.729i$                | $1.08 \cdot 10^{-11} - 0.629i$                 | $3.14 \cdot 10^{-14} - 0.629i$                  |
| 4.0 | $-5.55 \cdot 10^{-17} + 0.629i$       | $1.51 \cdot 10^{-15} + 3.3i$                 | $7.46 \cdot 10^{-11} + 2.67i$                  | $-7.54 \cdot 10^{-15} + 2.67i$                  |
| 5.0 | $-5.55 \cdot 10^{-17} - 0.629i$       | $2.43 \cdot 10^{-15} - 3.3i$                 | $7.4 \cdot 10^{-11} - 2.67i$                   | $1.54 \cdot 10^{-13} - 2.67i$                   |

Table 30: Case 1a (cpv MACI64 MR2022b),  $r = 5$ , Z4: poles.

|     | Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$ | LF, $\sigma_5 = 1 \cdot 10^{-05}$            | AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$ | AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$ |
|-----|---------------------------------------|----------------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | 0                                     | $5.62 \cdot 10^{-16} - 8.81 \cdot 10^{-17}i$ | $-7.91 \cdot 10^{-5} + 0.281i$                 | $-7.91 \cdot 10^{-5} + 0.281i$                  |
| 2.0 | $3.89 \cdot 10^{-16} + 2.67i$         | $7.72 \cdot 10^{-16} + 0.729i$               | $-7.96 \cdot 10^{-5} - 0.281i$                 | $-7.95 \cdot 10^{-5} - 0.281i$                  |
| 3.0 | $3.89 \cdot 10^{-16} - 2.67i$         | $2.77 \cdot 10^{-16} - 0.729i$               | $-3.61 \cdot 10^{-4} + 1.19i$                  | $-3.61 \cdot 10^{-4} + 1.19i$                   |
| 4.0 | $-5.55 \cdot 10^{-17} + 0.629i$       | $-2.67 \cdot 10^{-15} - 3.3i$                | $-3.65 \cdot 10^{-4} - 1.19i$                  | $-3.65 \cdot 10^{-4} - 1.19i$                   |
| 5.0 | $-5.55 \cdot 10^{-17} - 0.629i$       | $1.2 \cdot 10^{-15} + 3.3i$                  | $-4700.0 + 136.0i$                             | $-4700.0 + 134.0i$                              |

Table 31: Case 1a (cpv PCWIN64 MR2025b),  $r = 5$ , Z4: poles.

|     | Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$ | LF, $\sigma_5 = 1 \cdot 10^{-05}$            | AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$ | AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$ |
|-----|---------------------------------------|----------------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | 0                                     | $5.62 \cdot 10^{-16} - 8.81 \cdot 10^{-17}i$ | $-7.91 \cdot 10^{-5} + 0.281i$                 | $-7.91 \cdot 10^{-5} + 0.281i$                  |
| 2.0 | $3.89 \cdot 10^{-16} + 2.67i$         | $7.72 \cdot 10^{-16} + 0.729i$               | $-7.96 \cdot 10^{-5} - 0.281i$                 | $-7.95 \cdot 10^{-5} - 0.281i$                  |
| 3.0 | $3.89 \cdot 10^{-16} - 2.67i$         | $2.77 \cdot 10^{-16} - 0.729i$               | $-3.61 \cdot 10^{-4} + 1.19i$                  | $-3.61 \cdot 10^{-4} + 1.19i$                   |
| 4.0 | $-5.55 \cdot 10^{-17} + 0.629i$       | $-2.67 \cdot 10^{-15} - 3.3i$                | $-3.65 \cdot 10^{-4} - 1.19i$                  | $-3.65 \cdot 10^{-4} - 1.19i$                   |
| 5.0 | $-5.55 \cdot 10^{-17} - 0.629i$       | $1.2 \cdot 10^{-15} + 3.3i$                  | $-4700.0 + 136.0i$                             | $-4700.0 + 134.0i$                              |

Table 32: Case 1a (AlexReis PCWIN64 MR2025b),  $r = 5$ , Z4: poles.

### 4.3 Zeros

The zeros are given in the following tables. Each table gives the results obtained on different platform.

|     | Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$ | LF, $\sigma_5 = 1 \cdot 10^{-05}$ | AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$ | AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$ |
|-----|---------------------------------------|-----------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | NaN                                   | NaN                               | $2.08 \cdot 10^{-12} - 0.281z$                 | $-1.06 \cdot 10^{-13} + 0.281z$                 |
| 2.0 | $5.55 \cdot 10^{-17} + 1.19z$         | $2.26 \cdot 10^{-16} - 0.32z$     | $2.16 \cdot 10^{-12} + 0.281z$                 | $3.17 \cdot 10^{-14} - 0.281z$                  |
| 3.0 | $5.55 \cdot 10^{-17} - 1.19z$         | $1.36 \cdot 10^{-16} + 0.32z$     | $5.38 \cdot 10^{-12} - 1.19z$                  | $-4.15 \cdot 10^{-13} + 1.19z$                  |
| 4.0 | $6.94 \cdot 10^{-17} + 0.281z$        | $5.92 \cdot 10^{-16} + 1.42z$     | $5.74 \cdot 10^{-12} + 1.19z$                  | $2.15 \cdot 10^{-13} - 1.19z$                   |
| 5.0 | $6.94 \cdot 10^{-17} - 0.281z$        | $-1.68 \cdot 10^{-16} - 1.42z$    | $1.99 \cdot 10^{10} - 8.58 \cdot 10^{10}z$     | $-4.5 \cdot 10^8 + 9.58 \cdot 10^{10}z$         |

Table 33: Case 1a (cpv MACA64 MR2023b),  $r = 5$ , Z4: zeros.

|     | Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$ | LF, $\sigma_5 = 1 \cdot 10^{-05}$ | AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$ | AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$ |
|-----|---------------------------------------|-----------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | NaN                                   | NaN                               | $7.88 \cdot 10^{-12} + 0.281z$                 | $-7.88 \cdot 10^{-15} + 0.281z$                 |
| 2.0 | $5.55 \cdot 10^{-17} + 1.19z$         | $5.94 \cdot 10^{-16} + 0.32z$     | $7.81 \cdot 10^{-12} - 0.281z$                 | $1.46 \cdot 10^{-14} - 0.281z$                  |
| 3.0 | $5.55 \cdot 10^{-17} - 1.19z$         | $4.39 \cdot 10^{-16} - 0.32z$     | $2.07 \cdot 10^{-11} + 1.19z$                  | $-9.47 \cdot 10^{-15} + 1.19z$                  |
| 4.0 | $6.94 \cdot 10^{-17} + 0.281z$        | $1.88 \cdot 10^{-15} - 1.42z$     | $2.04 \cdot 10^{-11} - 1.19z$                  | $6.07 \cdot 10^{-14} - 1.19z$                   |
| 5.0 | $6.94 \cdot 10^{-17} - 0.281z$        | $9.89 \cdot 10^{-16} + 1.42z$     | $5.49 \cdot 10^{10} + 5.28 \cdot 10^{10}z$     | $8.85 \cdot 10^8 + 3.6 \cdot 10^{11}z$          |

Table 34: Case 1a (cpv MACI64 MR2022b),  $r = 5$ , Z4: zeros.

|     | Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$ | LF, $\sigma_5 = 1 \cdot 10^{-05}$ | AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$ | AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$ |
|-----|---------------------------------------|-----------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | NaN                                   | NaN                               | $-6.3 \cdot 10^{-5} - 4.98 \cdot 10^{-6}z$     | $-6.3 \cdot 10^{-5} - 4.9 \cdot 10^{-6}z$       |
| 2.0 | $5.55 \cdot 10^{-17} + 1.19z$         | $5.2 \cdot 10^{-16} - 0.32z$      | $-1.45 \cdot 10^{-4} + 0.629z$                 | $-1.45 \cdot 10^{-4} + 0.629z$                  |
| 3.0 | $5.55 \cdot 10^{-17} - 1.19z$         | $7.03 \cdot 10^{-16} + 0.32z$     | $-1.46 \cdot 10^{-4} - 0.629z$                 | $-1.46 \cdot 10^{-4} - 0.629z$                  |
| 4.0 | $6.94 \cdot 10^{-17} + 0.281z$        | $4.37 \cdot 10^{-16} + 1.42z$     | $-0.00156 + 2.66z$                             | $-0.00156 + 2.66z$                              |
| 5.0 | $6.94 \cdot 10^{-17} - 0.281z$        | $-4.16 \cdot 10^{-16} - 1.42z$    | $-0.00158 - 2.67z$                             | $-0.00158 - 2.67z$                              |

Table 35: Case 1a (cpv PCWIN64 MR2025b),  $r = 5$ , Z4: zeros.

|     | Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$ | LF, $\sigma_5 = 1 \cdot 10^{-05}$ | AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$ | AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$ |
|-----|---------------------------------------|-----------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | NaN                                   | NaN                               | $-6.3 \cdot 10^{-5} - 4.98 \cdot 10^{-6}z$     | $-6.3 \cdot 10^{-5} - 4.9 \cdot 10^{-6}z$       |
| 2.0 | $5.55 \cdot 10^{-17} + 1.19z$         | $5.2 \cdot 10^{-16} - 0.32z$      | $-1.45 \cdot 10^{-4} + 0.629z$                 | $-1.45 \cdot 10^{-4} + 0.629z$                  |
| 3.0 | $5.55 \cdot 10^{-17} - 1.19z$         | $7.03 \cdot 10^{-16} + 0.32z$     | $-1.46 \cdot 10^{-4} - 0.629z$                 | $-1.46 \cdot 10^{-4} - 0.629z$                  |
| 4.0 | $6.94 \cdot 10^{-17} + 0.281z$        | $4.37 \cdot 10^{-16} + 1.42z$     | $-0.00156 + 2.66z$                             | $-0.00156 + 2.66z$                              |
| 5.0 | $6.94 \cdot 10^{-17} - 0.281z$        | $-4.16 \cdot 10^{-16} - 1.42z$    | $-0.00158 - 2.67z$                             | $-0.00158 - 2.67z$                              |

Table 36: Case 1a (AlexReis PCWIN64 MR2025b),  $r = 5$ , Z4: zeros.

## 4.4 Poles and zeros visualisation

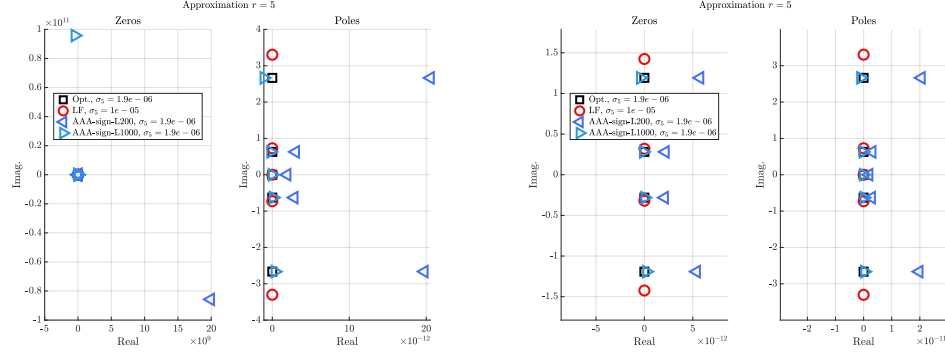


Figure 5: Case 1a (cpv MACA64 MR2023b),  $r = 5$ , Z4. Left frame: zeros (left) and poles (right). Right frame: same but discarding the largest real entry.

## 4.5 Approximation visualization

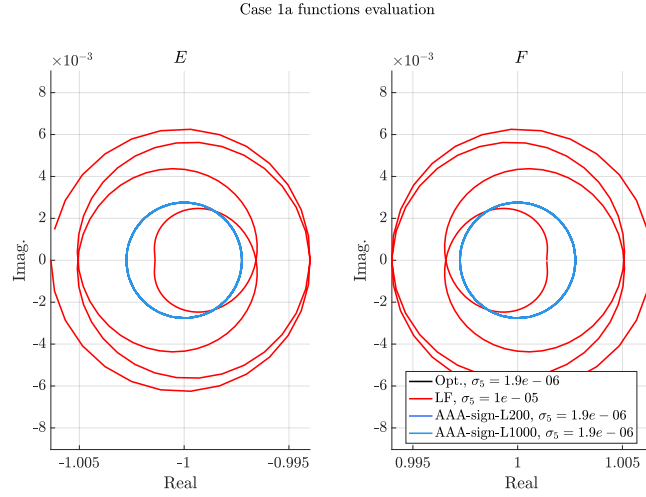


Figure 6: Case 1a (cpv MACA64 MR2023b),  $r = 5$ , Z4.  $E$  and  $F$  evaluations.

## 5 Approximation objective $r = 6$

### 5.1 Numerator and denominator coefficients

The numerator and denominator coefficients (both real and complex parts) are given in the following tables. Each table gives the results obtained on different platform.

| Basis | Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$ |       | LF, $\sigma_6 = 3.2 \cdot 10^{-07}$ |                       | AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$ |                       | AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$ |                       |
|-------|---------------------------------------|-------|-------------------------------------|-----------------------|------------------------------------------------|-----------------------|-------------------------------------------------|-----------------------|
|       | real                                  | imag. | real                                | imag.                 | real                                           | imag.                 | real                                            | imag.                 |
| $z^6$ | 0                                     | 0     | 0                                   | 0                     | $-1.2 \cdot 10^{-10}$                          | $-2.0 \cdot 10^{-10}$ | $1.9 \cdot 10^{-12}$                            | $-8.0 \cdot 10^{-11}$ |
| $z^5$ | 5.2                                   | 0     | 5.5                                 | $-5.7 \cdot 10^{-15}$ | 5.2                                            | $-6.3 \cdot 10^{-12}$ | 5.2                                             | $-4.6 \cdot 10^{-12}$ |
| $z^4$ | 0                                     | 0     | $-2.3 \cdot 10^{-14}$               | $-9.3 \cdot 10^{-15}$ | $-1.4 \cdot 10^{-9}$                           | $-2.3 \cdot 10^{-9}$  | $2.1 \cdot 10^{-11}$                            | $-9.2 \cdot 10^{-10}$ |
| $z^3$ | 13.0                                  | 0     | 15.0                                | $-1.1 \cdot 10^{-14}$ | 13.0                                           | $-4.7 \cdot 10^{-11}$ | 13.0                                            | $-3.5 \cdot 10^{-11}$ |
| $z^2$ | 0                                     | 0     | $-4.2 \cdot 10^{-14}$               | $-2.7 \cdot 10^{-14}$ | $-1.1 \cdot 10^{-9}$                           | $-1.7 \cdot 10^{-9}$  | $1.5 \cdot 10^{-11}$                            | $-7.0 \cdot 10^{-10}$ |
| $z$   | 2.9                                   | 0     | 3.7                                 | 0                     | 2.9                                            | $-1.7 \cdot 10^{-11}$ | 2.9                                             | $-1.3 \cdot 10^{-11}$ |
| 1.0   | 0                                     | 0     | $-4.1 \cdot 10^{-15}$               | $-3.2 \cdot 10^{-15}$ | $-5.3 \cdot 10^{-11}$                          | $-8.7 \cdot 10^{-11}$ | $7.8 \cdot 10^{-13}$                            | $-3.6 \cdot 10^{-11}$ |
| $z^6$ | 1.0                                   | 0     | 1.0                                 | 0                     | 1.0                                            | 0                     | 1.0                                             | 0                     |
| $z^5$ | 0                                     | 0     | $-6.9 \cdot 10^{-15}$               | $-2.9 \cdot 10^{-15}$ | $-6.5 \cdot 10^{-10}$                          | $-1.0 \cdot 10^{-9}$  | $9.8 \cdot 10^{-12}$                            | $-4.2 \cdot 10^{-10}$ |
| $z^4$ | 11.0                                  | 0     | 12.0                                | $-1.5 \cdot 10^{-14}$ | 11.0                                           | $-2.7 \cdot 10^{-11}$ | 11.0                                            | $-2.0 \cdot 10^{-11}$ |
| $z^3$ | 0                                     | 0     | $-4.3 \cdot 10^{-14}$               | $-2.8 \cdot 10^{-14}$ | $-1.6 \cdot 10^{-9}$                           | $-2.6 \cdot 10^{-9}$  | $2.4 \cdot 10^{-11}$                            | $-1.1 \cdot 10^{-9}$  |
| $z^2$ | 8.4                                   | 0     | 10.0                                | $-1.5 \cdot 10^{-15}$ | 8.4                                            | $-4.0 \cdot 10^{-11}$ | 8.4                                             | $-3.0 \cdot 10^{-11}$ |
| $z$   | 0                                     | 0     | $-2.0 \cdot 10^{-14}$               | $-1.1 \cdot 10^{-14}$ | $-3.6 \cdot 10^{-10}$                          | $-6.0 \cdot 10^{-10}$ | $5.3 \cdot 10^{-12}$                            | $-2.5 \cdot 10^{-10}$ |
| 1.0   | 0.42                                  | 0     | 0.56                                | $-5.3 \cdot 10^{-16}$ | 0.42                                           | $-3.0 \cdot 10^{-12}$ | 0.42                                            | $-2.3 \cdot 10^{-12}$ |

Table 37: Case 1a (cpv MACA64 MR2023b),  $r = 6$ , Z4: numerator (first lines) and denominator (last lines) coefficients.

| Basis | Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$ |       | LF, $\sigma_6 = 3.2 \cdot 10^{-07}$ |                       | AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$ |                       | AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$ |                      |
|-------|---------------------------------------|-------|-------------------------------------|-----------------------|------------------------------------------------|-----------------------|-------------------------------------------------|----------------------|
|       | real                                  | imag. | real                                | imag.                 | real                                           | imag.                 | real                                            | imag.                |
| $z^6$ | 0                                     | 0     | 0                                   | 0                     | $-1.2 \cdot 10^{-10}$                          | $-6.5 \cdot 10^{-11}$ | $-1.8 \cdot 10^{-12}$                           | $3.7 \cdot 10^{-10}$ |
| $z^5$ | 5.2                                   | 0     | 5.5                                 | $-8.5 \cdot 10^{-16}$ | 5.2                                            | $4.5 \cdot 10^{-12}$  | 5.2                                             | $5.7 \cdot 10^{-12}$ |
| $z^4$ | 0                                     | 0     | $-2.2 \cdot 10^{-14}$               | $-7.0 \cdot 10^{-15}$ | $-1.4 \cdot 10^{-9}$                           | $-7.1 \cdot 10^{-10}$ | $-2.0 \cdot 10^{-11}$                           | $4.2 \cdot 10^{-9}$  |
| $z^3$ | 13.0                                  | 0     | 15.0                                | $-2.3 \cdot 10^{-15}$ | 13.0                                           | $3.3 \cdot 10^{-11}$  | 13.0                                            | $4.2 \cdot 10^{-11}$ |
| $z^2$ | 0                                     | 0     | $-3.0 \cdot 10^{-14}$               | $-6.1 \cdot 10^{-15}$ | $-1.0 \cdot 10^{-9}$                           | $-5.2 \cdot 10^{-10}$ | $-1.5 \cdot 10^{-11}$                           | $3.2 \cdot 10^{-9}$  |
| $z$   | 2.9                                   | 0     | 3.7                                 | $-4.8 \cdot 10^{-15}$ | 2.9                                            | $1.2 \cdot 10^{-11}$  | 2.9                                             | $1.6 \cdot 10^{-11}$ |
| 1.0   | 0                                     | 0     | $-8.2 \cdot 10^{-16}$               | $-2.6 \cdot 10^{-16}$ | $-5.2 \cdot 10^{-11}$                          | $-2.5 \cdot 10^{-11}$ | $-7.2 \cdot 10^{-13}$                           | $1.6 \cdot 10^{-10}$ |
| $z^6$ | 1.0                                   | 0     | 1.0                                 | $3.5 \cdot 10^{-17}$  | 1.0                                            | $-3.6 \cdot 10^{-18}$ | 1.0                                             | 0                    |
| $z^5$ | 0                                     | 0     | $-4.6 \cdot 10^{-15}$               | $-1.1 \cdot 10^{-15}$ | $-6.4 \cdot 10^{-10}$                          | $-3.3 \cdot 10^{-10}$ | $-9.4 \cdot 10^{-12}$                           | $1.9 \cdot 10^{-9}$  |
| $z^4$ | 11.0                                  | 0     | 12.0                                | $5.7 \cdot 10^{-16}$  | 11.0                                           | $1.9 \cdot 10^{-11}$  | 11.0                                            | $2.4 \cdot 10^{-11}$ |
| $z^3$ | 0                                     | 0     | $-3.3 \cdot 10^{-14}$               | $-3.1 \cdot 10^{-15}$ | $-1.6 \cdot 10^{-9}$                           | $-8.1 \cdot 10^{-10}$ | $-2.3 \cdot 10^{-11}$                           | $4.9 \cdot 10^{-9}$  |
| $z^2$ | 8.4                                   | 0     | 10.0                                | $-9.1 \cdot 10^{-15}$ | 8.4                                            | $2.9 \cdot 10^{-11}$  | 8.4                                             | $3.6 \cdot 10^{-11}$ |
| $z$   | 0                                     | 0     | $-1.2 \cdot 10^{-14}$               | $-6.9 \cdot 10^{-15}$ | $-3.6 \cdot 10^{-10}$                          | $-1.8 \cdot 10^{-10}$ | $-5.0 \cdot 10^{-12}$                           | $1.1 \cdot 10^{-9}$  |
| 1.0   | 0.42                                  | 0     | 0.56                                | $3.5 \cdot 10^{-17}$  | 0.42                                           | $2.1 \cdot 10^{-12}$  | 0.42                                            | $2.7 \cdot 10^{-12}$ |

Table 38: Case 1a (cpv MACI64 MR2022b),  $r = 6$ , Z4: numerator (first lines) and denominator (last lines) coefficients.

| Basis | Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$ |       | LF, $\sigma_6 = 3.2 \cdot 10^{-07}$ |                       | AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$ |                      | AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$ |                      |
|-------|---------------------------------------|-------|-------------------------------------|-----------------------|------------------------------------------------|----------------------|-------------------------------------------------|----------------------|
|       | real                                  | imag. | real                                | imag.                 | real                                           | imag.                | real                                            | imag.                |
| $z^6$ | 0                                     | 0     | 0                                   | 0                     | $-4.4 \cdot 10^8$                              | $-2.2 \cdot 10^9$    | $-6.7 \cdot 10^7$                               | $-6.4 \cdot 10^9$    |
| $z^5$ | 5.2                                   | 0     | 5.5                                 | $-4.0 \cdot 10^{-15}$ | 5.2                                            | $-4.6 \cdot 10^{-4}$ | 5.2                                             | $5.3 \cdot 10^{-4}$  |
| $z^4$ | 0                                     | 0     | $-2.4 \cdot 10^{-16}$               | $1.1 \cdot 10^{-14}$  | $-5.0 \cdot 10^9$                              | $-2.5 \cdot 10^{10}$ | $-7.5 \cdot 10^8$                               | $-7.2 \cdot 10^{10}$ |
| $z^3$ | 13.0                                  | 0     | 15.0                                | $-1.9 \cdot 10^{-14}$ | 13.0                                           | $-3.2 \cdot 10^{-3}$ | 13.0                                            | $3.8 \cdot 10^{-3}$  |
| $z^2$ | 0                                     | 0     | $-1.8 \cdot 10^{-15}$               | $8.2 \cdot 10^{-15}$  | $-3.7 \cdot 10^9$                              | $-1.9 \cdot 10^{10}$ | $-5.6 \cdot 10^8$                               | $-5.4 \cdot 10^{10}$ |
| $z$   | 2.9                                   | 0     | 3.7                                 | $-5.4 \cdot 10^{-15}$ | 2.9                                            | $-1.3 \cdot 10^{-3}$ | 2.9                                             | $1.7 \cdot 10^{-3}$  |
| 1.0   | 0                                     | 0     | $-4.3 \cdot 10^{-16}$               | $3.7 \cdot 10^{-16}$  | $-1.9 \cdot 10^8$                              | $-9.4 \cdot 10^8$    | $-2.8 \cdot 10^7$                               | $-2.7 \cdot 10^9$    |
| $z^6$ | 1.0                                   | 0     | 1.0                                 | 0                     | 1.0                                            | 0                    | 1.0                                             | 0                    |
| $z^5$ | 0                                     | 0     | $1.8 \cdot 10^{-15}$                | $-9.9 \cdot 10^{-17}$ | $-2.3 \cdot 10^9$                              | $-1.2 \cdot 10^{10}$ | $-3.5 \cdot 10^8$                               | $-3.3 \cdot 10^{10}$ |
| $z^4$ | 11.0                                  | 0     | 12.0                                | $-1.1 \cdot 10^{-14}$ | 11.0                                           | $-1.9 \cdot 10^{-3}$ | 11.0                                            | $2.1 \cdot 10^{-3}$  |
| $z^3$ | 0                                     | 0     | $-4.6 \cdot 10^{-15}$               | $1.6 \cdot 10^{-14}$  | $-5.8 \cdot 10^9$                              | $-2.9 \cdot 10^{10}$ | $-8.7 \cdot 10^8$                               | $-8.3 \cdot 10^{10}$ |
| $z^2$ | 8.4                                   | 0     | 10.0                                | $-1.8 \cdot 10^{-14}$ | 8.5                                            | $-2.8 \cdot 10^{-3}$ | 8.4                                             | $3.6 \cdot 10^{-3}$  |
| $z$   | 0                                     | 0     | $7.3 \cdot 10^{-19}$                | $3.8 \cdot 10^{-15}$  | $-1.3 \cdot 10^9$                              | $-6.5 \cdot 10^9$    | $-2.0 \cdot 10^8$                               | $-1.9 \cdot 10^{10}$ |
| 1.0   | 0.42                                  | 0     | 0.56                                | $1.4 \cdot 10^{-16}$  | 0.42                                           | $-2.4 \cdot 10^{-4}$ | 0.42                                            | $2.7 \cdot 10^{-4}$  |

Table 39: Case 1a (cpv PCWIN64 MR2025b),  $r = 6$ , Z4: numerator (first lines) and denominator (last lines) coefficients.

| Basis | Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$ |       | LF, $\sigma_6 = 3.2 \cdot 10^{-07}$ |                       | AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$ |                      | AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$ |                      |
|-------|---------------------------------------|-------|-------------------------------------|-----------------------|------------------------------------------------|----------------------|-------------------------------------------------|----------------------|
|       | real                                  | imag. | real                                | imag.                 | real                                           | imag.                | real                                            | imag.                |
| $z^6$ | 0                                     | 0     | 0                                   | 0                     | $-4.4 \cdot 10^8$                              | $-2.2 \cdot 10^9$    | $-6.7 \cdot 10^7$                               | $-6.4 \cdot 10^9$    |
| $z^5$ | 5.2                                   | 0     | 5.5                                 | $-4.0 \cdot 10^{-15}$ | 5.2                                            | $-4.6 \cdot 10^{-4}$ | 5.2                                             | $5.3 \cdot 10^{-4}$  |
| $z^4$ | 0                                     | 0     | $-2.4 \cdot 10^{-16}$               | $1.1 \cdot 10^{-14}$  | $-5.0 \cdot 10^9$                              | $-2.5 \cdot 10^{10}$ | $-7.5 \cdot 10^8$                               | $-7.2 \cdot 10^{10}$ |
| $z^3$ | 13.0                                  | 0     | 15.0                                | $-1.9 \cdot 10^{-14}$ | 13.0                                           | $-3.2 \cdot 10^{-3}$ | 13.0                                            | $3.8 \cdot 10^{-3}$  |
| $z^2$ | 0                                     | 0     | $-1.8 \cdot 10^{-15}$               | $8.2 \cdot 10^{-15}$  | $-3.7 \cdot 10^9$                              | $-1.9 \cdot 10^{10}$ | $-5.6 \cdot 10^8$                               | $-5.4 \cdot 10^{10}$ |
| $z$   | 2.9                                   | 0     | 3.7                                 | $-5.4 \cdot 10^{-15}$ | 2.9                                            | $-1.3 \cdot 10^{-3}$ | 2.9                                             | $1.7 \cdot 10^{-3}$  |
| 1.0   | 0                                     | 0     | $-4.3 \cdot 10^{-16}$               | $3.7 \cdot 10^{-16}$  | $-1.9 \cdot 10^8$                              | $-9.4 \cdot 10^8$    | $-2.8 \cdot 10^7$                               | $-2.7 \cdot 10^9$    |
| $z^6$ | 1.0                                   | 0     | 1.0                                 | 0                     | 1.0                                            | 0                    | 1.0                                             | 0                    |
| $z^5$ | 0                                     | 0     | $1.8 \cdot 10^{-15}$                | $-9.9 \cdot 10^{-17}$ | $-2.3 \cdot 10^9$                              | $-1.2 \cdot 10^{10}$ | $-3.5 \cdot 10^8$                               | $-3.3 \cdot 10^{10}$ |
| $z^4$ | 11.0                                  | 0     | 12.0                                | $-1.1 \cdot 10^{-14}$ | 11.0                                           | $-1.9 \cdot 10^{-3}$ | 11.0                                            | $2.1 \cdot 10^{-3}$  |
| $z^3$ | 0                                     | 0     | $-4.6 \cdot 10^{-15}$               | $1.6 \cdot 10^{-14}$  | $-5.8 \cdot 10^9$                              | $-2.9 \cdot 10^{10}$ | $-8.7 \cdot 10^8$                               | $-8.3 \cdot 10^{10}$ |
| $z^2$ | 8.4                                   | 0     | 10.0                                | $-1.8 \cdot 10^{-14}$ | 8.5                                            | $-2.8 \cdot 10^{-3}$ | 8.4                                             | $3.6 \cdot 10^{-3}$  |
| $z$   | 0                                     | 0     | $7.3 \cdot 10^{-19}$                | $3.8 \cdot 10^{-15}$  | $-1.3 \cdot 10^9$                              | $-6.5 \cdot 10^9$    | $-2.0 \cdot 10^8$                               | $-1.9 \cdot 10^{10}$ |
| 1.0   | 0.42                                  | 0     | 0.56                                | $1.4 \cdot 10^{-16}$  | 0.42                                           | $-2.4 \cdot 10^{-4}$ | 0.42                                            | $2.7 \cdot 10^{-4}$  |

Table 40: Case 1a (AlexReis PCWIN64 MR2025b),  $r = 6$ , Z4: numerator (first lines) and denominator (last lines) coefficients.

## 5.2 Poles

The poles are given in the following tables. Each table gives the results obtained on different platform.

|     | Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$ | LF, $\sigma_6 = 3.2 \cdot 10^{-07}$ | AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$ | AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$ |
|-----|---------------------------------------|-------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | $5.55 \cdot 10^{-17} + 3.23i$         | $8.54 \cdot 10^{-16} - 0.243i$      | $1.9 \cdot 10^{-11} - 0.232i$                  | $-4.84 \cdot 10^{-13} - 0.232i$                 |
| 2.0 | $5.55 \cdot 10^{-17} - 3.23i$         | $1.13 \cdot 10^{-15} + 0.243i$      | $1.96 \cdot 10^{-11} + 0.232i$                 | $-7.24 \cdot 10^{-14} + 0.232i$                 |
| 3.0 | $-1.39 \cdot 10^{-16} + 0.866i$       | $1.4 \cdot 10^{-15} - 0.908i$       | $3.5 \cdot 10^{-11} - 0.866i$                  | $-1.3 \cdot 10^{-12} - 0.866i$                  |
| 4.0 | $-1.39 \cdot 10^{-16} - 0.866i$       | $1.03 \cdot 10^{-16} + 0.908i$      | $3.7 \cdot 10^{-11} + 0.866i$                  | $2.56 \cdot 10^{-13} + 0.866i$                  |
| 5.0 | $1.21 \cdot 10^{-17} + 0.232i$        | $-4.06 \cdot 10^{-16} - 3.4i$       | $2.65 \cdot 10^{-10} - 3.23i$                  | $-7.0 \cdot 10^{-12} - 3.23i$                   |
| 6.0 | $1.21 \cdot 10^{-17} - 0.232i$        | $3.94 \cdot 10^{-15} + 3.4i$        | $2.72 \cdot 10^{-10} + 3.23i$                  | $-1.24 \cdot 10^{-12} + 3.23i$                  |

Table 41: Case 1a (cpv MACA64 MR2023b),  $r = 6$ , Z4: poles.

|     | Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$ | LF, $\sigma_6 = 3.2 \cdot 10^{-07}$ | AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$ | AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$ |
|-----|---------------------------------------|-------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | $1.39 \cdot 10^{-16} + 3.23i$         | $7.01 \cdot 10^{-16} - 0.243i$      | $1.93 \cdot 10^{-11} - 0.232i$                 | $2.13 \cdot 10^{-13} + 0.232i$                  |
| 2.0 | $1.39 \cdot 10^{-16} - 3.23i$         | $3.33 \cdot 10^{-16} + 0.243i$      | $1.89 \cdot 10^{-11} + 0.232i$                 | $5.06 \cdot 10^{-13} - 0.232i$                  |
| 3.0 | $-2.78 \cdot 10^{-17} + 0.866i$       | $1.39 \cdot 10^{-16} - 0.908i$      | $3.64 \cdot 10^{-11} - 0.866i$                 | $-3.95 \cdot 10^{-13} + 0.866i$                 |
| 4.0 | $-2.78 \cdot 10^{-17} - 0.866i$       | $1.34 \cdot 10^{-15} + 0.908i$      | $3.49 \cdot 10^{-11} + 0.866i$                 | $1.43 \cdot 10^{-12} - 0.866i$                  |
| 5.0 | $1.73 \cdot 10^{-18} + 0.232i$        | $1.83 \cdot 10^{-15} - 3.4i$        | $2.69 \cdot 10^{-10} - 3.23i$                  | $4.22 \cdot 10^{-13} + 3.23i$                   |
| 6.0 | $1.73 \cdot 10^{-18} - 0.232i$        | $3.22 \cdot 10^{-16} + 3.4i$        | $2.63 \cdot 10^{-10} + 3.23i$                  | $7.46 \cdot 10^{-12} - 3.23i$                   |

Table 42: Case 1a (cpv MACI64 MR2022b),  $r = 6$ , Z4: poles.

|     | Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$ | LF, $\sigma_6 = 3.2 \cdot 10^{-07}$ | AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$ | AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$ |
|-----|---------------------------------------|-------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | $5.55 \cdot 10^{-17} + 3.23i$         | $-2.65 \cdot 10^{-16} + 0.243i$     | $1.25 \cdot 10^{-11} - 6.27 \cdot 10^{-11}i$   | $2.52 \cdot 10^{-13} - 2.27 \cdot 10^{-11}i$    |
| 2.0 | $5.55 \cdot 10^{-17} - 3.23i$         | $2.35 \cdot 10^{-16} - 0.243i$      | $1.66 \cdot 10^{-11} + 0.5i$                   | $3.42 \cdot 10^{-13} + 0.5i$                    |
| 3.0 | $-1.39 \cdot 10^{-16} + 0.866i$       | $7.55 \cdot 10^{-16} + 0.908i$      | $1.68 \cdot 10^{-11} - 0.5i$                   | $3.23 \cdot 10^{-13} - 0.5i$                    |
| 4.0 | $-1.39 \cdot 10^{-16} - 0.866i$       | $-1.43 \cdot 10^{-16} - 0.908i$     | $4.96 \cdot 10^{-11} + 1.5i$                   | $9.76 \cdot 10^{-13} + 1.5i$                    |
| 5.0 | $1.21 \cdot 10^{-17} + 0.232i$        | $3.11 \cdot 10^{-17} + 3.4i$        | $5.05 \cdot 10^{-11} - 1.5i$                   | $9.48 \cdot 10^{-13} - 1.5i$                    |
| 6.0 | $1.21 \cdot 10^{-17} - 0.232i$        | $-2.45 \cdot 10^{-15} - 3.4i$       | $2.31 \cdot 10^9 + 1.15 \cdot 10^{10}i$        | $3.47 \cdot 10^8 + 3.3 \cdot 10^{10}i$          |

Table 43: Case 1a (cpv PCWIN64 MR2025b),  $r = 6$ , Z4: poles.



|     | Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$ | LF, $\sigma_6 = 3.2 \cdot 10^{-07}$ | AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$ | AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$ |
|-----|---------------------------------------|-------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | $5.55 \cdot 10^{-17} + 3.23_z$        | $-2.65 \cdot 10^{-16} + 0.243_z$    | $1.25 \cdot 10^{-11} - 6.27 \cdot 10^{-11}_z$  | $2.52 \cdot 10^{-13} - 2.27 \cdot 10^{-11}_z$   |
| 2.0 | $5.55 \cdot 10^{-17} - 3.23_z$        | $2.35 \cdot 10^{-16} - 0.243_z$     | $1.66 \cdot 10^{-11} + 0.5_z$                  | $3.42 \cdot 10^{-13} + 0.5_z$                   |
| 3.0 | $-1.39 \cdot 10^{-16} + 0.866_z$      | $7.55 \cdot 10^{-16} + 0.908_z$     | $1.68 \cdot 10^{-11} - 0.5_z$                  | $3.23 \cdot 10^{-13} - 0.5_z$                   |
| 4.0 | $-1.39 \cdot 10^{-16} - 0.866_z$      | $-1.43 \cdot 10^{-16} - 0.908_z$    | $4.96 \cdot 10^{-11} + 1.5_z$                  | $9.76 \cdot 10^{-13} + 1.5_z$                   |
| 5.0 | $1.21 \cdot 10^{-17} + 0.232_z$       | $3.11 \cdot 10^{-17} + 3.4_z$       | $5.05 \cdot 10^{-11} - 1.5_z$                  | $9.48 \cdot 10^{-13} - 1.5_z$                   |
| 6.0 | $1.21 \cdot 10^{-17} - 0.232_z$       | $-2.45 \cdot 10^{-15} - 3.4_z$      | $2.31 \cdot 10^9 + 1.15 \cdot 10^{10}_z$       | $3.47 \cdot 10^8 + 3.3 \cdot 10^{10}_z$         |

Table 44: Case 1a (AlexReis PCWIN64 MR2025b),  $r = 6$ , Z4: poles.

### 5.3 Zeros

The zeros are given in the following tables. Each table gives the results obtained on different platform.

|     | Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$ | LF, $\sigma_6 = 3.2 \cdot 10^{-07}$         | AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$ | AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$ |
|-----|---------------------------------------|---------------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | NaN                                   | NaN                                         | $1.8 \cdot 10^{-11} + 2.97 \cdot 10^{-11}_z$   | $-2.66 \cdot 10^{-13} + 1.23 \cdot 10^{-11}_z$  |
| 2.0 | 0                                     | $1.1 \cdot 10^{-15} + 7.9 \cdot 10^{-16}_z$ | $2.34 \cdot 10^{-11} - 0.5_z$                  | $-7.85 \cdot 10^{-13} - 0.5_z$                  |
| 3.0 | $1.5_z$                               | $1.13 \cdot 10^{-15} - 0.523_z$             | $2.46 \cdot 10^{-11} + 0.5_z$                  | $1.08 \cdot 10^{-13} + 0.5_z$                   |
| 4.0 | $-1.5_z$                              | $6.11 \cdot 10^{-16} + 0.523_z$             | $7.02 \cdot 10^{-11} - 1.5_z$                  | $-2.43 \cdot 10^{-12} - 1.5_z$                  |
| 5.0 | $0.5_z$                               | $8.46 \cdot 10^{-16} - 1.57_z$              | $7.38 \cdot 10^{-11} + 1.5_z$                  | $2.62 \cdot 10^{-13} + 1.5_z$                   |
| 6.0 | $-0.5_z$                              | $5.09 \cdot 10^{-16} + 1.57_z$              | $1.18 \cdot 10^{10} - 1.88 \cdot 10^{10}_z$    | $-1.58 \cdot 10^9 - 6.51 \cdot 10^{10}_z$       |

Table 45: Case 1a (cpv MACA64 MR2023b),  $r = 6$ , Z4: zeros.

|     | Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$ | LF, $\sigma_6 = 3.2 \cdot 10^{-07}$          | AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$ | AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$ |
|-----|---------------------------------------|----------------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | NaN                                   | NaN                                          | $1.78 \cdot 10^{-11} + 8.65 \cdot 10^{-12}_z$  | $2.47 \cdot 10^{-13} - 5.44 \cdot 10^{-11}_z$   |
| 2.0 | 0                                     | $2.1 \cdot 10^{-16} + 1.53 \cdot 10^{-16}_z$ | $2.42 \cdot 10^{-11} - 0.5_z$                  | $-1.87 \cdot 10^{-13} + 0.5_z$                  |
| 3.0 | $1.5_z$                               | $5.51 \cdot 10^{-16} - 0.523_z$              | $2.33 \cdot 10^{-11} + 0.5_z$                  | $8.55 \cdot 10^{-13} - 0.5_z$                   |
| 4.0 | $-1.5_z$                              | $1.27 \cdot 10^{-15} + 0.523_z$              | $7.25 \cdot 10^{-11} - 1.5_z$                  | $-5.55 \cdot 10^{-13} + 1.5_z$                  |
| 5.0 | $0.5_z$                               | $9.91 \cdot 10^{-16} - 1.57_z$               | $7.0 \cdot 10^{-11} + 1.5_z$                   | $2.66 \cdot 10^{-12} - 1.5_z$                   |
| 6.0 | $-0.5_z$                              | $8.9 \cdot 10^{-16} + 1.57_z$                | $3.3 \cdot 10^{10} - 1.73 \cdot 10^{10}_z$     | $6.91 \cdot 10^7 + 1.4 \cdot 10^{10}_z$         |

Table 46: Case 1a (cpv MACI64 MR2022b),  $r = 6$ , Z4: zeros.

|     | Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$ | LF, $\sigma_6 = 3.2 \cdot 10^{-07}$           | AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$ | AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$ |
|-----|---------------------------------------|-----------------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | NaN                                   | NaN                                           | $1.34 \cdot 10^{-11} + 0.232_z$                | $2.75 \cdot 10^{-13} + 0.232_z$                 |
| 2.0 | 0                                     | $1.18 \cdot 10^{-16} - 5.87 \cdot 10^{-17}_z$ | $1.35 \cdot 10^{-11} - 0.232_z$                | $2.66 \cdot 10^{-13} - 0.232_z$                 |
| 3.0 | $1.5_z$                               | $3.18 \cdot 10^{-17} + 0.523_z$               | $2.48 \cdot 10^{-11} + 0.866_z$                | $5.0 \cdot 10^{-13} + 0.866_z$                  |
| 4.0 | $-1.5_z$                              | $-1.45 \cdot 10^{-17} - 0.523_z$              | $2.53 \cdot 10^{-11} - 0.866_z$                | $4.74 \cdot 10^{-13} - 0.866_z$                 |
| 5.0 | $0.5_z$                               | $4.28 \cdot 10^{-16} + 1.57_z$                | $1.86 \cdot 10^{-10} + 3.23_z$                 | $3.6 \cdot 10^{-12} + 3.23_z$                   |
| 6.0 | $-0.5_z$                              | $-5.15 \cdot 10^{-16} - 1.57_z$               | $1.88 \cdot 10^{-10} - 3.23_z$                 | $3.54 \cdot 10^{-12} - 3.23_z$                  |

Table 47: Case 1a (cpv PCWIN64 MR2025b),  $r = 6$ , Z4: zeros.

|     | Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$ | LF, $\sigma_6 = 3.2 \cdot 10^{-07}$           | AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$ | AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$ |
|-----|---------------------------------------|-----------------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | NaN                                   | NaN                                           | $1.34 \cdot 10^{-11} + 0.232_z$                | $2.75 \cdot 10^{-13} + 0.232_z$                 |
| 2.0 | 0                                     | $1.18 \cdot 10^{-16} - 5.87 \cdot 10^{-17}_z$ | $1.35 \cdot 10^{-11} - 0.232_z$                | $2.66 \cdot 10^{-13} - 0.232_z$                 |
| 3.0 | $1.5_z$                               | $3.18 \cdot 10^{-17} + 0.523_z$               | $2.48 \cdot 10^{-11} + 0.866_z$                | $5.0 \cdot 10^{-13} + 0.866_z$                  |
| 4.0 | $-1.5_z$                              | $-1.45 \cdot 10^{-17} - 0.523_z$              | $2.53 \cdot 10^{-11} - 0.866_z$                | $4.74 \cdot 10^{-13} - 0.866_z$                 |
| 5.0 | $0.5_z$                               | $4.28 \cdot 10^{-16} + 1.57_z$                | $1.86 \cdot 10^{-10} + 3.23_z$                 | $3.6 \cdot 10^{-12} + 3.23_z$                   |
| 6.0 | $-0.5_z$                              | $-5.15 \cdot 10^{-16} - 1.57_z$               | $1.88 \cdot 10^{-10} - 3.23_z$                 | $3.54 \cdot 10^{-12} - 3.23_z$                  |

Table 48: Case 1a (AlexReis PCWIN64 MR2025b),  $r = 6$ , Z4: zeros.

## 5.4 Poles and zeros visualisation

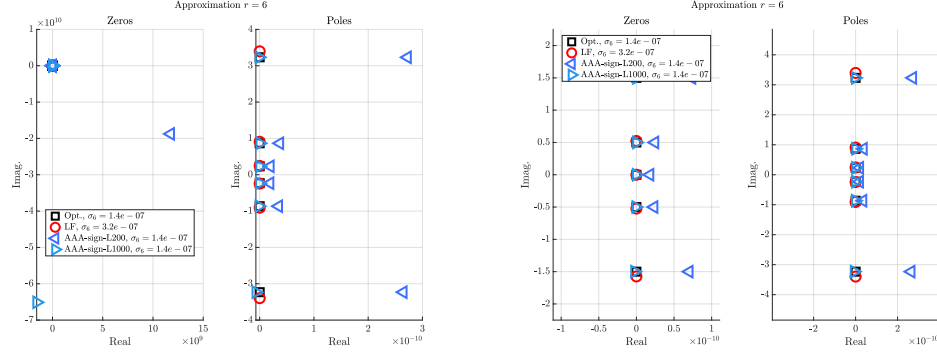


Figure 7: Case 1a (cpv MACA64 MR2023b),  $r = 6$ , Z4. Left frame: zeros (left) and poles (right). Right frame: same but discarding the largest real entry.

## 5.5 Approximation visualization

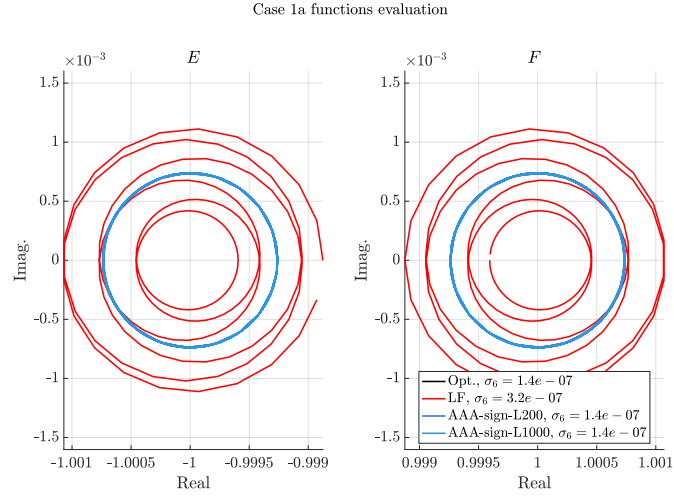


Figure 8: Case 1a (cpv MACA64 MR2023b),  $r = 6$ , Z4.  $E$  and  $F$  evaluations.

## 6 Approximation objective $r = 7$

### 6.1 Numerator and denominator coefficients

The numerator and denominator coefficients (both real and complex parts) are given in the following tables. Each table gives the results obtained on different platform.

| Basis | Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$ |       | LF, $\sigma_7 = 5.2 \cdot 10^{-08}$ |                       | AAA-sign-L200, $\sigma_7 = 9.8 \cdot 10^{-09}$ |                      | AAA-sign-L1000, $\sigma_7 = 9.8 \cdot 10^{-09}$ |                     |
|-------|---------------------------------------|-------|-------------------------------------|-----------------------|------------------------------------------------|----------------------|-------------------------------------------------|---------------------|
|       | real                                  | imag. | real                                | imag.                 | real                                           | imag.                | real                                            | imag.               |
| $z^7$ | 0                                     | 0     | 0                                   | 0                     | $-2.5 \cdot 10^6$                              | $-1.8 \cdot 10^8$    | $-5.6 \cdot 10^5$                               | $-2.6 \cdot 10^8$   |
| $z^6$ | 6.1                                   | 0     | 7.2                                 | $-1.9 \cdot 10^{-14}$ | 6.0                                            | $1.1 \cdot 10^{-3}$  | 6.0                                             | $7.4 \cdot 10^{-4}$ |
| $z^5$ | 0                                     | 0     | $-1.5 \cdot 10^{-13}$               | $-5.8 \cdot 10^{-15}$ | $-4.0 \cdot 10^7$                              | $-2.8 \cdot 10^9$    | $-8.8 \cdot 10^6$                               | $-4.0 \cdot 10^9$   |
| $z^4$ | 23.0                                  | 0     | 36.0                                | $-7.5 \cdot 10^{-14}$ | 22.0                                           | 0.012                | 22.0                                            | $8.5 \cdot 10^{-3}$ |
| $z^3$ | 0                                     | 0     | $-3.2 \cdot 10^{-13}$               | $-4.4 \cdot 10^{-14}$ | $-5.0 \cdot 10^7$                              | $-3.5 \cdot 10^9$    | $-1.1 \cdot 10^7$                               | $-5.0 \cdot 10^9$   |
| $z^2$ | 10.0                                  | 0     | 20.0                                | $-1.2 \cdot 10^{-13}$ | 10.0                                           | $8.7 \cdot 10^{-3}$  | 10.0                                            | $6.6 \cdot 10^{-3}$ |
| $z$   | 0                                     | 0     | $-5.3 \cdot 10^{-14}$               | $-3.8 \cdot 10^{-14}$ | $-7.5 \cdot 10^6$                              | $-5.2 \cdot 10^8$    | $-1.6 \cdot 10^6$                               | $-7.6 \cdot 10^8$   |
| 1.0   | 0.37                                  | 0     | 0.85                                | $-1.2 \cdot 10^{-14}$ | 0.35                                           | $4.3 \cdot 10^{-4}$  | 0.35                                            | $3.4 \cdot 10^{-4}$ |
| $z^7$ | 1.0                                   | 0     | 1.0                                 | 0                     | 1.0                                            | $1.7 \cdot 10^{-17}$ | 1.0                                             | 0                   |
| $z^6$ | 0                                     | 0     | $-2.9 \cdot 10^{-14}$               | $-1.6 \cdot 10^{-14}$ | $-1.5 \cdot 10^7$                              | $-1.1 \cdot 10^9$    | $-3.4 \cdot 10^6$                               | $-1.6 \cdot 10^9$   |
| $z^5$ | 16.0                                  | 0     | 22.0                                | $-5.2 \cdot 10^{-14}$ | 16.0                                           | $5.5 \cdot 10^{-3}$  | 16.0                                            | $3.9 \cdot 10^{-3}$ |
| $z^4$ | 0                                     | 0     | $-3.1 \cdot 10^{-13}$               | $1.8 \cdot 10^{-16}$  | $-5.8 \cdot 10^7$                              | $-4.0 \cdot 10^9$    | $-1.3 \cdot 10^7$                               | $-5.8 \cdot 10^9$   |
| $z^3$ | 20.0                                  | 0     | 35.0                                | $-1.1 \cdot 10^{-13}$ | 19.0                                           | 0.014                | 19.0                                            | 0.01                |
| $z^2$ | 0                                     | 0     | $-1.8 \cdot 10^{-13}$               | $-7.8 \cdot 10^{-14}$ | $-2.6 \cdot 10^7$                              | $-1.8 \cdot 10^9$    | $-5.7 \cdot 10^6$                               | $-2.6 \cdot 10^9$   |
| $z$   | 3.0                                   | 0     | 6.3                                 | $-6.4 \cdot 10^{-14}$ | 2.9                                            | $3.0 \cdot 10^{-3}$  | 2.9                                             | $2.3 \cdot 10^{-3}$ |
| 1.0   | 0                                     | 0     | $-6.8 \cdot 10^{-15}$               | $-3.7 \cdot 10^{-16}$ | $-9.3 \cdot 10^5$                              | $-6.5 \cdot 10^7$    | $-2.0 \cdot 10^5$                               | $-9.4 \cdot 10^7$   |

Table 49: Case 1a (cpv MACA64 MR2023b),  $r = 7$ , Z4: numerator (first lines) and denominator (last lines) coefficients.

| Basis | Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$ |       | LF, $\sigma_7 = 5.2 \cdot 10^{-08}$ |                      | AAA-sign-L200, $\sigma_7 = 1.1 \cdot 10^{-08}$ |                     | AAA-sign-L1000, $\sigma_7 = 9.9 \cdot 10^{-09}$ |                      |
|-------|---------------------------------------|-------|-------------------------------------|----------------------|------------------------------------------------|---------------------|-------------------------------------------------|----------------------|
|       | real                                  | imag. | real                                | imag.                | real                                           | imag.               | real                                            | imag.                |
| $z^7$ | 0                                     | 0     | 0                                   | 0                    | -93.0                                          | 0.99                | $1.5 \cdot 10^{-4}$                             | $-7.1 \cdot 10^{-5}$ |
| $z^6$ | 6.1                                   | 0     | 7.2                                 | $1.3 \cdot 10^{-14}$ | 4.5                                            | $8.6 \cdot 10^{-3}$ | 6.1                                             | $-1.5 \cdot 10^{-5}$ |
| $z^5$ | 0                                     | 0     | $-2.3 \cdot 10^{-15}$               | $5.2 \cdot 10^{-14}$ | $-1.5 \cdot 10^3$                              | 16.0                | $-3.9 \cdot 10^{-4}$                            | $-1.0 \cdot 10^{-3}$ |
| $z^4$ | 23.0                                  | 0     | 36.0                                | $9.9 \cdot 10^{-14}$ | 5.9                                            | 0.097               | 23.0                                            | $-1.1 \cdot 10^{-4}$ |
| $z^3$ | 0                                     | 0     | $6.8 \cdot 10^{-15}$                | $1.2 \cdot 10^{-13}$ | $-1.8 \cdot 10^3$                              | 19.0                | $-3.7 \cdot 10^{-3}$                            | $-1.3 \cdot 10^{-3}$ |
| $z^2$ | 10.0                                  | 0     | 20.0                                | $1.1 \cdot 10^{-13}$ | -2.2                                           | 0.074               | 10.0                                            | $-4.2 \cdot 10^{-5}$ |
| $z$   | 0                                     | 0     | $1.1 \cdot 10^{-15}$                | $3.6 \cdot 10^{-14}$ | -270.0                                         | 2.9                 | $-9.8 \cdot 10^{-4}$                            | $-1.9 \cdot 10^{-4}$ |
| 1.0   | 0.37                                  | 0     | 0.85                                | $3.8 \cdot 10^{-15}$ | -0.25                                          | $3.8 \cdot 10^{-3}$ | 0.37                                            | $5.3 \cdot 10^{-7}$  |
| $z^7$ | 1.0                                   | 0     | 1.0                                 | $5.2 \cdot 10^{-17}$ | 1.0                                            | 0                   | 1.0                                             | 0                    |
| $z^6$ | 0                                     | 0     | $-4.1 \cdot 10^{-15}$               | $9.4 \cdot 10^{-15}$ | -560.0                                         | 6.0                 | $3.7 \cdot 10^{-4}$                             | $-4.1 \cdot 10^{-4}$ |
| $z^5$ | 16.0                                  | 0     | 22.0                                | $5.5 \cdot 10^{-14}$ | 7.9                                            | 0.044               | 16.0                                            | $-6.1 \cdot 10^{-5}$ |
| $z^4$ | 0                                     | 0     | $1.6 \cdot 10^{-14}$                | $1.3 \cdot 10^{-13}$ | $-2.1 \cdot 10^3$                              | 22.0                | $-2.4 \cdot 10^{-3}$                            | $-1.5 \cdot 10^{-3}$ |
| $z^3$ | 20.0                                  | 0     | 35.0                                | $1.3 \cdot 10^{-13}$ | 0.34                                           | 0.11                | 20.0                                            | $-9.6 \cdot 10^{-5}$ |
| $z^2$ | 0                                     | 0     | $1.6 \cdot 10^{-15}$                | $7.5 \cdot 10^{-14}$ | -940.0                                         | 10.0                | $-2.7 \cdot 10^{-3}$                            | $-6.5 \cdot 10^{-4}$ |
| $z$   | 3.0                                   | 0     | 6.3                                 | $4.2 \cdot 10^{-14}$ | -1.3                                           | 0.026               | 3.0                                             | $-5.5 \cdot 10^{-6}$ |
| 1.0   | 0                                     | 0     | $-4.0 \cdot 10^{-16}$               | $4.8 \cdot 10^{-15}$ | -33.0                                          | 0.36                | $-1.4 \cdot 10^{-4}$                            | $-2.5 \cdot 10^{-5}$ |

Table 50: Case 1a (cpv MACI64 MR2022b),  $r = 7$ , Z4: numerator (first lines) and denominator (last lines) coefficients.

| Basis | Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$ |       | LF, $\sigma_7 = 5.2 \cdot 10^{-08}$ |                       | AAA-sign-L200, $\sigma_7 = 9.8 \cdot 10^{-09}$ |                      | AAA-sign-L1000, $\sigma_7 = 9.8 \cdot 10^{-09}$ |                     |
|-------|---------------------------------------|-------|-------------------------------------|-----------------------|------------------------------------------------|----------------------|-------------------------------------------------|---------------------|
|       | real                                  | imag. | real                                | imag.                 | real                                           | imag.                | real                                            | imag.               |
| $z^7$ | 0                                     | 0     | 0                                   | 0                     | $-2.0 \cdot 10^6$                              | $-1.5 \cdot 10^8$    | $6.3 \cdot 10^4$                                | $-1.7 \cdot 10^8$   |
| $z^6$ | 6.1                                   | 0     | 7.2                                 | $-1.5 \cdot 10^{-14}$ | 6.1                                            | $-3.2 \cdot 10^{-5}$ | 6.1                                             | $4.4 \cdot 10^{-4}$ |
| $z^5$ | 0                                     | 0     | $-4.4 \cdot 10^{-14}$               | $1.0 \cdot 10^{-13}$  | $-3.1 \cdot 10^7$                              | $-2.3 \cdot 10^9$    | $9.9 \cdot 10^5$                                | $-2.7 \cdot 10^9$   |
| $z^4$ | 23.0                                  | 0     | 36.0                                | $-1.1 \cdot 10^{-13}$ | 23.0                                           | $-3.5 \cdot 10^{-4}$ | 23.0                                            | $4.4 \cdot 10^{-3}$ |
| $z^3$ | 0                                     | 0     | $-4.8 \cdot 10^{-14}$               | $2.0 \cdot 10^{-13}$  | $-3.9 \cdot 10^7$                              | $-2.9 \cdot 10^9$    | $1.2 \cdot 10^6$                                | $-3.4 \cdot 10^9$   |
| $z^2$ | 10.0                                  | 0     | 20.0                                | $-6.7 \cdot 10^{-14}$ | 10.0                                           | $-3.1 \cdot 10^{-4}$ | 10.0                                            | $3.0 \cdot 10^{-3}$ |
| $z$   | 0                                     | 0     | $-2.2 \cdot 10^{-15}$               | $4.1 \cdot 10^{-14}$  | $-5.8 \cdot 10^6$                              | $-4.4 \cdot 10^8$    | $1.9 \cdot 10^5$                                | $-5.1 \cdot 10^8$   |
| 1.0   | 0.37                                  | 0     | 0.85                                | $-9.7 \cdot 10^{-15}$ | 0.37                                           | $-1.6 \cdot 10^{-5}$ | 0.37                                            | $1.3 \cdot 10^{-4}$ |
| $z^7$ | 1.0                                   | 0     | 1.0                                 | $-5.2 \cdot 10^{-17}$ | 1.0                                            | 0                    | 1.0                                             | 0                   |
| $z^6$ | 0                                     | 0     | $-7.7 \cdot 10^{-15}$               | $1.5 \cdot 10^{-14}$  | $-1.2 \cdot 10^7$                              | $-9.0 \cdot 10^8$    | $3.8 \cdot 10^5$                                | $-1.0 \cdot 10^9$   |
| $z^5$ | 16.0                                  | 0     | 22.0                                | $-6.0 \cdot 10^{-14}$ | 16.0                                           | $-1.7 \cdot 10^{-4}$ | 16.0                                            | $2.1 \cdot 10^{-3}$ |
| $z^4$ | 0                                     | 0     | $-7.1 \cdot 10^{-14}$               | $2.0 \cdot 10^{-13}$  | $-4.5 \cdot 10^7$                              | $-3.4 \cdot 10^9$    | $1.4 \cdot 10^6$                                | $-3.9 \cdot 10^9$   |
| $z^3$ | 20.0                                  | 0     | 35.0                                | $-1.1 \cdot 10^{-13}$ | 20.0                                           | $-4.2 \cdot 10^{-4}$ | 20.0                                            | $4.9 \cdot 10^{-3}$ |
| $z^2$ | 0                                     | 0     | $-1.6 \cdot 10^{-14}$               | $1.2 \cdot 10^{-13}$  | $-2.0 \cdot 10^7$                              | $-1.5 \cdot 10^9$    | $6.4 \cdot 10^5$                                | $-1.7 \cdot 10^9$   |
| $z$   | 3.0                                   | 0     | 6.3                                 | $-3.5 \cdot 10^{-14}$ | 3.0                                            | $-1.1 \cdot 10^{-4}$ | 3.0                                             | $9.7 \cdot 10^{-4}$ |
| 1.0   | 0                                     | 0     | $1.0 \cdot 10^{-15}$                | $7.7 \cdot 10^{-15}$  | $-7.2 \cdot 10^5$                              | $-5.4 \cdot 10^7$    | $2.3 \cdot 10^4$                                | $-6.2 \cdot 10^7$   |

Table 51: Case 1a (cpv PCWIN64 MR2025b),  $r = 7$ , Z4: numerator (first lines) and denominator (last lines) coefficients.

| Basis | Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$ |       | LF, $\sigma_7 = 5.2 \cdot 10^{-08}$ |                       | AAA-sign-L200, $\sigma_7 = 9.8 \cdot 10^{-09}$ |                      | AAA-sign-L1000, $\sigma_7 = 9.8 \cdot 10^{-09}$ |                     |
|-------|---------------------------------------|-------|-------------------------------------|-----------------------|------------------------------------------------|----------------------|-------------------------------------------------|---------------------|
|       | real                                  | imag. | real                                | imag.                 | real                                           | imag.                | real                                            | imag.               |
| $z^7$ | 0                                     | 0     | 0                                   | 0                     | $-2.0 \cdot 10^6$                              | $-1.5 \cdot 10^8$    | $6.3 \cdot 10^4$                                | $-1.7 \cdot 10^8$   |
| $z^6$ | 6.1                                   | 0     | 7.2                                 | $-1.5 \cdot 10^{-14}$ | 6.1                                            | $-3.2 \cdot 10^{-5}$ | 6.1                                             | $4.4 \cdot 10^{-4}$ |
| $z^5$ | 0                                     | 0     | $-4.4 \cdot 10^{-14}$               | $1.0 \cdot 10^{-13}$  | $-3.1 \cdot 10^7$                              | $-2.3 \cdot 10^9$    | $9.9 \cdot 10^5$                                | $-2.7 \cdot 10^9$   |
| $z^4$ | 23.0                                  | 0     | 36.0                                | $-1.1 \cdot 10^{-13}$ | 23.0                                           | $-3.5 \cdot 10^{-4}$ | 23.0                                            | $4.4 \cdot 10^{-3}$ |
| $z^3$ | 0                                     | 0     | $-4.8 \cdot 10^{-14}$               | $2.0 \cdot 10^{-13}$  | $-3.9 \cdot 10^7$                              | $-2.9 \cdot 10^9$    | $1.2 \cdot 10^6$                                | $-3.4 \cdot 10^9$   |
| $z^2$ | 10.0                                  | 0     | 20.0                                | $-6.7 \cdot 10^{-14}$ | 10.0                                           | $-3.1 \cdot 10^{-4}$ | 10.0                                            | $3.0 \cdot 10^{-3}$ |
| $z$   | 0                                     | 0     | $-2.2 \cdot 10^{-15}$               | $4.1 \cdot 10^{-14}$  | $-5.8 \cdot 10^6$                              | $-4.4 \cdot 10^8$    | $1.9 \cdot 10^5$                                | $-5.1 \cdot 10^8$   |
| 1.0   | 0.37                                  | 0     | 0.85                                | $-9.7 \cdot 10^{-15}$ | 0.37                                           | $-1.6 \cdot 10^{-5}$ | 0.37                                            | $1.3 \cdot 10^{-4}$ |
| $z^7$ | 1.0                                   | 0     | 1.0                                 | $-5.2 \cdot 10^{-17}$ | 1.0                                            | 0                    | 1.0                                             | 0                   |
| $z^6$ | 0                                     | 0     | $-7.7 \cdot 10^{-15}$               | $1.5 \cdot 10^{-14}$  | $-1.2 \cdot 10^7$                              | $-9.0 \cdot 10^8$    | $3.8 \cdot 10^5$                                | $-1.0 \cdot 10^9$   |
| $z^5$ | 16.0                                  | 0     | 22.0                                | $-6.0 \cdot 10^{-14}$ | 16.0                                           | $-1.7 \cdot 10^{-4}$ | 16.0                                            | $2.1 \cdot 10^{-3}$ |
| $z^4$ | 0                                     | 0     | $-7.1 \cdot 10^{-14}$               | $2.0 \cdot 10^{-13}$  | $-4.5 \cdot 10^7$                              | $-3.4 \cdot 10^9$    | $1.4 \cdot 10^6$                                | $-3.9 \cdot 10^9$   |
| $z^3$ | 20.0                                  | 0     | 35.0                                | $-1.1 \cdot 10^{-13}$ | 20.0                                           | $-4.2 \cdot 10^{-4}$ | 20.0                                            | $4.9 \cdot 10^{-3}$ |
| $z^2$ | 0                                     | 0     | $-1.6 \cdot 10^{-14}$               | $1.2 \cdot 10^{-13}$  | $-2.0 \cdot 10^7$                              | $-1.5 \cdot 10^9$    | $6.4 \cdot 10^5$                                | $-1.7 \cdot 10^9$   |
| $z$   | 3.0                                   | 0     | 6.3                                 | $-3.5 \cdot 10^{-14}$ | 3.0                                            | $-1.1 \cdot 10^{-4}$ | 3.0                                             | $9.7 \cdot 10^{-4}$ |
| 1.0   | 0                                     | 0     | $1.0 \cdot 10^{-15}$                | $7.7 \cdot 10^{-15}$  | $-7.2 \cdot 10^5$                              | $-5.4 \cdot 10^7$    | $2.3 \cdot 10^4$                                | $-6.2 \cdot 10^7$   |

Table 52: Case 1a (AlexReis PCWIN64 MR2025b),  $r = 7$ , Z4: numerator (first lines) and denominator (last lines) coefficients.

## 6.2 Poles

The poles are given in the following tables. Each table gives the results obtained on different platform.

|     | Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$ | LF, $\sigma_7 = 5.2 \cdot 10^{-08}$          | AAA-sign-L200, $\sigma_7 = 9.8 \cdot 10^{-09}$ | AAA-sign-L1000, $\sigma_7 = 9.8 \cdot 10^{-09}$ |
|-----|---------------------------------------|----------------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | 0                                     | $1.08 \cdot 10^{-15} + 5.86 \cdot 10^{-17}i$ | $1.63 \cdot 10^{-11} + 0.198i$                 | $5.75 \cdot 10^{-12} + 0.198i$                  |
| 2.0 | $-7.63 \cdot 10^{-17} + 3.79i$        | $-1.46 \cdot 10^{-16} - 0.458i$              | $5.67 \cdot 10^{-12} - 0.198i$                 | $-2.73 \cdot 10^{-12} - 0.198i$                 |
| 3.0 | $-7.63 \cdot 10^{-17} - 3.79i$        | $3.4 \cdot 10^{-15} + 0.458i$                | $3.53 \cdot 10^{-11} + 0.691i$                 | $1.68 \cdot 10^{-11} + 0.691i$                  |
| 4.0 | $2.78 \cdot 10^{-17} + 1.09i$         | $3.32 \cdot 10^{-15} + 1.23i$                | $-1.59 \cdot 10^{-12} - 0.691i$                | $-1.26 \cdot 10^{-11} - 0.691i$                 |
| 5.0 | $2.78 \cdot 10^{-17} - 1.09i$         | $5.36 \cdot 10^{-15} - 1.23i$                | $1.01 \cdot 10^{-10} + 1.8i$                   | $4.36 \cdot 10^{-11} + 1.8i$                    |
| 6.0 | $7.63 \cdot 10^{-17} + 0.417i$        | $1.82 \cdot 10^{-15} - 4.47i$                | $6.14 \cdot 10^{-12} - 1.8i$                   | $-3.18 \cdot 10^{-11} - 1.8i$                   |
| 7.0 | $7.63 \cdot 10^{-17} - 0.417i$        | $1.53 \cdot 10^{-14} + 4.47i$                | $1.54 \cdot 10^7 + 1.07 \cdot 10^9i$           | $3.38 \cdot 10^6 + 1.55 \cdot 10^9i$            |

Table 53: Case 1a (cpv MACA64 MR2023b),  $r = 7$ , Z4: poles.

|     | Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$ | LF, $\sigma_7 = 5.2 \cdot 10^{-08}$         | AAA-sign-L200, $\sigma_7 = 1.1 \cdot 10^{-08}$ | AAA-sign-L1000, $\sigma_7 = 9.9 \cdot 10^{-09}$ |
|-----|---------------------------------------|---------------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | 0                                     | $6.4 \cdot 10^{-17} - 7.54 \cdot 10^{-16}z$ | $-8.57 \cdot 10^{-4} - 0.197z$                 | $4.89 \cdot 10^{-5} + 8.52 \cdot 10^{-6}z$      |
| 2.0 | $-1.46 \cdot 10^{-16} + 3.79z$        | $-7.57 \cdot 10^{-16} + 0.458z$             | $-8.53 \cdot 10^{-4} + 0.197z$                 | $4.97 \cdot 10^{-5} - 0.417z$                   |
| 3.0 | $-1.46 \cdot 10^{-16} - 3.79z$        | $6.74 \cdot 10^{-16} - 0.458z$              | $-1.38 \cdot 10^{-4} - 0.689z$                 | $4.82 \cdot 10^{-5} + 0.417z$                   |
| 4.0 | $2.78 \cdot 10^{-17} + 1.09z$         | $-7.92 \cdot 10^{-16} + 1.23z$              | $-1.19 \cdot 10^{-4} + 0.689z$                 | $3.1 \cdot 10^{-5} - 1.09z$                     |
| 5.0 | $2.78 \cdot 10^{-17} - 1.09z$         | $-2.55 \cdot 10^{-16} - 1.23z$              | $0.00463 - 1.79z$                              | $3.26 \cdot 10^{-5} + 1.09z$                    |
| 6.0 | $-3.47 \cdot 10^{-17} + 0.417z$       | $-4.93 \cdot 10^{-15} + 4.47z$              | $0.0047 + 1.79z$                               | $-2.98 \cdot 10^{-4} - 3.79z$                   |
| 7.0 | $-3.47 \cdot 10^{-17} - 0.417z$       | $1.15 \cdot 10^{-14} - 4.47z$               | $565.0 - 6.0z$                                 | $-2.83 \cdot 10^{-4} + 3.8z$                    |

Table 54: Case 1a (cpv MACI64 MR2022b),  $r = 7$ , Z4: poles.

|     | Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$ | LF, $\sigma_7 = 5.2 \cdot 10^{-08}$           | AAA-sign-L200, $\sigma_7 = 9.8 \cdot 10^{-09}$ | AAA-sign-L1000, $\sigma_7 = 9.8 \cdot 10^{-09}$ |
|-----|---------------------------------------|-----------------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | 0                                     | $-1.61 \cdot 10^{-16} - 1.23 \cdot 10^{-15}z$ | $1.16 \cdot 10^{-11} + 0.198z$                 | $-2.97 \cdot 10^{-12} + 0.198z$                 |
| 2.0 | $-7.63 \cdot 10^{-17} + 3.79z$        | $6.42 \cdot 10^{-16} + 0.458z$                | $1.16 \cdot 10^{-11} - 0.198z$                 | $2.94 \cdot 10^{-12} - 0.198z$                  |
| 3.0 | $-7.63 \cdot 10^{-17} - 3.79z$        | $-4.86 \cdot 10^{-16} - 0.458z$               | $1.81 \cdot 10^{-11} + 0.691z$                 | $-1.0 \cdot 10^{-11} + 0.691z$                  |
| 4.0 | $2.78 \cdot 10^{-17} + 1.09z$         | $1.24 \cdot 10^{-15} + 1.23z$                 | $1.81 \cdot 10^{-11} - 0.691z$                 | $9.84 \cdot 10^{-12} - 0.691z$                  |
| 5.0 | $2.78 \cdot 10^{-17} - 1.09z$         | $1.68 \cdot 10^{-15} - 1.23z$                 | $5.9 \cdot 10^{-11} + 1.8z$                    | $-2.56 \cdot 10^{-11} + 1.8z$                   |
| 6.0 | $7.63 \cdot 10^{-17} + 0.417z$        | $8.32 \cdot 10^{-15} + 4.47z$                 | $5.86 \cdot 10^{-11} - 1.8z$                   | $2.36 \cdot 10^{-11} - 1.8z$                    |
| 7.0 | $7.63 \cdot 10^{-17} - 0.417z$        | $-9.17 \cdot 10^{-16} - 4.47z$                | $1.2 \cdot 10^7 + 9.01 \cdot 10^8z$            | $-3.8 \cdot 10^5 + 1.04 \cdot 10^9z$            |

Table 55: Case 1a (cpv PCWIN64 MR2025b),  $r = 7$ , Z4: poles.

|     | Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$ | LF, $\sigma_7 = 5.2 \cdot 10^{-08}$           | AAA-sign-L200, $\sigma_7 = 9.8 \cdot 10^{-09}$ | AAA-sign-L1000, $\sigma_7 = 9.8 \cdot 10^{-09}$ |
|-----|---------------------------------------|-----------------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | 0                                     | $-1.61 \cdot 10^{-16} - 1.23 \cdot 10^{-15}z$ | $1.16 \cdot 10^{-11} + 0.198z$                 | $-2.97 \cdot 10^{-12} + 0.198z$                 |
| 2.0 | $-7.63 \cdot 10^{-17} + 3.79z$        | $6.42 \cdot 10^{-16} + 0.458z$                | $1.16 \cdot 10^{-11} - 0.198z$                 | $2.94 \cdot 10^{-12} - 0.198z$                  |
| 3.0 | $-7.63 \cdot 10^{-17} - 3.79z$        | $-4.86 \cdot 10^{-16} - 0.458z$               | $1.81 \cdot 10^{-11} + 0.691z$                 | $-1.0 \cdot 10^{-11} + 0.691z$                  |
| 4.0 | $2.78 \cdot 10^{-17} + 1.09z$         | $1.24 \cdot 10^{-15} + 1.23z$                 | $1.81 \cdot 10^{-11} - 0.691z$                 | $9.84 \cdot 10^{-12} - 0.691z$                  |
| 5.0 | $2.78 \cdot 10^{-17} - 1.09z$         | $1.68 \cdot 10^{-15} - 1.23z$                 | $5.9 \cdot 10^{-11} + 1.8z$                    | $-2.56 \cdot 10^{-11} + 1.8z$                   |
| 6.0 | $7.63 \cdot 10^{-17} + 0.417z$        | $8.32 \cdot 10^{-15} + 4.47z$                 | $5.86 \cdot 10^{-11} - 1.8z$                   | $2.36 \cdot 10^{-11} - 1.8z$                    |
| 7.0 | $7.63 \cdot 10^{-17} - 0.417z$        | $-9.17 \cdot 10^{-16} - 4.47z$                | $1.2 \cdot 10^7 + 9.01 \cdot 10^8z$            | $-3.8 \cdot 10^5 + 1.04 \cdot 10^9z$            |

Table 56: Case 1a (AlexReis PCWIN64 MR2025b),  $r = 7$ , Z4: poles.

### 6.3 Zeros

The zeros are given in the following tables. Each table gives the results obtained on different platform.

|     | Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$ | LF, $\sigma_7 = 5.2 \cdot 10^{-08}$ | AAA-sign-L200, $\sigma_7 = 9.8 \cdot 10^{-09}$ | AAA-sign-L1000, $\sigma_7 = 9.8 \cdot 10^{-09}$ |
|-----|---------------------------------------|-------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | NaN                                   | NaN                                 | $1.05 \cdot 10^{-11} - 6.74 \cdot 10^{-10}z$   | $1.46 \cdot 10^{-12} - 4.66 \cdot 10^{-10}z$    |
| 2.0 | $1.73 \cdot 10^{-17} + 1.8z$          | $2.11 \cdot 10^{-16} - 0.216z$      | $2.4 \cdot 10^{-11} + 0.417z$                  | $1.06 \cdot 10^{-11} + 0.417z$                  |
| 3.0 | $1.73 \cdot 10^{-17} - 1.8z$          | $2.13 \cdot 10^{-15} + 0.216z$      | $1.61 \cdot 10^{-12} - 0.417z$                 | $-7.23 \cdot 10^{-12} - 0.417z$                 |
| 4.0 | $5.55 \cdot 10^{-17} + 0.691z$        | $1.4 \cdot 10^{-15} - 0.768z$       | $5.5 \cdot 10^{-11} + 1.09z$                   | $2.6 \cdot 10^{-11} + 1.09z$                    |
| 5.0 | $5.55 \cdot 10^{-17} - 0.691z$        | $4.64 \cdot 10^{-15} + 0.768z$      | $-2.6 \cdot 10^{-12} - 1.09z$                  | $-1.99 \cdot 10^{-11} - 1.09z$                  |
| 6.0 | $1.21 \cdot 10^{-17} + 0.198z$        | $5.1 \cdot 10^{-15} + 2.07z$        | $3.02 \cdot 10^{-10} + 3.79z$                  | $1.01 \cdot 10^{-10} + 3.79z$                   |
| 7.0 | $1.21 \cdot 10^{-17} - 0.198z$        | $7.22 \cdot 10^{-15} - 2.07z$       | $1.02 \cdot 10^{-10} - 3.79z$                  | $-5.76 \cdot 10^{-11} - 3.79z$                  |

Table 57: Case 1a (cpv MACA64 MR2023b),  $r = 7$ , Z4: zeros.

|     | Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$ | LF, $\sigma_7 = 5.2 \cdot 10^{-08}$ | AAA-sign-L200, $\sigma_7 = 1.1 \cdot 10^{-08}$ | AAA-sign-L1000, $\sigma_7 = 9.9 \cdot 10^{-09}$ |
|-----|---------------------------------------|-------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | NaN                                   | NaN                                 | $-9.17 \cdot 10^{-4} + 4.3 \cdot 10^{-6}z$     | $4.98 \cdot 10^{-5} - 0.198z$                   |
| 2.0 | $-3.12 \cdot 10^{-17} + 1.8z$         | $2.07 \cdot 10^{-16} + 0.216z$      | $-6.4 \cdot 10^{-4} - 0.416z$                  | $4.86 \cdot 10^{-5} + 0.198z$                   |
| 3.0 | $-3.12 \cdot 10^{-17} - 1.8z$         | $-1.59 \cdot 10^{-16} - 0.216z$     | $-6.31 \cdot 10^{-4} + 0.416z$                 | $4.58 \cdot 10^{-5} - 0.691z$                   |
| 4.0 | $4.16 \cdot 10^{-17} + 0.691z$        | $-1.16 \cdot 10^{-15} + 0.768z$     | $0.00106 - 1.08z$                              | $4.52 \cdot 10^{-5} + 0.691z$                   |
| 5.0 | $4.16 \cdot 10^{-17} - 0.691z$        | $1.19 \cdot 10^{-15} - 0.768z$      | $0.0011 + 1.08z$                               | $-1.89 \cdot 10^{-5} - 1.8z$                    |
| 6.0 | $5.2 \cdot 10^{-18} + 0.198z$         | $-1.59 \cdot 10^{-16} + 2.07z$      | $0.0242 - 3.78z$                               | $-1.35 \cdot 10^{-5} + 1.8z$                    |
| 7.0 | $5.2 \cdot 10^{-18} - 0.198z$         | $8.43 \cdot 10^{-16} - 2.07z$       | $0.0244 + 3.78z$                               | $-3.31 \cdot 10^4 - 1.58 \cdot 10^4z$           |

Table 58: Case 1a (cpv MACI64 MR2022b),  $r = 7$ , Z4: zeros.

|     | Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$ | LF, $\sigma_7 = 5.2 \cdot 10^{-08}$ | AAA-sign-L200, $\sigma_7 = 9.8 \cdot 10^{-09}$ | AAA-sign-L1000, $\sigma_7 = 9.8 \cdot 10^{-09}$ |
|-----|---------------------------------------|-------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | NaN                                   | NaN                                 | $1.11 \cdot 10^{-11} - 8.34 \cdot 10^{-10}i$   | $-2.09 \cdot 10^{-14} - 7.35 \cdot 10^{-10}i$   |
| 2.0 | $1.73 \cdot 10^{-17} + 1.8i$          | $8.87 \cdot 10^{-16} + 0.216i$      | $1.36 \cdot 10^{-11} + 0.417i$                 | $-6.16 \cdot 10^{-12} + 0.417i$                 |
| 3.0 | $1.73 \cdot 10^{-17} - 1.8i$          | $-8.68 \cdot 10^{-16} - 0.216i$     | $1.36 \cdot 10^{-11} - 0.417i$                 | $6.12 \cdot 10^{-12} - 0.417i$                  |
| 4.0 | $5.55 \cdot 10^{-17} + 0.691i$        | $2.39 \cdot 10^{-16} + 0.768i$      | $2.86 \cdot 10^{-11} + 1.09i$                  | $-1.55 \cdot 10^{-11} + 1.09i$                  |
| 5.0 | $5.55 \cdot 10^{-17} - 0.691i$        | $4.76 \cdot 10^{-16} - 0.768i$      | $2.84 \cdot 10^{-11} - 1.09i$                  | $1.49 \cdot 10^{-11} - 1.09i$                   |
| 6.0 | $1.21 \cdot 10^{-17} + 0.198i$        | $3.76 \cdot 10^{-15} + 2.07i$       | $2.24 \cdot 10^{-10} + 3.79i$                  | $-5.59 \cdot 10^{-11} + 3.79i$                  |
| 7.0 | $1.21 \cdot 10^{-17} - 0.198i$        | $1.55 \cdot 10^{-15} - 2.07i$       | $2.23 \cdot 10^{-10} - 3.79i$                  | $4.61 \cdot 10^{-11} - 3.79i$                   |

Table 59: Case 1a (cpv PCWIN64 MR2025b),  $r = 7$ , Z4: zeros.

|     | Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$ | LF, $\sigma_7 = 5.2 \cdot 10^{-08}$ | AAA-sign-L200, $\sigma_7 = 9.8 \cdot 10^{-09}$ | AAA-sign-L1000, $\sigma_7 = 9.8 \cdot 10^{-09}$ |
|-----|---------------------------------------|-------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | NaN                                   | NaN                                 | $1.11 \cdot 10^{-11} - 8.34 \cdot 10^{-10}i$   | $-2.09 \cdot 10^{-14} - 7.35 \cdot 10^{-10}i$   |
| 2.0 | $1.73 \cdot 10^{-17} + 1.8i$          | $8.87 \cdot 10^{-16} + 0.216i$      | $1.36 \cdot 10^{-11} + 0.417i$                 | $-6.16 \cdot 10^{-12} + 0.417i$                 |
| 3.0 | $1.73 \cdot 10^{-17} - 1.8i$          | $-8.68 \cdot 10^{-16} - 0.216i$     | $1.36 \cdot 10^{-11} - 0.417i$                 | $6.12 \cdot 10^{-12} - 0.417i$                  |
| 4.0 | $5.55 \cdot 10^{-17} + 0.691i$        | $2.39 \cdot 10^{-16} + 0.768i$      | $2.86 \cdot 10^{-11} + 1.09i$                  | $-1.55 \cdot 10^{-11} + 1.09i$                  |
| 5.0 | $5.55 \cdot 10^{-17} - 0.691i$        | $4.76 \cdot 10^{-16} - 0.768i$      | $2.84 \cdot 10^{-11} - 1.09i$                  | $1.49 \cdot 10^{-11} - 1.09i$                   |
| 6.0 | $1.21 \cdot 10^{-17} + 0.198i$        | $3.76 \cdot 10^{-15} + 2.07i$       | $2.24 \cdot 10^{-10} + 3.79i$                  | $-5.59 \cdot 10^{-11} + 3.79i$                  |
| 7.0 | $1.21 \cdot 10^{-17} - 0.198i$        | $1.55 \cdot 10^{-15} - 2.07i$       | $2.23 \cdot 10^{-10} - 3.79i$                  | $4.61 \cdot 10^{-11} - 3.79i$                   |

Table 60: Case 1a (AlexReis PCWIN64 MR2025b),  $r = 7$ , Z4: zeros.

## 6.4 Poles and zeros visualisation

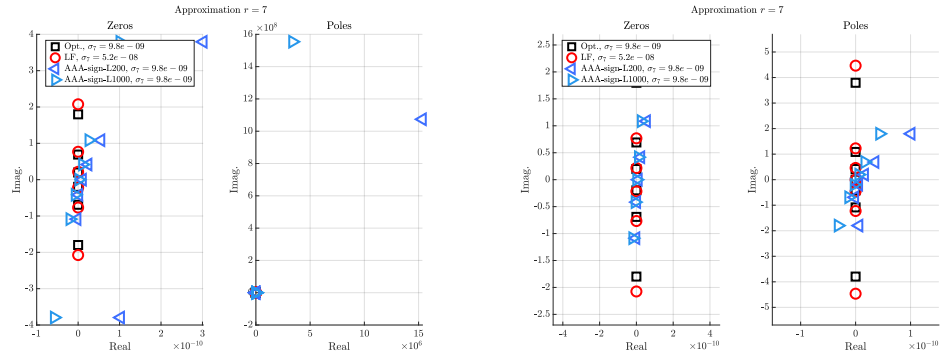


Figure 9: Case 1a (cpv MACA64 MR2023b),  $r = 7$ , Z4. Left frame: zeros (left) and poles (right). Right frame: same but discarding the largest real entry.

## 6.5 Approximation visualization

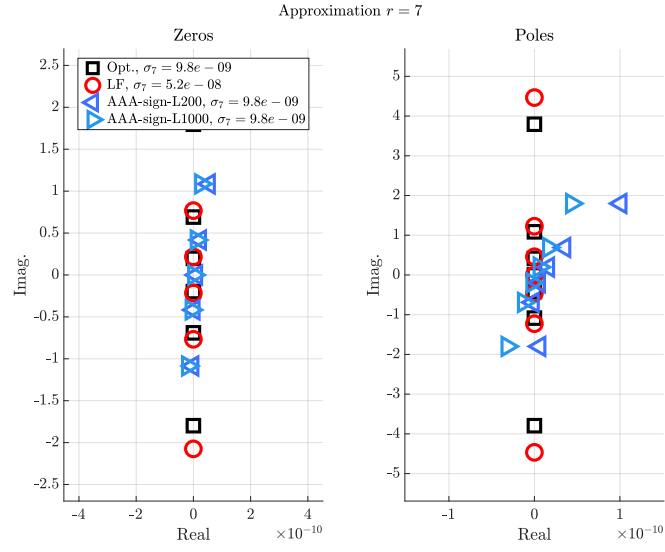


Figure 10: Case 1a (cpv MACA64 MR2023b),  $r = 7$ , Z4.  $E$  and  $F$  evaluations.

## 7 Approximation objective $r = 8$

### 7.1 Numerator and denominator coefficients

The numerator and denominator coefficients (both real and complex parts) are given in the following tables. Each table gives the results obtained on different platform.

| Basis | Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$ |       | LF, $\sigma_8 = 1.6 \cdot 10^{-09}$ |                       | AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$ |                      | AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$ |                      |
|-------|---------------------------------------|-------|-------------------------------------|-----------------------|------------------------------------------------|----------------------|-------------------------------------------------|----------------------|
|       | real                                  | imag. | real                                | imag.                 | real                                           | imag.                | real                                            | imag.                |
| $z^8$ | 0                                     | 0     | 0                                   | 0                     | $-8.7 \cdot 10^5$                              | $1.7 \cdot 10^8$     | $-5.1 \cdot 10^6$                               | $-7.9 \cdot 10^8$    |
| $z^7$ | 6.9                                   | 0     | 7.2                                 | $-1.9 \cdot 10^{-14}$ | 7.0                                            | $-4.9 \cdot 10^{-4}$ | 7.0                                             | $-3.4 \cdot 10^{-3}$ |
| $z^6$ | 0                                     | 0     | $-5.3 \cdot 10^{-14}$               | $-1.4 \cdot 10^{-13}$ | $-1.8 \cdot 10^7$                              | $3.7 \cdot 10^9$     | $-1.1 \cdot 10^8$                               | $-1.7 \cdot 10^{10}$ |
| $z^5$ | 36.0                                  | 0     | 41.0                                | $-7.4 \cdot 10^{-14}$ | 37.0                                           | $-9.8 \cdot 10^{-3}$ | 37.0                                            | -0.046               |
| $z^4$ | 0                                     | 0     | $7.7 \cdot 10^{-14}$                | $-3.2 \cdot 10^{-13}$ | $-3.4 \cdot 10^7$                              | $6.9 \cdot 10^9$     | $-2.0 \cdot 10^8$                               | $-3.1 \cdot 10^{10}$ |
| $z^3$ | 27.0                                  | 0     | 33.0                                | $-2.2 \cdot 10^{-14}$ | 28.0                                           | -0.014               | 28.0                                            | -0.051               |
| $z^2$ | 0                                     | 0     | $1.6 \cdot 10^{-13}$                | $-7.9 \cdot 10^{-14}$ | $-1.0 \cdot 10^7$                              | $2.1 \cdot 10^9$     | $-6.0 \cdot 10^7$                               | $-9.3 \cdot 10^9$    |
| $z$   | 2.9                                   | 0     | 3.7                                 | $3.3 \cdot 10^{-14}$  | 3.0                                            | $-2.4 \cdot 10^{-3}$ | 3.1                                             | $-7.0 \cdot 10^{-3}$ |
| 1.0   | 0                                     | 0     | $1.0 \cdot 10^{-14}$                | $1.0 \cdot 10^{-14}$  | $-2.8 \cdot 10^5$                              | $5.5 \cdot 10^7$     | $-1.6 \cdot 10^6$                               | $-2.5 \cdot 10^8$    |
| $z^8$ | 1.0                                   | 0     | 1.0                                 | 0                     | 1.0                                            | 0                    | 1.0                                             | 0                    |
| $z^7$ | 0                                     | 0     | $-1.4 \cdot 10^{-14}$               | $-4.1 \cdot 10^{-14}$ | $-6.0 \cdot 10^6$                              | $1.2 \cdot 10^9$     | $-3.5 \cdot 10^7$                               | $-5.4 \cdot 10^9$    |
| $z^6$ | 21.0                                  | 0     | 23.0                                | $-5.5 \cdot 10^{-14}$ | 21.0                                           | $-3.4 \cdot 10^{-3}$ | 21.0                                            | -0.019               |
| $z^5$ | 0                                     | 0     | $-5.6 \cdot 10^{-14}$               | $-2.7 \cdot 10^{-13}$ | $-3.2 \cdot 10^7$                              | $6.3 \cdot 10^9$     | $-1.9 \cdot 10^8$                               | $-2.9 \cdot 10^{10}$ |
| $z^4$ | 39.0                                  | 0     | 46.0                                | $-6.5 \cdot 10^{-14}$ | 40.0                                           | -0.015               | 40.0                                            | -0.062               |
| $z^3$ | 0                                     | 0     | $1.9 \cdot 10^{-13}$                | $-2.4 \cdot 10^{-13}$ | $-2.4 \cdot 10^7$                              | $4.7 \cdot 10^9$     | $-1.4 \cdot 10^8$                               | $-2.1 \cdot 10^{10}$ |
| $z^2$ | 12.0                                  | 0     | 15.0                                | $3.1 \cdot 10^{-14}$  | 12.0                                           | $-8.0 \cdot 10^{-3}$ | 12.0                                            | -0.025               |
| $z$   | 0                                     | 0     | $6.2 \cdot 10^{-14}$                | $1.9 \cdot 10^{-14}$  | $-2.5 \cdot 10^6$                              | $5.1 \cdot 10^8$     | $-1.5 \cdot 10^7$                               | $-2.3 \cdot 10^9$    |
| 1.0   | 0.32                                  | 0     | 0.42                                | $8.3 \cdot 10^{-15}$  | 0.33                                           | $-3.2 \cdot 10^{-4}$ | 0.33                                            | $-8.5 \cdot 10^{-4}$ |

Table 61: Case 1a (cpv MACA64 MR2023b),  $r = 8$ , Z4: numerator (first lines) and denominator (last lines) coefficients.

| Basis | Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$ |       | LF, $\sigma_8 = 1.6 \cdot 10^{-09}$ |                       | AAA-sign-L200, $\sigma_8 = 7.4 \cdot 10^{-10}$ |                      | AAA-sign-L1000, $\sigma_8 = 7.4 \cdot 10^{-10}$ |                      |
|-------|---------------------------------------|-------|-------------------------------------|-----------------------|------------------------------------------------|----------------------|-------------------------------------------------|----------------------|
|       | real                                  | imag. | real                                | imag.                 | real                                           | imag.                | real                                            | imag.                |
| $z^8$ | 0                                     | 0     | 0                                   | 0                     | $4.6 \cdot 10^{-3}$                            | $1.0 \cdot 10^{-4}$  | $4.6 \cdot 10^{-3}$                             | $1.3 \cdot 10^{-4}$  |
| $z^7$ | 6.9                                   | 0     | 7.2                                 | $1.3 \cdot 10^{-14}$  | 6.9                                            | $-8.5 \cdot 10^{-5}$ | 6.9                                             | $-8.5 \cdot 10^{-5}$ |
| $z^6$ | 0                                     | 0     | $6.3 \cdot 10^{-14}$                | $-1.2 \cdot 10^{-13}$ | 0.12                                           | $1.5 \cdot 10^{-3}$  | 0.12                                            | $2.0 \cdot 10^{-3}$  |
| $z^5$ | 36.0                                  | 0     | 41.0                                | $1.9 \cdot 10^{-13}$  | 36.0                                           | $-1.3 \cdot 10^{-3}$ | 36.0                                            | $-1.3 \cdot 10^{-3}$ |
| $z^4$ | 0                                     | 0     | $3.4 \cdot 10^{-13}$                | $-3.0 \cdot 10^{-13}$ | 0.25                                           | $1.8 \cdot 10^{-3}$  | 0.25                                            | $2.8 \cdot 10^{-3}$  |
| $z^3$ | 27.0                                  | 0     | 33.0                                | $3.6 \cdot 10^{-13}$  | 27.0                                           | $-1.5 \cdot 10^{-3}$ | 27.0                                            | $-1.6 \cdot 10^{-3}$ |
| $z^2$ | 0                                     | 0     | $1.6 \cdot 10^{-13}$                | $-1.4 \cdot 10^{-13}$ | 0.086                                          | $2.6 \cdot 10^{-4}$  | 0.086                                           | $5.5 \cdot 10^{-4}$  |
| $z$   | 2.9                                   | 0     | 3.7                                 | $3.8 \cdot 10^{-14}$  | 2.9                                            | $-2.2 \cdot 10^{-4}$ | 2.9                                             | $-2.3 \cdot 10^{-4}$ |
| 1.0   | 0                                     | 0     | $6.1 \cdot 10^{-15}$                | $-6.4 \cdot 10^{-15}$ | $2.6 \cdot 10^{-3}$                            | $-3.5 \cdot 10^{-7}$ | $2.6 \cdot 10^{-3}$                             | $7.3 \cdot 10^{-6}$  |
| $z^8$ | 1.0                                   | 0     | 1.0                                 | 0                     | 1.0                                            | 0                    | 1.0                                             | 0                    |
| $z^7$ | 0                                     | 0     | $3.0 \cdot 10^{-15}$                | $-2.1 \cdot 10^{-14}$ | 0.035                                          | $6.0 \cdot 10^{-4}$  | 0.035                                           | $7.7 \cdot 10^{-4}$  |
| $z^6$ | 21.0                                  | 0     | 23.0                                | $7.1 \cdot 10^{-14}$  | 21.0                                           | $-5.0 \cdot 10^{-4}$ | 21.0                                            | $-5.1 \cdot 10^{-4}$ |
| $z^5$ | 0                                     | 0     | $2.2 \cdot 10^{-13}$                | $-2.3 \cdot 10^{-13}$ | 0.22                                           | $2.2 \cdot 10^{-3}$  | 0.22                                            | $3.0 \cdot 10^{-3}$  |
| $z^4$ | 39.0                                  | 0     | 46.0                                | $3.4 \cdot 10^{-13}$  | 39.0                                           | $-1.8 \cdot 10^{-3}$ | 39.0                                            | $-1.8 \cdot 10^{-3}$ |
| $z^3$ | 0                                     | 0     | $3.1 \cdot 10^{-13}$                | $-2.5 \cdot 10^{-13}$ | 0.19                                           | $9.3 \cdot 10^{-4}$  | 0.19                                            | $1.6 \cdot 10^{-3}$  |
| $z^2$ | 12.0                                  | 0     | 15.0                                | $1.9 \cdot 10^{-13}$  | 12.0                                           | $-7.9 \cdot 10^{-4}$ | 12.0                                            | $-7.9 \cdot 10^{-4}$ |
| $z$   | 0                                     | 0     | $4.5 \cdot 10^{-14}$                | $-5.3 \cdot 10^{-14}$ | 0.023                                          | $3.0 \cdot 10^{-5}$  | 0.023                                           | $1.0 \cdot 10^{-4}$  |
| 1.0   | 0.32                                  | 0     | 0.42                                | $1.8 \cdot 10^{-15}$  | 0.32                                           | $-2.7 \cdot 10^{-5}$ | 0.32                                            | $-2.7 \cdot 10^{-5}$ |

Table 62: Case 1a (cpv MACI64 MR2022b),  $r = 8$ , Z4: numerator (first lines) and denominator (last lines) coefficients.



| Basis | Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$ |       | LF, $\sigma_8 = 1.6 \cdot 10^{-09}$ |                       | AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$ |                       | AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$ |                       |
|-------|---------------------------------------|-------|-------------------------------------|-----------------------|------------------------------------------------|-----------------------|-------------------------------------------------|-----------------------|
|       | real                                  | imag. | real                                | imag.                 | real                                           | imag.                 | real                                            | imag.                 |
| $z^8$ | 0                                     | 0     | 0                                   | 0                     | $-5.8 \cdot 10^{-11}$                          | $-1.2 \cdot 10^{-8}$  | $2.5 \cdot 10^{-12}$                            | $1.4 \cdot 10^{-9}$   |
| $z^7$ | 6.9                                   | 0     | 7.2                                 | $-1.5 \cdot 10^{-14}$ | 6.9                                            | $-2.1 \cdot 10^{-10}$ | 6.9                                             | $-5.3 \cdot 10^{-11}$ |
| $z^6$ | 0                                     | 0     | $-5.0 \cdot 10^{-14}$               | $3.3 \cdot 10^{-14}$  | $-1.2 \cdot 10^{-9}$                           | $-2.6 \cdot 10^{-7}$  | $6.3 \cdot 10^{-11}$                            | $2.9 \cdot 10^{-8}$   |
| $z^5$ | 36.0                                  | 0     | 41.0                                | $-2.6 \cdot 10^{-14}$ | 36.0                                           | $-3.3 \cdot 10^{-9}$  | 36.0                                            | $-8.3 \cdot 10^{-10}$ |
| $z^4$ | 0                                     | 0     | $-1.7 \cdot 10^{-13}$               | $1.5 \cdot 10^{-13}$  | $-2.1 \cdot 10^{-9}$                           | $-4.8 \cdot 10^{-7}$  | $1.3 \cdot 10^{-10}$                            | $5.5 \cdot 10^{-8}$   |
| $z^3$ | 27.0                                  | 0     | 33.0                                | $1.7 \cdot 10^{-13}$  | 27.0                                           | $-4.1 \cdot 10^{-9}$  | 27.0                                            | $-1.0 \cdot 10^{-9}$  |
| $z^2$ | 0                                     | 0     | $-8.5 \cdot 10^{-14}$               | $9.5 \cdot 10^{-14}$  | $-6.3 \cdot 10^{-10}$                          | $-1.5 \cdot 10^{-7}$  | $4.3 \cdot 10^{-11}$                            | $1.6 \cdot 10^{-8}$   |
| $z$   | 2.9                                   | 0     | 3.7                                 | $4.3 \cdot 10^{-14}$  | 2.9                                            | $-6.2 \cdot 10^{-10}$ | 2.9                                             | $-1.6 \cdot 10^{-10}$ |
| 1.0   | 0                                     | 0     | $-2.2 \cdot 10^{-15}$               | $4.0 \cdot 10^{-15}$  | $-1.7 \cdot 10^{-11}$                          | $-3.9 \cdot 10^{-9}$  | $1.3 \cdot 10^{-12}$                            | $4.3 \cdot 10^{-10}$  |
| $z^8$ | 1.0                                   | 0     | 1.0                                 | 0                     | 1.0                                            | 0                     | 1.0                                             | 0                     |
| $z^7$ | 0                                     | 0     | $-7.1 \cdot 10^{-15}$               | $-2.0 \cdot 10^{-15}$ | $-3.9 \cdot 10^{-10}$                          | $-8.4 \cdot 10^{-8}$  | $1.9 \cdot 10^{-11}$                            | $9.7 \cdot 10^{-9}$   |
| $z^6$ | 21.0                                  | 0     | 23.0                                | $-4.0 \cdot 10^{-14}$ | 21.0                                           | $-1.3 \cdot 10^{-9}$  | 21.0                                            | $-3.2 \cdot 10^{-10}$ |
| $z^5$ | 0                                     | 0     | $-1.2 \cdot 10^{-13}$               | $9.2 \cdot 10^{-14}$  | $-2.0 \cdot 10^{-9}$                           | $-4.5 \cdot 10^{-7}$  | $1.2 \cdot 10^{-10}$                            | $5.1 \cdot 10^{-8}$   |
| $z^4$ | 39.0                                  | 0     | 46.0                                | $9.0 \cdot 10^{-14}$  | 39.0                                           | $-4.8 \cdot 10^{-9}$  | 39.0                                            | $-1.2 \cdot 10^{-9}$  |
| $z^3$ | 0                                     | 0     | $-1.6 \cdot 10^{-13}$               | $1.6 \cdot 10^{-13}$  | $-1.5 \cdot 10^{-9}$                           | $-3.4 \cdot 10^{-7}$  | $9.7 \cdot 10^{-11}$                            | $3.8 \cdot 10^{-8}$   |
| $z^2$ | 12.0                                  | 0     | 15.0                                | $1.2 \cdot 10^{-13}$  | 12.0                                           | $-2.1 \cdot 10^{-9}$  | 12.0                                            | $-5.4 \cdot 10^{-10}$ |
| $z$   | 0                                     | 0     | $-2.1 \cdot 10^{-14}$               | $3.3 \cdot 10^{-14}$  | $-1.5 \cdot 10^{-10}$                          | $-3.6 \cdot 10^{-8}$  | $1.1 \cdot 10^{-11}$                            | $4.0 \cdot 10^{-9}$   |
| 1.0   | 0.32                                  | 0     | 0.42                                | $6.7 \cdot 10^{-15}$  | 0.32                                           | $-7.6 \cdot 10^{-11}$ | 0.32                                            | $-1.9 \cdot 10^{-11}$ |

Table 63: Case 1a (cpv PCWIN64 MR2025b),  $r = 8$ , Z4: numerator (first lines) and denominator (last lines) coefficients.

| Basis | Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$ |       | LF, $\sigma_8 = 1.6 \cdot 10^{-09}$ |                       | AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$ |                       | AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$ |                       |
|-------|---------------------------------------|-------|-------------------------------------|-----------------------|------------------------------------------------|-----------------------|-------------------------------------------------|-----------------------|
|       | real                                  | imag. | real                                | imag.                 | real                                           | imag.                 | real                                            | imag.                 |
| $z^8$ | 0                                     | 0     | 0                                   | 0                     | $-5.8 \cdot 10^{-11}$                          | $-1.2 \cdot 10^{-8}$  | $2.5 \cdot 10^{-12}$                            | $1.4 \cdot 10^{-9}$   |
| $z^7$ | 6.9                                   | 0     | 7.2                                 | $-1.5 \cdot 10^{-14}$ | 6.9                                            | $-2.1 \cdot 10^{-10}$ | 6.9                                             | $-5.3 \cdot 10^{-11}$ |
| $z^6$ | 0                                     | 0     | $-5.0 \cdot 10^{-14}$               | $3.3 \cdot 10^{-14}$  | $-1.2 \cdot 10^{-9}$                           | $-2.6 \cdot 10^{-7}$  | $6.3 \cdot 10^{-11}$                            | $2.9 \cdot 10^{-8}$   |
| $z^5$ | 36.0                                  | 0     | 41.0                                | $-2.6 \cdot 10^{-14}$ | 36.0                                           | $-3.3 \cdot 10^{-9}$  | 36.0                                            | $-8.3 \cdot 10^{-10}$ |
| $z^4$ | 0                                     | 0     | $-1.7 \cdot 10^{-13}$               | $1.5 \cdot 10^{-13}$  | $-2.1 \cdot 10^{-9}$                           | $-4.8 \cdot 10^{-7}$  | $1.3 \cdot 10^{-10}$                            | $5.5 \cdot 10^{-8}$   |
| $z^3$ | 27.0                                  | 0     | 33.0                                | $1.7 \cdot 10^{-13}$  | 27.0                                           | $-4.1 \cdot 10^{-9}$  | 27.0                                            | $-1.0 \cdot 10^{-9}$  |
| $z^2$ | 0                                     | 0     | $-8.5 \cdot 10^{-14}$               | $9.5 \cdot 10^{-14}$  | $-6.3 \cdot 10^{-10}$                          | $-1.5 \cdot 10^{-7}$  | $4.3 \cdot 10^{-11}$                            | $1.6 \cdot 10^{-8}$   |
| $z$   | 2.9                                   | 0     | 3.7                                 | $4.3 \cdot 10^{-14}$  | 2.9                                            | $-6.2 \cdot 10^{-10}$ | 2.9                                             | $-1.6 \cdot 10^{-10}$ |
| 1.0   | 0                                     | 0     | $-2.2 \cdot 10^{-15}$               | $4.0 \cdot 10^{-15}$  | $-1.7 \cdot 10^{-11}$                          | $-3.9 \cdot 10^{-9}$  | $1.3 \cdot 10^{-12}$                            | $4.3 \cdot 10^{-10}$  |
| $z^8$ | 1.0                                   | 0     | 1.0                                 | 0                     | 1.0                                            | 0                     | 1.0                                             | 0                     |
| $z^7$ | 0                                     | 0     | $-7.1 \cdot 10^{-15}$               | $-2.0 \cdot 10^{-15}$ | $-3.9 \cdot 10^{-10}$                          | $-8.4 \cdot 10^{-8}$  | $1.9 \cdot 10^{-11}$                            | $9.7 \cdot 10^{-9}$   |
| $z^6$ | 21.0                                  | 0     | 23.0                                | $-4.0 \cdot 10^{-14}$ | 21.0                                           | $-1.3 \cdot 10^{-9}$  | 21.0                                            | $-3.2 \cdot 10^{-10}$ |
| $z^5$ | 0                                     | 0     | $-1.2 \cdot 10^{-13}$               | $9.2 \cdot 10^{-14}$  | $-2.0 \cdot 10^{-9}$                           | $-4.5 \cdot 10^{-7}$  | $1.2 \cdot 10^{-10}$                            | $5.1 \cdot 10^{-8}$   |
| $z^4$ | 39.0                                  | 0     | 46.0                                | $9.0 \cdot 10^{-14}$  | 39.0                                           | $-4.8 \cdot 10^{-9}$  | 39.0                                            | $-1.2 \cdot 10^{-9}$  |
| $z^3$ | 0                                     | 0     | $-1.6 \cdot 10^{-13}$               | $1.6 \cdot 10^{-13}$  | $-1.5 \cdot 10^{-9}$                           | $-3.4 \cdot 10^{-7}$  | $9.7 \cdot 10^{-11}$                            | $3.8 \cdot 10^{-8}$   |
| $z^2$ | 12.0                                  | 0     | 15.0                                | $1.2 \cdot 10^{-13}$  | 12.0                                           | $-2.1 \cdot 10^{-9}$  | 12.0                                            | $-5.4 \cdot 10^{-10}$ |
| $z$   | 0                                     | 0     | $-2.1 \cdot 10^{-14}$               | $3.3 \cdot 10^{-14}$  | $-1.5 \cdot 10^{-10}$                          | $-3.6 \cdot 10^{-8}$  | $1.1 \cdot 10^{-11}$                            | $4.0 \cdot 10^{-9}$   |
| 1.0   | 0.32                                  | 0     | 0.42                                | $6.7 \cdot 10^{-15}$  | 0.32                                           | $-7.6 \cdot 10^{-11}$ | 0.32                                            | $-1.9 \cdot 10^{-11}$ |

Table 64: Case 1a (AlexReis PCWIN64 MR2025b),  $r = 8$ , Z4: numerator (first lines) and denominator (last lines) coefficients.

## 7.2 Poles

The poles are given in the following tables. Each table gives the results obtained on different platform.

|     | Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$ | LF, $\sigma_8 = 1.6 \cdot 10^{-09}$ | AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$ | AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$ |
|-----|---------------------------------------|-------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | $6.66 \cdot 10^{-16} + 4.35z$         | $-4.06 \cdot 10^{-15} + 0.178z$     | $3.84 \cdot 10^{-12} + 6.42 \cdot 10^{-10}z$   | $5.64 \cdot 10^{-13} - 1.45 \cdot 10^{-10}z$    |
| 2.0 | $6.66 \cdot 10^{-16} - 4.35z$         | $-6.53 \cdot 10^{-16} - 0.178z$     | $-4.23 \cdot 10^{-12} - 0.359z$                | $-2.48 \cdot 10^{-12} + 0.359z$                 |
| 3.0 | $2.22 \cdot 10^{-16} + 1.3z$          | $-7.33 \cdot 10^{-17} - 0.599z$     | $1.28 \cdot 10^{-11} + 0.359z$                 | $3.91 \cdot 10^{-12} - 0.359z$                  |
| 4.0 | $2.22 \cdot 10^{-16} - 1.3z$          | $-1.58 \cdot 10^{-15} + 0.599z$     | $-1.39 \cdot 10^{-11} - 0.866z$                | $-6.57 \cdot 10^{-12} + 0.866z$                 |
| 5.0 | $1.39 \cdot 10^{-17} + 0.579z$        | $4.61 \cdot 10^{-15} - 1.34z$       | $2.73 \cdot 10^{-11} + 0.866z$                 | $9.26 \cdot 10^{-12} - 0.866z$                  |
| 6.0 | $1.39 \cdot 10^{-17} - 0.579z$        | $3.38 \cdot 10^{-15} + 1.34z$       | $-2.86 \cdot 10^{-11} - 2.09z$                 | $-1.37 \cdot 10^{-11} + 2.09z$                  |
| 7.0 | $1.47 \cdot 10^{-17} + 0.172z$        | $-4.63 \cdot 10^{-16} - 4.52z$      | $7.16 \cdot 10^{-11} + 2.09z$                  | $2.47 \cdot 10^{-11} - 2.09z$                   |
| 8.0 | $1.47 \cdot 10^{-17} - 0.172z$        | $1.25 \cdot 10^{-14} + 4.52z$       | $6.04 \cdot 10^6 - 1.21 \cdot 10^9z$           | $3.54 \cdot 10^7 + 5.44 \cdot 10^9z$            |

Table 65: Case 1a (cpv MACA64 MR2023b),  $r = 8$ , Z4: poles.

|     | Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$ | LF, $\sigma_8 = 1.6 \cdot 10^{-09}$ | AAA-sign-L200, $\sigma_8 = 7.4 \cdot 10^{-10}$ | AAA-sign-L1000, $\sigma_8 = 7.4 \cdot 10^{-10}$ |
|-----|---------------------------------------|-------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | $8.88 \cdot 10^{-16} + 4.35_z$        | $-2.44 \cdot 10^{-15} - 0.178_z$    | $-8.97 \cdot 10^{-4} + 0.172_z$                | $-8.97 \cdot 10^{-4} + 0.172_z$                 |
| 2.0 | $8.88 \cdot 10^{-16} - 4.35_z$        | $-5.6 \cdot 10^{-16} + 0.178_z$     | $-9.0 \cdot 10^{-4} - 0.172_z$                 | $-9.0 \cdot 10^{-4} - 0.172_z$                  |
| 3.0 | $8.33 \cdot 10^{-17} + 1.3_z$         | $-5.87 \cdot 10^{-17} - 0.599_z$    | $-0.0011 + 0.579_z$                            | $-0.0011 + 0.579_z$                             |
| 4.0 | $8.33 \cdot 10^{-17} - 1.3_z$         | $-3.93 \cdot 10^{-15} + 0.599_z$    | $-0.00112 - 0.579_z$                           | $-0.00112 - 0.579_z$                            |
| 5.0 | $4.16 \cdot 10^{-17} + 0.579_z$       | $1.3 \cdot 10^{-15} - 1.34_z$       | $-0.002 + 1.3_z$                               | $-0.002 + 1.3_z$                                |
| 6.0 | $4.16 \cdot 10^{-17} - 0.579_z$       | $-5.03 \cdot 10^{-15} + 1.34_z$     | $-0.00203 - 1.3_z$                             | $-0.00203 - 1.3_z$                              |
| 7.0 | $-2.6 \cdot 10^{-18} + 0.172_z$       | $1.06 \cdot 10^{-14} - 4.52_z$      | $-0.0135 + 4.36_z$                             | $-0.0135 + 4.36_z$                              |
| 8.0 | $-2.6 \cdot 10^{-18} - 0.172_z$       | $-2.57 \cdot 10^{-15} + 4.52_z$     | $-0.0136 - 4.36_z$                             | $-0.0136 - 4.36_z$                              |

Table 66: Case 1a (cpv MACI64 MR2022b),  $r = 8$ , Z4: poles.

|     | Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$ | LF, $\sigma_8 = 1.6 \cdot 10^{-09}$ | AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$ | AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$ |
|-----|---------------------------------------|-------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | $7.22 \cdot 10^{-16} + 4.35_z$        | $1.37 \cdot 10^{-17} + 0.178_z$     | $7.54 \cdot 10^{-13} - 0.172_z$                | $8.64 \cdot 10^{-13} + 0.172_z$                 |
| 2.0 | $7.22 \cdot 10^{-16} - 4.35_z$        | $1.31 \cdot 10^{-15} - 0.178_z$     | $1.1 \cdot 10^{-11} + 0.172_z$                 | $-1.73 \cdot 10^{-12} - 0.172_z$                |
| 3.0 | $1.94 \cdot 10^{-16} + 1.3_z$         | $-1.07 \cdot 10^{-15} + 0.599_z$    | $-9.06 \cdot 10^{-12} - 0.579_z$               | $3.82 \cdot 10^{-12} + 0.579_z$                 |
| 4.0 | $1.94 \cdot 10^{-16} - 1.3_z$         | $3.39 \cdot 10^{-15} - 0.599_z$     | $2.57 \cdot 10^{-11} + 0.579_z$                | $-4.98 \cdot 10^{-12} - 0.579_z$                |
| 5.0 | $-4.16 \cdot 10^{-17} + 0.579_z$      | $-1.82 \cdot 10^{-15} + 1.34_z$     | $-2.0 \cdot 10^{-11} - 1.3_z$                  | $8.73 \cdot 10^{-12} + 1.3_z$                   |
| 6.0 | $-4.16 \cdot 10^{-17} - 0.579_z$      | $3.47 \cdot 10^{-15} - 1.34_z$      | $5.85 \cdot 10^{-11} + 1.3_z$                  | $-1.1 \cdot 10^{-11} - 1.3_z$                   |
| 7.0 | $1.73 \cdot 10^{-18} + 0.172_z$       | $-5.1 \cdot 10^{-15} - 4.52_z$      | $2.98 \cdot 10^{-11} - 4.35_z$                 | $2.57 \cdot 10^{-11} + 4.35_z$                  |
| 8.0 | $1.73 \cdot 10^{-18} - 0.172_z$       | $6.85 \cdot 10^{-15} + 4.52_z$      | $2.95 \cdot 10^{-10} + 4.35_z$                 | $-4.05 \cdot 10^{-11} - 4.35_z$                 |

Table 67: Case 1a (cpv PCWIN64 MR2025b),  $r = 8$ , Z4: poles.

|     | Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$ | LF, $\sigma_8 = 1.6 \cdot 10^{-09}$ | AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$ | AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$ |
|-----|---------------------------------------|-------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | $7.22 \cdot 10^{-16} + 4.35_z$        | $1.37 \cdot 10^{-17} + 0.178_z$     | $7.54 \cdot 10^{-13} - 0.172_z$                | $8.64 \cdot 10^{-13} + 0.172_z$                 |
| 2.0 | $7.22 \cdot 10^{-16} - 4.35_z$        | $1.31 \cdot 10^{-15} - 0.178_z$     | $1.1 \cdot 10^{-11} + 0.172_z$                 | $-1.73 \cdot 10^{-12} - 0.172_z$                |
| 3.0 | $1.94 \cdot 10^{-16} + 1.3_z$         | $-1.07 \cdot 10^{-15} + 0.599_z$    | $-9.06 \cdot 10^{-12} - 0.579_z$               | $3.82 \cdot 10^{-12} + 0.579_z$                 |
| 4.0 | $1.94 \cdot 10^{-16} - 1.3_z$         | $3.39 \cdot 10^{-15} - 0.599_z$     | $2.57 \cdot 10^{-11} + 0.579_z$                | $-4.98 \cdot 10^{-12} - 0.579_z$                |
| 5.0 | $-4.16 \cdot 10^{-17} + 0.579_z$      | $-1.82 \cdot 10^{-15} + 1.34_z$     | $-2.0 \cdot 10^{-11} - 1.3_z$                  | $8.73 \cdot 10^{-12} + 1.3_z$                   |
| 6.0 | $-4.16 \cdot 10^{-17} - 0.579_z$      | $3.47 \cdot 10^{-15} - 1.34_z$      | $5.85 \cdot 10^{-11} + 1.3_z$                  | $-1.1 \cdot 10^{-11} - 1.3_z$                   |
| 7.0 | $1.73 \cdot 10^{-18} + 0.172_z$       | $-5.1 \cdot 10^{-15} - 4.52_z$      | $2.98 \cdot 10^{-11} - 4.35_z$                 | $2.57 \cdot 10^{-11} + 4.35_z$                  |
| 8.0 | $1.73 \cdot 10^{-18} - 0.172_z$       | $6.85 \cdot 10^{-15} + 4.52_z$      | $2.95 \cdot 10^{-10} + 4.35_z$                 | $-4.05 \cdot 10^{-11} - 4.35_z$                 |

Table 68: Case 1a (AlexReis PCWIN64 MR2025b),  $r = 8$ , Z4: poles.

### 7.3 Zeros

The zeros are given in the following tables. Each table gives the results obtained on different platform.

|     | Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$ | LF, $\sigma_8 = 1.6 \cdot 10^{-09}$            | AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$ | AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$ |
|-----|---------------------------------------|------------------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | NaN                                   | NaN                                            | $-1.46 \cdot 10^{-13} - 0.172_z$               | $-9.08 \cdot 10^{-13} + 0.172_z$                |
| 2.0 | 0                                     | $-2.76 \cdot 10^{-15} - 2.73 \cdot 10^{-15}_z$ | $8.04 \cdot 10^{-12} + 0.172_z$                | $2.11 \cdot 10^{-12} - 0.172_z$                 |
| 3.0 | $1.39 \cdot 10^{-16} + 2.09_z$        | $4.64 \cdot 10^{-16} - 0.371_z$                | $-8.67 \cdot 10^{-12} - 0.579_z$               | $-4.32 \cdot 10^{-12} + 0.579_z$                |
| 4.0 | $1.39 \cdot 10^{-16} - 2.09_z$        | $-3.67 \cdot 10^{-15} + 0.371_z$               | $1.88 \cdot 10^{-11} + 0.579_z$                | $6.15 \cdot 10^{-12} - 0.579_z$                 |
| 5.0 | $-5.55 \cdot 10^{-17} + 0.866_z$      | $8.79 \cdot 10^{-16} - 0.897_z$                | $-2.05 \cdot 10^{-11} - 1.3_z$                 | $-9.53 \cdot 10^{-12} + 1.3_z$                  |
| 6.0 | $-5.55 \cdot 10^{-17} - 0.866_z$      | $1.64 \cdot 10^{-15} + 0.897_z$                | $4.14 \cdot 10^{-11} + 1.3_z$                  | $1.43 \cdot 10^{-11} - 1.3_z$                   |
| 7.0 | $-6.94 \cdot 10^{-18} + 0.359_z$      | $6.16 \cdot 10^{-15} - 2.17_z$                 | $-2.26 \cdot 10^{-11} - 4.35_z$                | $-1.71 \cdot 10^{-11} + 4.35_z$                 |
| 8.0 | $-6.94 \cdot 10^{-18} - 0.359_z$      | $4.41 \cdot 10^{-15} + 2.17_z$                 | $1.86 \cdot 10^{-10} + 4.35_z$                 | $6.27 \cdot 10^{-11} - 4.35_z$                  |

Table 69: Case 1a (cpv MACA64 MR2023b),  $r = 8$ , Z4: zeros.

|     | Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$ | LF, $\sigma_8 = 1.6 \cdot 10^{-09}$           | AAA-sign-L200, $\sigma_8 = 7.4 \cdot 10^{-10}$ | AAA-sign-L1000, $\sigma_8 = 7.4 \cdot 10^{-10}$ |
|-----|---------------------------------------|-----------------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | NaN                                   | NaN                                           | $-8.78 \cdot 10^{-4} + 5.09 \cdot 10^{-8}i$    | $-8.78 \cdot 10^{-4} - 2.54 \cdot 10^{-6}i$     |
| 2.0 | 0                                     | $-1.63 \cdot 10^{-15} + 1.75 \cdot 10^{-15}i$ | $-9.64 \cdot 10^{-4} + 0.359i$                 | $-9.64 \cdot 10^{-4} + 0.359i$                  |
| 3.0 | $-3.33 \cdot 10^{-16} + 2.09i$        | $-2.19 \cdot 10^{-15} - 0.371i$               | $-9.71 \cdot 10^{-4} - 0.359i$                 | $-9.71 \cdot 10^{-4} - 0.359i$                  |
| 4.0 | $-3.33 \cdot 10^{-16} - 2.09i$        | $-1.01 \cdot 10^{-15} + 0.371i$               | $-0.00138 + 0.866i$                            | $-0.00138 + 0.866i$                             |
| 5.0 | $-5.55 \cdot 10^{-17} + 0.866i$       | $1.73 \cdot 10^{-15} - 0.897i$                | $-0.0014 - 0.866i$                             | $-0.0014 - 0.866i$                              |
| 6.0 | $-5.55 \cdot 10^{-17} - 0.866i$       | $-5.71 \cdot 10^{-15} + 0.897i$               | $-0.0038 + 2.09i$                              | $-0.0038 + 2.09i$                               |
| 7.0 | $-6.25 \cdot 10^{-17} + 0.359i$       | $2.2 \cdot 10^{-15} - 2.17i$                  | $-0.00385 - 2.09i$                             | $-0.00385 - 2.09i$                              |
| 8.0 | $-6.25 \cdot 10^{-17} - 0.359i$       | $-2.01 \cdot 10^{-15} + 2.17i$                | $-1500.0 + 32.5i$                              | $-1500.0 + 40.5i$                               |

Table 70: Case 1a (cpv MACI64 MR2022b),  $r = 8$ , Z4: zeros.

|     | Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$ | LF, $\sigma_8 = 1.6 \cdot 10^{-09}$          | AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$ | AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$ |
|-----|---------------------------------------|----------------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | NaN                                   | NaN                                          | $5.66 \cdot 10^{-12} + 1.34 \cdot 10^{-9}i$    | $-4.28 \cdot 10^{-13} - 1.47 \cdot 10^{-10}i$   |
| 2.0 | 0                                     | $5.49 \cdot 10^{-16} - 1.11 \cdot 10^{-15}i$ | $-4.07 \cdot 10^{-12} - 0.359i$                | $2.24 \cdot 10^{-12} + 0.359i$                  |
| 3.0 | $1.39 \cdot 10^{-16} + 2.09i$         | $-4.33 \cdot 10^{-17} + 0.371i$              | $1.74 \cdot 10^{-11} + 0.359i$                 | $-3.19 \cdot 10^{-12} - 0.359i$                 |
| 4.0 | $1.39 \cdot 10^{-16} - 2.09i$         | $2.25 \cdot 10^{-15} - 0.371i$               | $-1.44 \cdot 10^{-11} - 0.866i$                | $5.83 \cdot 10^{-12} + 0.866i$                  |
| 5.0 | $-5.55 \cdot 10^{-17} + 0.866i$       | $-2.49 \cdot 10^{-15} + 0.897i$              | $3.78 \cdot 10^{-11} + 0.866i$                 | $-7.39 \cdot 10^{-12} - 0.866i$                 |
| 6.0 | $-5.55 \cdot 10^{-17} - 0.866i$       | $3.87 \cdot 10^{-15} - 0.897i$               | $-2.21 \cdot 10^{-11} - 2.09i$                 | $1.38 \cdot 10^{-11} + 2.09i$                   |
| 7.0 | $-6.94 \cdot 10^{-18} + 0.359i$       | $8.84 \cdot 10^{-16} + 2.17i$                | $1.05 \cdot 10^{-10} + 2.09i$                  | $-1.81 \cdot 10^{-11} - 2.09i$                  |
| 8.0 | $-6.94 \cdot 10^{-18} - 0.359i$       | $2.13 \cdot 10^{-15} - 2.17i$                | $2.7 \cdot 10^6 - 5.7 \cdot 10^8i$             | $-8.64 \cdot 10^6 + 4.91 \cdot 10^9i$           |

Table 71: Case 1a (cpv PCWIN64 MR2025b),  $r = 8$ , Z4: zeros.

|     | Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$ | LF, $\sigma_8 = 1.6 \cdot 10^{-09}$          | AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$ | AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$ |
|-----|---------------------------------------|----------------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | NaN                                   | NaN                                          | $5.66 \cdot 10^{-12} + 1.34 \cdot 10^{-9}i$    | $-4.28 \cdot 10^{-13} - 1.47 \cdot 10^{-10}i$   |
| 2.0 | 0                                     | $5.49 \cdot 10^{-16} - 1.11 \cdot 10^{-15}i$ | $-4.07 \cdot 10^{-12} - 0.359i$                | $2.24 \cdot 10^{-12} + 0.359i$                  |
| 3.0 | $1.39 \cdot 10^{-16} + 2.09i$         | $-4.33 \cdot 10^{-17} + 0.371i$              | $1.74 \cdot 10^{-11} + 0.359i$                 | $-3.19 \cdot 10^{-12} - 0.359i$                 |
| 4.0 | $1.39 \cdot 10^{-16} - 2.09i$         | $2.25 \cdot 10^{-15} - 0.371i$               | $-1.44 \cdot 10^{-11} - 0.866i$                | $5.83 \cdot 10^{-12} + 0.866i$                  |
| 5.0 | $-5.55 \cdot 10^{-17} + 0.866i$       | $-2.49 \cdot 10^{-15} + 0.897i$              | $3.78 \cdot 10^{-11} + 0.866i$                 | $-7.39 \cdot 10^{-12} - 0.866i$                 |
| 6.0 | $-5.55 \cdot 10^{-17} - 0.866i$       | $3.87 \cdot 10^{-15} - 0.897i$               | $-2.21 \cdot 10^{-11} - 2.09i$                 | $1.38 \cdot 10^{-11} + 2.09i$                   |
| 7.0 | $-6.94 \cdot 10^{-18} + 0.359i$       | $8.84 \cdot 10^{-16} + 2.17i$                | $1.05 \cdot 10^{-10} + 2.09i$                  | $-1.81 \cdot 10^{-11} - 2.09i$                  |
| 8.0 | $-6.94 \cdot 10^{-18} - 0.359i$       | $2.13 \cdot 10^{-15} - 2.17i$                | $2.7 \cdot 10^6 - 5.7 \cdot 10^8i$             | $-8.64 \cdot 10^6 + 4.91 \cdot 10^9i$           |

Table 72: Case 1a (AlexReis PCWIN64 MR2025b),  $r = 8$ , Z4: zeros.

## 7.4 Poles and zeros visualisation

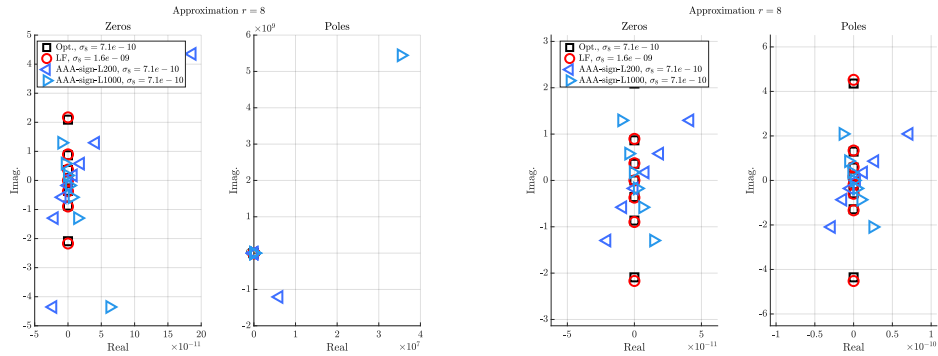


Figure 11: Case 1a (cpv MACA64 MR2023b),  $r = 8$ , Z4. Left frame: zeros (left) and poles (right). Right frame: same but discarding the largest real entry.

## 7.5 Approximation visualization

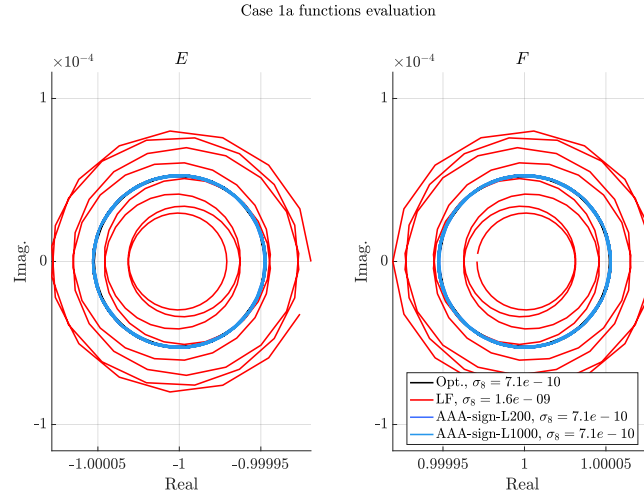


Figure 12: Case 1a (cpv MACA64 MR2023b),  $r = 8$ , Z4.  $E$  and  $F$  evaluations.

## 8 Approximation objective $r = 9$

### 8.1 Numerator and denominator coefficients

The numerator and denominator coefficients (both real and complex parts) are given in the following tables. Each table gives the results obtained on different platform.

| Basis | Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$ |       | LF, $\sigma_9 = 2.7 \cdot 10^{-10}$ |                       | AAA-sign-L200, $\sigma_9 = 5.1 \cdot 10^{-11}$ |                      | AAA-sign-L1000, $\sigma_9 = 5.1 \cdot 10^{-11}$ |                     |
|-------|---------------------------------------|-------|-------------------------------------|-----------------------|------------------------------------------------|----------------------|-------------------------------------------------|---------------------|
|       | real                                  | imag. | real                                | imag.                 | real                                           | imag.                | real                                            | imag.               |
| $z^9$ | 0                                     | 0     | 0                                   | 0                     | $-2.1 \cdot 10^3$                              | $1.4 \cdot 10^6$     | -940.0                                          | $-5.1 \cdot 10^5$   |
| $z^8$ | 7.8                                   | 0     | 8.9                                 | $5.1 \cdot 10^{-14}$  | 7.8                                            | $-8.0 \cdot 10^{-4}$ | 7.8                                             | $3.7 \cdot 10^{-4}$ |
| $z^7$ | 0                                     | 0     | $9.2 \cdot 10^{-13}$                | $1.1 \cdot 10^{-12}$  | $-5.7 \cdot 10^4$                              | $3.7 \cdot 10^7$     | $-2.5 \cdot 10^4$                               | $-1.4 \cdot 10^7$   |
| $z^6$ | 55.0                                  | 0     | 78.0                                | $1.6 \cdot 10^{-13}$  | 54.0                                           | -0.016               | 55.0                                            | $7.1 \cdot 10^{-3}$ |
| $z^5$ | 0                                     | 0     | $4.0 \cdot 10^{-12}$                | $6.7 \cdot 10^{-12}$  | $-1.5 \cdot 10^5$                              | $9.8 \cdot 10^7$     | $-6.6 \cdot 10^4$                               | $-3.6 \cdot 10^7$   |
| $z^4$ | 61.0                                  | 0     | 110.0                               | $3.2 \cdot 10^{-13}$  | 60.0                                           | -0.027               | 62.0                                            | 0.012               |
| $z^3$ | 0                                     | 0     | $2.2 \cdot 10^{-12}$                | $4.8 \cdot 10^{-12}$  | $-7.4 \cdot 10^4$                              | $4.9 \cdot 10^7$     | $-3.3 \cdot 10^4$                               | $-1.8 \cdot 10^7$   |
| $z^2$ | 13.0                                  | 0     | 27.0                                | $3.1 \cdot 10^{-13}$  | 13.0                                           | $-7.6 \cdot 10^{-3}$ | 13.0                                            | $3.1 \cdot 10^{-3}$ |
| $z$   | 0                                     | 0     | $6.9 \cdot 10^{-14}$                | $5.3 \cdot 10^{-13}$  | $-6.0 \cdot 10^3$                              | $3.9 \cdot 10^6$     | $-2.7 \cdot 10^3$                               | $-1.5 \cdot 10^6$   |
| 1.0   | 0.27                                  | 0     | 0.64                                | $-4.9 \cdot 10^{-17}$ | 0.27                                           | $-1.9 \cdot 10^{-4}$ | 0.28                                            | $7.3 \cdot 10^{-5}$ |
| $z^9$ | 1.0                                   | 0     | 1.0                                 | 0                     | 1.0                                            | 0                    | 1.0                                             | 0                   |
| $z^8$ | 0                                     | 0     | $1.5 \cdot 10^{-13}$                | $1.3 \cdot 10^{-13}$  | $-1.6 \cdot 10^4$                              | $1.1 \cdot 10^7$     | $-7.3 \cdot 10^3$                               | $-4.0 \cdot 10^6$   |
| $z^7$ | 27.0                                  | 0     | 35.0                                | $1.7 \cdot 10^{-13}$  | 27.0                                           | $-5.3 \cdot 10^{-3}$ | 27.0                                            | $2.4 \cdot 10^{-3}$ |
| $z^6$ | 0                                     | 0     | $2.5 \cdot 10^{-12}$                | $3.7 \cdot 10^{-12}$  | $-1.1 \cdot 10^5$                              | $7.5 \cdot 10^7$     | $-5.1 \cdot 10^4$                               | $-2.8 \cdot 10^7$   |
| $z^5$ | 71.0                                  | 0     | 110.0                               | $1.0 \cdot 10^{-13}$  | 70.0                                           | -0.026               | 72.0                                            | 0.012               |
| $z^4$ | 0                                     | 0     | $3.8 \cdot 10^{-12}$                | $7.1 \cdot 10^{-12}$  | $-1.3 \cdot 10^5$                              | $8.5 \cdot 10^7$     | $-5.8 \cdot 10^4$                               | $-3.2 \cdot 10^7$   |
| $z^3$ | 35.0                                  | 0     | 67.0                                | $5.1 \cdot 10^{-13}$  | 35.0                                           | -0.018               | 36.0                                            | $7.7 \cdot 10^{-3}$ |
| $z^2$ | 0                                     | 0     | $6.6 \cdot 10^{-13}$                | $2.1 \cdot 10^{-12}$  | $-2.8 \cdot 10^4$                              | $1.8 \cdot 10^7$     | $-1.2 \cdot 10^4$                               | $-6.8 \cdot 10^6$   |
| $z$   | 2.8                                   | 0     | 6.2                                 | $6.5 \cdot 10^{-14}$  | 2.8                                            | $-1.8 \cdot 10^{-3}$ | 2.9                                             | $7.3 \cdot 10^{-4}$ |
| 1.0   | 0                                     | 0     | $-6.9 \cdot 10^{-15}$               | $6.8 \cdot 10^{-14}$  | -570.0                                         | $3.8 \cdot 10^5$     | -260.0                                          | $-1.4 \cdot 10^5$   |

Table 73: Case 1a (cpv MACA64 MR2023b),  $r = 9$ , Z4: numerator (first lines) and denominator (last lines) coefficients.

| Basis | Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$ |       | LF, $\sigma_9 = 2.7 \cdot 10^{-10}$ |                       | AAA-sign-L200, $\sigma_9 = 5.2 \cdot 10^{-11}$ |                  | AAA-sign-L1000, $\sigma_9 = 5.2 \cdot 10^{-11}$ |                  |
|-------|---------------------------------------|-------|-------------------------------------|-----------------------|------------------------------------------------|------------------|-------------------------------------------------|------------------|
|       | real                                  | imag. | real                                | imag.                 | real                                           | imag.            | real                                            | imag.            |
| $z^9$ | 0                                     | 0     | 0                                   | 0                     | 190.0                                          | 32.0             | 190.0                                           | 32.0             |
| $z^8$ | 7.8                                   | 0     | 8.9                                 | $-1.1 \cdot 10^{-13}$ | 8.3                                            | 0.29             | 8.3                                             | 0.29             |
| $z^7$ | 0                                     | 0     | $-1.7 \cdot 10^{-13}$               | $3.8 \cdot 10^{-13}$  | $5.2 \cdot 10^3$                               | 850.0            | $5.2 \cdot 10^3$                                | 850.0            |
| $z^6$ | 55.0                                  | 0     | 78.0                                | $-3.4 \cdot 10^{-12}$ | 65.0                                           | 6.0              | 65.0                                            | 6.0              |
| $z^5$ | 0                                     | 0     | $-3.2 \cdot 10^{-13}$               | $3.3 \cdot 10^{-12}$  | $1.4 \cdot 10^4$                               | $2.2 \cdot 10^3$ | $1.4 \cdot 10^4$                                | $2.2 \cdot 10^3$ |
| $z^4$ | 61.0                                  | 0     | 110.0                               | $-6.6 \cdot 10^{-12}$ | 81.0                                           | 11.0             | 81.0                                            | 11.0             |
| $z^3$ | 0                                     | 0     | $-7.6 \cdot 10^{-14}$               | $2.8 \cdot 10^{-12}$  | $6.8 \cdot 10^3$                               | $1.1 \cdot 10^3$ | $6.8 \cdot 10^3$                                | $1.1 \cdot 10^3$ |
| $z^2$ | 13.0                                  | 0     | 27.0                                | $-2.2 \cdot 10^{-12}$ | 19.0                                           | 3.2              | 19.0                                            | 3.2              |
| $z$   | 0                                     | 0     | $3.2 \cdot 10^{-14}$                | $3.4 \cdot 10^{-13}$  | 550.0                                          | 89.0             | 550.0                                           | 89.0             |
| 1.0   | 0.27                                  | 0     | 0.64                                | $-4.4 \cdot 10^{-14}$ | 0.43                                           | 0.085            | 0.43                                            | 0.085            |
| $z^9$ | 1.0                                   | 0     | 1.0                                 | 0                     | 1.0                                            | 0                | 1.0                                             | 0                |
| $z^8$ | 0                                     | 0     | $-4.9 \cdot 10^{-14}$               | $2.4 \cdot 10^{-14}$  | $1.5 \cdot 10^3$                               | 250.0            | $1.5 \cdot 10^3$                                | 250.0            |
| $z^7$ | 27.0                                  | 0     | 35.0                                | $-1.0 \cdot 10^{-12}$ | 30.0                                           | 2.0              | 30.0                                            | 2.0              |
| $z^6$ | 0                                     | 0     | $-2.7 \cdot 10^{-13}$               | $1.6 \cdot 10^{-12}$  | $1.0 \cdot 10^4$                               | $1.7 \cdot 10^3$ | $1.0 \cdot 10^4$                                | $1.7 \cdot 10^3$ |
| $z^5$ | 71.0                                  | 0     | 110.0                               | $-6.0 \cdot 10^{-12}$ | 89.0                                           | 10.0             | 89.0                                            | 10.0             |
| $z^4$ | 0                                     | 0     | $-2.2 \cdot 10^{-13}$               | $3.9 \cdot 10^{-12}$  | $1.2 \cdot 10^4$                               | $1.9 \cdot 10^3$ | $1.2 \cdot 10^4$                                | $1.9 \cdot 10^3$ |
| $z^3$ | 35.0                                  | 0     | 67.0                                | $-4.8 \cdot 10^{-12}$ | 49.0                                           | 7.6              | 49.0                                            | 7.6              |
| $z^2$ | 0                                     | 0     | $2.4 \cdot 10^{-14}$                | $1.3 \cdot 10^{-12}$  | $2.5 \cdot 10^3$                               | 410.0            | $2.5 \cdot 10^3$                                | 410.0            |
| $z$   | 2.8                                   | 0     | 6.2                                 | $-5.1 \cdot 10^{-13}$ | 4.3                                            | 0.79             | 4.3                                             | 0.79             |
| 1.0   | 0                                     | 0     | $6.7 \cdot 10^{-15}$                | $3.6 \cdot 10^{-14}$  | 52.0                                           | 8.6              | 52.0                                            | 8.6              |

Table 74: Case 1a (cpv MACI64 MR2022b),  $r = 9$ , Z4: numerator (first lines) and denominator (last lines) coefficients.

| Basis | Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$ |       | LF, $\sigma_9 = 2.7 \cdot 10^{-10}$ |                       | AAA-sign-L200, $\sigma_9 = 5.1 \cdot 10^{-11}$ |                      | AAA-sign-L1000, $\sigma_9 = 5.1 \cdot 10^{-11}$ |                      |
|-------|---------------------------------------|-------|-------------------------------------|-----------------------|------------------------------------------------|----------------------|-------------------------------------------------|----------------------|
|       | real                                  | imag. | real                                | imag.                 | real                                           | imag.                | real                                            | imag.                |
| $z^9$ | 0                                     | 0     | 0                                   | 0                     | $2.5 \cdot 10^{-9}$                            | $-2.9 \cdot 10^{-6}$ | $-3.5 \cdot 10^{-10}$                           | $-7.9 \cdot 10^{-7}$ |
| $z^8$ | 7.8                                   | 0     | 8.9                                 | $-2.6 \cdot 10^{-13}$ | 7.8                                            | $3.9 \cdot 10^{-8}$  | 7.8                                             | $1.9 \cdot 10^{-9}$  |
| $z^7$ | 0                                     | 0     | $3.5 \cdot 10^{-13}$                | $-1.3 \cdot 10^{-13}$ | $7.7 \cdot 10^{-8}$                            | $-7.9 \cdot 10^{-5}$ | $-8.9 \cdot 10^{-9}$                            | $-2.1 \cdot 10^{-5}$ |
| $z^6$ | 55.0                                  | 0     | 78.0                                | $-4.4 \cdot 10^{-12}$ | 55.0                                           | $8.3 \cdot 10^{-7}$  | 55.0                                            | $4.1 \cdot 10^{-8}$  |
| $z^5$ | 0                                     | 0     | $2.0 \cdot 10^{-12}$                | $3.8 \cdot 10^{-13}$  | $2.2 \cdot 10^{-7}$                            | $-2.1 \cdot 10^{-4}$ | $-2.2 \cdot 10^{-8}$                            | $-5.6 \cdot 10^{-5}$ |
| $z^4$ | 61.0                                  | 0     | 110.0                               | $-7.9 \cdot 10^{-12}$ | 61.0                                           | $1.6 \cdot 10^{-6}$  | 61.0                                            | $7.7 \cdot 10^{-8}$  |
| $z^3$ | 0                                     | 0     | $1.7 \cdot 10^{-12}$                | $6.9 \cdot 10^{-13}$  | $1.2 \cdot 10^{-7}$                            | $-1.0 \cdot 10^{-4}$ | $-1.1 \cdot 10^{-8}$                            | $-2.8 \cdot 10^{-5}$ |
| $z^2$ | 13.0                                  | 0     | 27.0                                | $-2.4 \cdot 10^{-12}$ | 13.0                                           | $4.7 \cdot 10^{-7}$  | 13.0                                            | $2.3 \cdot 10^{-8}$  |
| $z$   | 0                                     | 0     | $2.1 \cdot 10^{-13}$                | $1.2 \cdot 10^{-13}$  | $9.6 \cdot 10^{-9}$                            | $-8.2 \cdot 10^{-6}$ | $-8.4 \cdot 10^{-10}$                           | $-2.2 \cdot 10^{-6}$ |
| 1.0   | 0.27                                  | 0     | 0.64                                | $-6.1 \cdot 10^{-14}$ | 0.27                                           | $1.3 \cdot 10^{-8}$  | 0.27                                            | $6.3 \cdot 10^{-10}$ |
| $z^9$ | 1.0                                   | 0     | 1.0                                 | 0                     | 1.0                                            | 0                    | 1.0                                             | 0                    |
| $z^8$ | 0                                     | 0     | $5.0 \cdot 10^{-14}$                | $-5.1 \cdot 10^{-14}$ | $2.1 \cdot 10^{-8}$                            | $-2.3 \cdot 10^{-5}$ | $-2.6 \cdot 10^{-9}$                            | $-6.1 \cdot 10^{-6}$ |
| $z^7$ | 27.0                                  | 0     | 35.0                                | $-1.6 \cdot 10^{-12}$ | 27.0                                           | $2.7 \cdot 10^{-7}$  | 27.0                                            | $1.3 \cdot 10^{-8}$  |
| $z^6$ | 0                                     | 0     | $1.1 \cdot 10^{-12}$                | $-4.6 \cdot 10^{-14}$ | $1.6 \cdot 10^{-7}$                            | $-1.6 \cdot 10^{-4}$ | $-1.8 \cdot 10^{-8}$                            | $-4.3 \cdot 10^{-5}$ |
| $z^5$ | 71.0                                  | 0     | 110.0                               | $-7.4 \cdot 10^{-12}$ | 71.0                                           | $1.4 \cdot 10^{-6}$  | 71.0                                            | $7.1 \cdot 10^{-8}$  |
| $z^4$ | 0                                     | 0     | $2.3 \cdot 10^{-12}$                | $7.6 \cdot 10^{-13}$  | $2.0 \cdot 10^{-7}$                            | $-1.8 \cdot 10^{-4}$ | $-1.9 \cdot 10^{-8}$                            | $-4.8 \cdot 10^{-5}$ |
| $z^3$ | 35.0                                  | 0     | 67.0                                | $-5.6 \cdot 10^{-12}$ | 35.0                                           | $1.1 \cdot 10^{-6}$  | 35.0                                            | $5.4 \cdot 10^{-8}$  |
| $z^2$ | 0                                     | 0     | $8.0 \cdot 10^{-13}$                | $3.8 \cdot 10^{-13}$  | $4.4 \cdot 10^{-8}$                            | $-3.8 \cdot 10^{-5}$ | $-3.9 \cdot 10^{-9}$                            | $-1.0 \cdot 10^{-5}$ |
| $z$   | 2.8                                   | 0     | 6.2                                 | $-5.9 \cdot 10^{-13}$ | 2.8                                            | $1.2 \cdot 10^{-7}$  | 2.8                                             | $5.8 \cdot 10^{-9}$  |
| 1.0   | 0                                     | 0     | $2.3 \cdot 10^{-14}$                | $1.9 \cdot 10^{-14}$  | $9.3 \cdot 10^{-10}$                           | $-7.9 \cdot 10^{-7}$ | $-8.0 \cdot 10^{-11}$                           | $-2.1 \cdot 10^{-7}$ |

Table 75: Case 1a (cpv PCWIN64 MR2025b),  $r = 9$ , Z4: numerator (first lines) and denominator (last lines) coefficients.

| Basis | Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$ |       | LF, $\sigma_9 = 2.7 \cdot 10^{-10}$ |                       | AAA-sign-L200, $\sigma_9 = 5.1 \cdot 10^{-11}$ |                      | AAA-sign-L1000, $\sigma_9 = 5.1 \cdot 10^{-11}$ |                      |
|-------|---------------------------------------|-------|-------------------------------------|-----------------------|------------------------------------------------|----------------------|-------------------------------------------------|----------------------|
|       | real                                  | imag. | real                                | imag.                 | real                                           | imag.                | real                                            | imag.                |
| $z^9$ | 0                                     | 0     | 0                                   | 0                     | $2.5 \cdot 10^{-9}$                            | $-2.9 \cdot 10^{-6}$ | $-3.5 \cdot 10^{-10}$                           | $-7.9 \cdot 10^{-7}$ |
| $z^8$ | 7.8                                   | 0     | 8.9                                 | $-2.6 \cdot 10^{-13}$ | 7.8                                            | $3.9 \cdot 10^{-8}$  | 7.8                                             | $1.9 \cdot 10^{-9}$  |
| $z^7$ | 0                                     | 0     | $3.5 \cdot 10^{-13}$                | $-1.3 \cdot 10^{-13}$ | $7.7 \cdot 10^{-8}$                            | $-7.9 \cdot 10^{-5}$ | $-8.9 \cdot 10^{-9}$                            | $-2.1 \cdot 10^{-5}$ |
| $z^6$ | 55.0                                  | 0     | 78.0                                | $-4.4 \cdot 10^{-12}$ | 55.0                                           | $8.3 \cdot 10^{-7}$  | 55.0                                            | $4.1 \cdot 10^{-8}$  |
| $z^5$ | 0                                     | 0     | $2.0 \cdot 10^{-12}$                | $3.8 \cdot 10^{-13}$  | $2.2 \cdot 10^{-7}$                            | $-2.1 \cdot 10^{-4}$ | $-2.2 \cdot 10^{-8}$                            | $-5.6 \cdot 10^{-5}$ |
| $z^4$ | 61.0                                  | 0     | 110.0                               | $-7.9 \cdot 10^{-12}$ | 61.0                                           | $1.6 \cdot 10^{-6}$  | 61.0                                            | $7.7 \cdot 10^{-8}$  |
| $z^3$ | 0                                     | 0     | $1.7 \cdot 10^{-12}$                | $6.9 \cdot 10^{-13}$  | $1.2 \cdot 10^{-7}$                            | $-1.0 \cdot 10^{-4}$ | $-1.1 \cdot 10^{-8}$                            | $-2.8 \cdot 10^{-5}$ |
| $z^2$ | 13.0                                  | 0     | 27.0                                | $-2.4 \cdot 10^{-12}$ | 13.0                                           | $4.7 \cdot 10^{-7}$  | 13.0                                            | $2.3 \cdot 10^{-8}$  |
| $z$   | 0                                     | 0     | $2.1 \cdot 10^{-13}$                | $1.2 \cdot 10^{-13}$  | $9.6 \cdot 10^{-9}$                            | $-8.2 \cdot 10^{-6}$ | $-8.4 \cdot 10^{-10}$                           | $-2.2 \cdot 10^{-6}$ |
| 1.0   | 0.27                                  | 0     | 0.64                                | $-6.1 \cdot 10^{-14}$ | 0.27                                           | $1.3 \cdot 10^{-8}$  | 0.27                                            | $6.3 \cdot 10^{-10}$ |
| $z^9$ | 1.0                                   | 0     | 1.0                                 | 0                     | 1.0                                            | 0                    | 1.0                                             | 0                    |
| $z^8$ | 0                                     | 0     | $5.0 \cdot 10^{-14}$                | $-5.1 \cdot 10^{-14}$ | $2.1 \cdot 10^{-8}$                            | $-2.3 \cdot 10^{-5}$ | $-2.6 \cdot 10^{-9}$                            | $-6.1 \cdot 10^{-6}$ |
| $z^7$ | 27.0                                  | 0     | 35.0                                | $-1.6 \cdot 10^{-12}$ | 27.0                                           | $2.7 \cdot 10^{-7}$  | 27.0                                            | $1.3 \cdot 10^{-8}$  |
| $z^6$ | 0                                     | 0     | $1.1 \cdot 10^{-12}$                | $-4.6 \cdot 10^{-14}$ | $1.6 \cdot 10^{-7}$                            | $-1.6 \cdot 10^{-4}$ | $-1.8 \cdot 10^{-8}$                            | $-4.3 \cdot 10^{-5}$ |
| $z^5$ | 71.0                                  | 0     | 110.0                               | $-7.4 \cdot 10^{-12}$ | 71.0                                           | $1.4 \cdot 10^{-6}$  | 71.0                                            | $7.1 \cdot 10^{-8}$  |
| $z^4$ | 0                                     | 0     | $2.3 \cdot 10^{-12}$                | $7.6 \cdot 10^{-13}$  | $2.0 \cdot 10^{-7}$                            | $-1.8 \cdot 10^{-4}$ | $-1.9 \cdot 10^{-8}$                            | $-4.8 \cdot 10^{-5}$ |
| $z^3$ | 35.0                                  | 0     | 67.0                                | $-5.6 \cdot 10^{-12}$ | 35.0                                           | $1.1 \cdot 10^{-6}$  | 35.0                                            | $5.4 \cdot 10^{-8}$  |
| $z^2$ | 0                                     | 0     | $8.0 \cdot 10^{-13}$                | $3.8 \cdot 10^{-13}$  | $4.4 \cdot 10^{-8}$                            | $-3.8 \cdot 10^{-5}$ | $-3.9 \cdot 10^{-9}$                            | $-1.0 \cdot 10^{-5}$ |
| $z$   | 2.8                                   | 0     | 6.2                                 | $-5.9 \cdot 10^{-13}$ | 2.8                                            | $1.2 \cdot 10^{-7}$  | 2.8                                             | $5.8 \cdot 10^{-9}$  |
| 1.0   | 0                                     | 0     | $2.3 \cdot 10^{-14}$                | $1.9 \cdot 10^{-14}$  | $9.3 \cdot 10^{-10}$                           | $-7.9 \cdot 10^{-7}$ | $-8.0 \cdot 10^{-11}$                           | $-2.1 \cdot 10^{-7}$ |

Table 76: Case 1a (AlexReis PCWIN64 MR2025b),  $r = 9$ , Z4: numerator (first lines) and denominator (last lines) coefficients.

## 8.2 Poles

The poles are given in the following tables. Each table gives the results obtained on different platform.

|     | Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$ | LF, $\sigma_9 = 2.7 \cdot 10^{-10}$          | AAA-sign-L200, $\sigma_9 = 5.1 \cdot 10^{-11}$ | AAA-sign-L1000, $\sigma_9 = 5.1 \cdot 10^{-11}$ |
|-----|---------------------------------------|----------------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | 0                                     | $1.11 \cdot 10^{-15} - 1.09 \cdot 10^{-14}i$ | $5.97 \cdot 10^{-10} - 0.153i$                 | $-1.86 \cdot 10^{-10} + 0.153i$                 |
| 2.0 | $-1.11 \cdot 10^{-15} + 4.91i$        | $-4.13 \cdot 10^{-15} + 0.338i$              | $-2.87 \cdot 10^{-10} + 0.153i$                | $1.01 \cdot 10^{-9} - 0.153i$                   |
| 3.0 | $-1.11 \cdot 10^{-15} - 4.91i$        | $-3.5 \cdot 10^{-15} - 0.338i$               | $1.65 \cdot 10^{-9} - 0.5i$                    | $-1.45 \cdot 10^{-9} + 0.5i$                    |
| 4.0 | $-5.55 \cdot 10^{-17} + 1.5i$         | $-1.64 \cdot 10^{-14} + 0.789i$              | $-1.26 \cdot 10^{-9} + 0.5i$                   | $2.51 \cdot 10^{-9} - 0.5i$                     |
| 5.0 | $-5.55 \cdot 10^{-17} - 1.5i$         | $-6.96 \cdot 10^{-15} - 0.789i$              | $3.35 \cdot 10^{-9} - 1.03i$                   | $-3.19 \cdot 10^{-9} + 1.03i$                   |
| 6.0 | $-1.39 \cdot 10^{-17} + 0.727i$       | $-1.08 \cdot 10^{-14} + 1.67i$               | $-2.7 \cdot 10^{-9} + 1.03i$                   | $5.04 \cdot 10^{-9} - 1.03i$                    |
| 7.0 | $-1.39 \cdot 10^{-17} - 0.727i$       | $-2.77 \cdot 10^{-14} - 1.67i$               | $7.99 \cdot 10^{-9} - 2.38i$                   | $-6.5 \cdot 10^{-9} + 2.38i$                    |
| 8.0 | $1.73 \cdot 10^{-18} + 0.315i$        | $-5.92 \cdot 10^{-14} + 5.6i$                | $-6.02 \cdot 10^{-9} + 2.38i$                  | $1.26 \cdot 10^{-8} - 2.38i$                    |
| 9.0 | $1.73 \cdot 10^{-18} - 0.315i$        | $-2.58 \cdot 10^{-14} - 5.6i$                | $1.63 \cdot 10^4 - 1.07 \cdot 10^7i$           | $7310.0 + 4.0 \cdot 10^6i$                      |

Table 77: Case 1a (cpv MACA64 MR2023b),  $r = 9$ , Z4: poles.

|     | Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$ | LF, $\sigma_9 = 2.7 \cdot 10^{-10}$           | AAA-sign-L200, $\sigma_9 = 5.2 \cdot 10^{-11}$ | AAA-sign-L1000, $\sigma_9 = 5.2 \cdot 10^{-11}$ |
|-----|---------------------------------------|-----------------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | 0                                     | $-1.08 \cdot 10^{-15} - 5.74 \cdot 10^{-15}i$ | $-7.88 \cdot 10^{-4} + 0.153i$                 | $-7.88 \cdot 10^{-4} + 0.153i$                  |
| 2.0 | $5.55 \cdot 10^{-16} + 4.91i$         | $1.48 \cdot 10^{-15} + 0.338i$                | $-8.16 \cdot 10^{-4} - 0.153i$                 | $-8.16 \cdot 10^{-4} - 0.153i$                  |
| 3.0 | $5.55 \cdot 10^{-16} - 4.91i$         | $-1.07 \cdot 10^{-15} - 0.338i$               | $-9.03 \cdot 10^{-4} + 0.5i$                   | $-9.03 \cdot 10^{-4} + 0.5i$                    |
| 4.0 | $1.11 \cdot 10^{-16} + 1.5i$          | $8.32 \cdot 10^{-15} + 0.789i$                | $-0.001 - 0.5i$                                | $-0.001 - 0.5i$                                 |
| 5.0 | $1.11 \cdot 10^{-16} - 1.5i$          | $-6.52 \cdot 10^{-15} - 0.789i$               | $-0.0016 - 1.03i$                              | $-0.0016 - 1.03i$                               |
| 6.0 | $2.78 \cdot 10^{-17} + 0.727i$        | $2.26 \cdot 10^{-14} + 1.67i$                 | $-0.00138 + 1.03i$                             | $-0.00138 + 1.03i$                              |
| 7.0 | $2.78 \cdot 10^{-17} - 0.727i$        | $-1.99 \cdot 10^{-14} - 1.67i$                | $-0.00476 - 2.38i$                             | $-0.00476 - 2.38i$                              |
| 8.0 | $1.73 \cdot 10^{-18} + 0.315i$        | $-6.04 \cdot 10^{-14} - 5.6i$                 | $-0.0042 + 2.38i$                              | $-0.0042 + 2.38i$                               |
| 9.0 | $1.73 \cdot 10^{-18} - 0.315i$        | $1.06 \cdot 10^{-13} + 5.6i$                  | $-1500.0 - 247.0i$                             | $-1500.0 - 246.0i$                              |

Table 78: Case 1a (cpv MACI64 MR2022b),  $r = 9$ , Z4: poles.

|     | Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$ | LF, $\sigma_9 = 2.7 \cdot 10^{-10}$           | AAA-sign-L200, $\sigma_9 = 5.1 \cdot 10^{-11}$ | AAA-sign-L1000, $\sigma_9 = 5.1 \cdot 10^{-11}$ |
|-----|---------------------------------------|-----------------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | 0                                     | $-3.67 \cdot 10^{-15} - 3.11 \cdot 10^{-15}i$ | $-3.27 \cdot 10^{-10} + 2.77 \cdot 10^{-7}i$   | $2.81 \cdot 10^{-11} + 7.54 \cdot 10^{-8}i$     |
| 2.0 | $-1.11 \cdot 10^{-15} + 4.91i$        | $-2.8 \cdot 10^{-15} + 0.338i$                | $1.28 \cdot 10^{-9} - 0.315i$                  | $1.14 \cdot 10^{-10} - 0.315i$                  |
| 3.0 | $-1.11 \cdot 10^{-15} - 4.91i$        | $-5.57 \cdot 10^{-15} - 0.338i$               | $-2.03 \cdot 10^{-9} + 0.315i$                 | $-4.97 \cdot 10^{-11} + 0.315i$                 |
| 4.0 | $-5.55 \cdot 10^{-17} + 1.5i$         | $3.63 \cdot 10^{-15} + 0.789i$                | $3.22 \cdot 10^{-9} - 0.727i$                  | $2.35 \cdot 10^{-10} - 0.727i$                  |
| 5.0 | $-5.55 \cdot 10^{-17} - 1.5i$         | $-1.15 \cdot 10^{-14} - 0.789i$               | $-4.32 \cdot 10^{-9} + 0.727i$                 | $-1.36 \cdot 10^{-10} + 0.727i$                 |
| 6.0 | $-1.39 \cdot 10^{-17} + 0.727i$       | $-2.13 \cdot 10^{-14} - 1.67i$                | $6.52 \cdot 10^{-9} - 1.5i$                    | $5.02 \cdot 10^{-10} - 1.5i$                    |
| 7.0 | $-1.39 \cdot 10^{-17} - 0.727i$       | $9.08 \cdot 10^{-15} + 1.67i$                 | $-8.83 \cdot 10^{-9} + 1.5i$                   | $-2.53 \cdot 10^{-10} + 1.5i$                   |
| 8.0 | $1.73 \cdot 10^{-18} + 0.315i$        | $-1.42 \cdot 10^{-13} - 5.6i$                 | $1.66 \cdot 10^{-8} - 4.91i$                   | $2.31 \cdot 10^{-9} - 4.91i$                    |
| 9.0 | $1.73 \cdot 10^{-18} - 0.315i$        | $1.23 \cdot 10^{-13} + 5.6i$                  | $-3.32 \cdot 10^{-8} + 4.91i$                  | $-1.3 \cdot 10^{-10} + 4.91i$                   |

Table 79: Case 1a (cpv PCWIN64 MR2025b),  $r = 9$ , Z4: poles.

|     | Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$ | LF, $\sigma_9 = 2.7 \cdot 10^{-10}$           | AAA-sign-L200, $\sigma_9 = 5.1 \cdot 10^{-11}$ | AAA-sign-L1000, $\sigma_9 = 5.1 \cdot 10^{-11}$ |
|-----|---------------------------------------|-----------------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | 0                                     | $-3.67 \cdot 10^{-15} - 3.11 \cdot 10^{-15}i$ | $-3.27 \cdot 10^{-10} + 2.77 \cdot 10^{-7}i$   | $2.81 \cdot 10^{-11} + 7.54 \cdot 10^{-8}i$     |
| 2.0 | $-1.11 \cdot 10^{-15} + 4.91i$        | $-2.8 \cdot 10^{-15} + 0.338i$                | $1.28 \cdot 10^{-9} - 0.315i$                  | $1.14 \cdot 10^{-10} - 0.315i$                  |
| 3.0 | $-1.11 \cdot 10^{-15} - 4.91i$        | $-5.57 \cdot 10^{-15} - 0.338i$               | $-2.03 \cdot 10^{-9} + 0.315i$                 | $-4.97 \cdot 10^{-11} + 0.315i$                 |
| 4.0 | $-5.55 \cdot 10^{-17} + 1.5i$         | $3.63 \cdot 10^{-15} + 0.789i$                | $3.22 \cdot 10^{-9} - 0.727i$                  | $2.35 \cdot 10^{-10} - 0.727i$                  |
| 5.0 | $-5.55 \cdot 10^{-17} - 1.5i$         | $-1.15 \cdot 10^{-14} - 0.789i$               | $-4.32 \cdot 10^{-9} + 0.727i$                 | $-1.36 \cdot 10^{-10} + 0.727i$                 |
| 6.0 | $-1.39 \cdot 10^{-17} + 0.727i$       | $-2.13 \cdot 10^{-14} - 1.67i$                | $6.52 \cdot 10^{-9} - 1.5i$                    | $5.02 \cdot 10^{-10} - 1.5i$                    |
| 7.0 | $-1.39 \cdot 10^{-17} - 0.727i$       | $9.08 \cdot 10^{-15} + 1.67i$                 | $-8.83 \cdot 10^{-9} + 1.5i$                   | $-2.53 \cdot 10^{-10} + 1.5i$                   |
| 8.0 | $1.73 \cdot 10^{-18} + 0.315i$        | $-1.42 \cdot 10^{-13} - 5.6i$                 | $1.66 \cdot 10^{-8} - 4.91i$                   | $2.31 \cdot 10^{-9} - 4.91i$                    |
| 9.0 | $1.73 \cdot 10^{-18} - 0.315i$        | $1.23 \cdot 10^{-13} + 5.6i$                  | $-3.32 \cdot 10^{-8} + 4.91i$                  | $-1.3 \cdot 10^{-10} + 4.91i$                   |

Table 80: Case 1a (AlexReis PCWIN64 MR2025b),  $r = 9$ , Z4: poles.

### 8.3 Zeros

The zeros are given in the following tables. Each table gives the results obtained on different platform.

|     | Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$ | LF, $\sigma_9 = 2.7 \cdot 10^{-10}$ | AAA-sign-L200, $\sigma_9 = 5.1 \cdot 10^{-11}$ | AAA-sign-L1000, $\sigma_9 = 5.1 \cdot 10^{-11}$ |
|-----|---------------------------------------|-------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | NaN                                   | NaN                                 | $1.51 \cdot 10^{-10} + 6.79 \cdot 10^{-8}i$    | $4.01 \cdot 10^{-10} - 1.92 \cdot 10^{-7}i$     |
| 2.0 | $-8.33 \cdot 10^{-17} + 2.38i$        | $8.36 \cdot 10^{-16} + 0.163i$      | $1.08 \cdot 10^{-9} - 0.315i$                  | $-7.87 \cdot 10^{-10} + 0.315i$                 |
| 3.0 | $-8.33 \cdot 10^{-17} - 2.38i$        | $-1.5 \cdot 10^{-15} - 0.163i$      | $-7.46 \cdot 10^{-10} + 0.315i$                | $1.69 \cdot 10^{-9} - 0.315i$                   |
| 4.0 | $2.78 \cdot 10^{-17} + 1.03i$         | $-1.15 \cdot 10^{-14} + 0.539i$     | $2.36 \cdot 10^{-9} - 0.727i$                  | $-2.22 \cdot 10^{-9} + 0.727i$                  |
| 5.0 | $2.78 \cdot 10^{-17} - 1.03i$         | $-3.68 \cdot 10^{-15} - 0.539i$     | $-1.88 \cdot 10^{-9} + 0.727i$                 | $3.55 \cdot 10^{-9} - 0.727i$                   |
| 6.0 | $-6.07 \cdot 10^{-18} + 0.5i$         | $-1.44 \cdot 10^{-14} + 1.13i$      | $4.9 \cdot 10^{-9} - 1.5i$                     | $-4.53 \cdot 10^{-9} + 1.5i$                    |
| 7.0 | $-6.07 \cdot 10^{-18} - 0.5i$         | $-1.54 \cdot 10^{-14} - 1.13i$      | $-3.91 \cdot 10^{-9} + 1.5i$                   | $7.48 \cdot 10^{-9} - 1.5i$                     |
| 8.0 | $1.99 \cdot 10^{-17} + 0.153i$        | $-2.2 \cdot 10^{-14} + 2.68i$       | $1.81 \cdot 10^{-8} - 4.91i$                   | $-8.23 \cdot 10^{-9} + 4.91i$                   |
| 9.0 | $1.99 \cdot 10^{-17} - 0.153i$        | $-3.49 \cdot 10^{-14} - 2.68i$      | $-1.09 \cdot 10^{-8} + 4.91i$                  | $3.12 \cdot 10^{-8} - 4.91i$                    |

Table 81: Case 1a (cpv MACA64 MR2023b),  $r = 9$ , Z4: zeros.

|     | Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$ | LF, $\sigma_9 = 2.7 \cdot 10^{-10}$ | AAA-sign-L200, $\sigma_9 = 5.2 \cdot 10^{-11}$ | AAA-sign-L1000, $\sigma_9 = 5.2 \cdot 10^{-11}$ |
|-----|---------------------------------------|-------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | NaN                                   | NaN                                 | $-7.86 \cdot 10^{-4} - 2.65 \cdot 10^{-5}i$    | $-7.86 \cdot 10^{-4} - 2.68 \cdot 10^{-5}i$     |
| 2.0 | $-2.78 \cdot 10^{-16} + 2.38i$        | $-2.12 \cdot 10^{-15} + 0.163i$     | $-8.23 \cdot 10^{-4} + 0.315i$                 | $-8.23 \cdot 10^{-4} + 0.315i$                  |
| 3.0 | $-2.78 \cdot 10^{-16} - 2.38i$        | $4.75 \cdot 10^{-16} - 0.163i$      | $-8.82 \cdot 10^{-4} - 0.315i$                 | $-8.82 \cdot 10^{-4} - 0.315i$                  |
| 4.0 | $-4.16 \cdot 10^{-17} + 1.03i$        | $6.99 \cdot 10^{-15} + 0.539i$      | $-0.00121 - 0.726i$                            | $-0.00121 - 0.726i$                             |
| 5.0 | $-4.16 \cdot 10^{-17} - 1.03i$        | $-5.34 \cdot 10^{-15} - 0.539i$     | $-0.00106 + 0.726i$                            | $-0.00106 + 0.726i$                             |
| 6.0 | $1.73 \cdot 10^{-18} + 0.5i$          | $8.08 \cdot 10^{-15} + 1.13i$       | $-0.00243 - 1.5i$                              | $-0.00243 - 1.5i$                               |
| 7.0 | $1.73 \cdot 10^{-18} - 0.5i$          | $-8.11 \cdot 10^{-15} - 1.13i$      | $-0.00209 + 1.5i$                              | $-0.00209 + 1.5i$                               |
| 8.0 | $-8.67 \cdot 10^{-19} + 0.153i$       | $-3.84 \cdot 10^{-14} - 2.68i$      | $-0.017 - 4.91i$                               | $-0.017 - 4.91i$                                |
| 9.0 | $-8.67 \cdot 10^{-19} - 0.153i$       | $5.6 \cdot 10^{-14} + 2.68i$        | $-0.0159 + 4.91i$                              | $-0.0159 + 4.91i$                               |

Table 82: Case 1a (cpv MACI64 MR2022b),  $r = 9$ , Z4: zeros.

|     | Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$ | LF, $\sigma_9 = 2.7 \cdot 10^{-10}$ | AAA-sign-L200, $\sigma_9 = 5.1 \cdot 10^{-11}$ | AAA-sign-L1000, $\sigma_9 = 5.1 \cdot 10^{-11}$ |
|-----|---------------------------------------|-------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | NaN                                   | NaN                                 | $4.67 \cdot 10^{-10} - 0.153i$                 | $6.88 \cdot 10^{-11} - 0.153i$                  |
| 2.0 | $1.11 \cdot 10^{-16} + 2.38i$         | $-3.53 \cdot 10^{-15} + 0.163i$     | $-1.14 \cdot 10^{-9} + 0.153i$                 | $-1.08 \cdot 10^{-11} + 0.153i$                 |
| 3.0 | $1.11 \cdot 10^{-16} - 2.38i$         | $-4.0 \cdot 10^{-15} - 0.163i$      | $2.17 \cdot 10^{-9} - 0.5i$                    | $1.67 \cdot 10^{-10} - 0.5i$                    |
| 4.0 | $6.94 \cdot 10^{-17} + 1.03i$         | $2.91 \cdot 10^{-16} + 0.539i$      | $-3.05 \cdot 10^{-9} + 0.5i$                   | $-9.07 \cdot 10^{-11} + 0.5i$                   |
| 5.0 | $6.94 \cdot 10^{-17} - 1.03i$         | $-8.48 \cdot 10^{-15} - 0.539i$     | $4.57 \cdot 10^{-9} - 1.03i$                   | $3.34 \cdot 10^{-10} - 1.03i$                   |
| 6.0 | $2.08 \cdot 10^{-17} + 0.5i$          | $4.72 \cdot 10^{-15} + 1.13i$       | $-6.06 \cdot 10^{-9} + 1.03i$                  | $-1.89 \cdot 10^{-10} + 1.03i$                  |
| 7.0 | $2.08 \cdot 10^{-17} - 0.5i$          | $-1.45 \cdot 10^{-14} - 1.13i$      | $9.83 \cdot 10^{-9} - 2.38i$                   | $8.7 \cdot 10^{-10} - 2.38i$                    |
| 8.0 | $5.2 \cdot 10^{-18} + 0.153i$         | $-4.7 \cdot 10^{-14} - 2.68i$       | $-1.44 \cdot 10^{-8} + 2.38i$                  | $-3.2 \cdot 10^{-10} + 2.38i$                   |
| 9.0 | $5.2 \cdot 10^{-18} - 0.153i$         | $3.31 \cdot 10^{-14} + 2.68i$       | $-2310.0 - 2.66 \cdot 10^6i$                   | $4370.0 - 9.93 \cdot 10^6i$                     |

Table 83: Case 1a (cpv PCWIN64 MR2025b),  $r = 9$ , Z4: zeros.

|     | Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$ | LF, $\sigma_9 = 2.7 \cdot 10^{-10}$ | AAA-sign-L200, $\sigma_9 = 5.1 \cdot 10^{-11}$ | AAA-sign-L1000, $\sigma_9 = 5.1 \cdot 10^{-11}$ |
|-----|---------------------------------------|-------------------------------------|------------------------------------------------|-------------------------------------------------|
| 1.0 | NaN                                   | NaN                                 | $4.67 \cdot 10^{-10} - 0.153i$                 | $6.88 \cdot 10^{-11} - 0.153i$                  |
| 2.0 | $1.11 \cdot 10^{-16} + 2.38i$         | $-3.53 \cdot 10^{-15} + 0.163i$     | $-1.14 \cdot 10^{-9} + 0.153i$                 | $-1.08 \cdot 10^{-11} + 0.153i$                 |
| 3.0 | $1.11 \cdot 10^{-16} - 2.38i$         | $-4.0 \cdot 10^{-15} - 0.163i$      | $2.17 \cdot 10^{-9} - 0.5i$                    | $1.67 \cdot 10^{-10} - 0.5i$                    |
| 4.0 | $6.94 \cdot 10^{-17} + 1.03i$         | $2.91 \cdot 10^{-16} + 0.539i$      | $-3.05 \cdot 10^{-9} + 0.5i$                   | $-9.07 \cdot 10^{-11} + 0.5i$                   |
| 5.0 | $6.94 \cdot 10^{-17} - 1.03i$         | $-8.48 \cdot 10^{-15} - 0.539i$     | $4.57 \cdot 10^{-9} - 1.03i$                   | $3.34 \cdot 10^{-10} - 1.03i$                   |
| 6.0 | $2.08 \cdot 10^{-17} + 0.5i$          | $4.72 \cdot 10^{-15} + 1.13i$       | $-6.06 \cdot 10^{-9} + 1.03i$                  | $-1.89 \cdot 10^{-10} + 1.03i$                  |
| 7.0 | $2.08 \cdot 10^{-17} - 0.5i$          | $-1.45 \cdot 10^{-14} - 1.13i$      | $9.83 \cdot 10^{-9} - 2.38i$                   | $8.7 \cdot 10^{-10} - 2.38i$                    |
| 8.0 | $5.2 \cdot 10^{-18} + 0.153i$         | $-4.7 \cdot 10^{-14} - 2.68i$       | $-1.44 \cdot 10^{-8} + 2.38i$                  | $-3.2 \cdot 10^{-10} + 2.38i$                   |
| 9.0 | $5.2 \cdot 10^{-18} - 0.153i$         | $3.31 \cdot 10^{-14} + 2.68i$       | $-2310.0 - 2.66 \cdot 10^6i$                   | $4370.0 - 9.93 \cdot 10^6i$                     |

Table 84: Case 1a (AlexReis PCWIN64 MR2025b),  $r = 9$ , Z4: zeros.

## 8.4 Poles and zeros visualisation

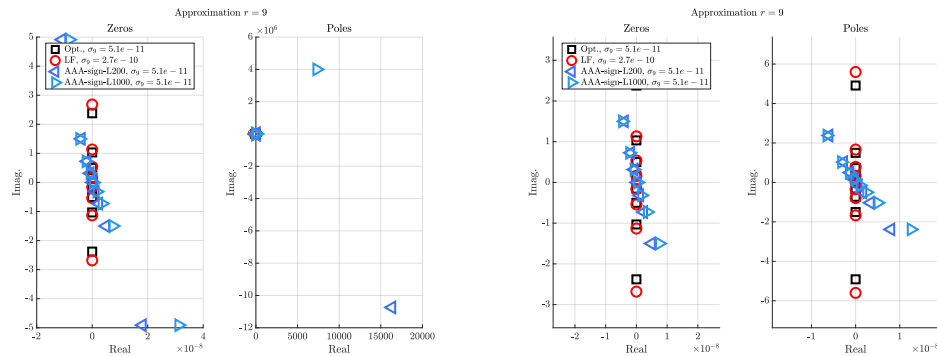


Figure 13: Case 1a (cpv MACA64 MR2023b),  $r = 9$ , Z4. Left frame: zeros (left) and poles (right). Right frame: same but discarding the largest real entry.



## 8.5 Approximation visualization

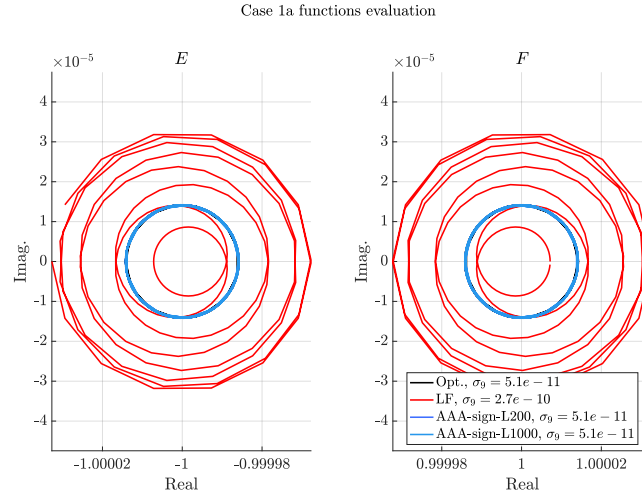


Figure 14: Case 1a (cpv MACA64 MR2023b),  $r = 9$ , Z4.  $E$  and  $F$  evaluations.

## 9 Approximation objective $r = 10$

### 9.1 Numerator and denominator coefficients

The numerator and denominator coefficients (both real and complex parts) are given in the following tables. Each table gives the results obtained on different platform.

|          | Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$ |       | LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$ |       | AAA-sign-L200, $\sigma_{10} = 4.3 \cdot 10^{-12}$ |                       | AAA-sign-L1000, $\sigma_{10} = 4.3 \cdot 10^{-12}$ |                      |
|----------|------------------------------------------|-------|----------------------------------------|-------|---------------------------------------------------|-----------------------|----------------------------------------------------|----------------------|
| Basis    | real                                     | imag. | real                                   | imag. | real                                              | imag.                 | real                                               | imag.                |
| $z^{10}$ | 0                                        | 0     | 0                                      | 0     | $7.7 \cdot 10^{-3}$                               | $3.1 \cdot 10^{-3}$   | $7.7 \cdot 10^{-3}$                                | $3.1 \cdot 10^{-3}$  |
| $z^9$    | 8.7                                      | 0     | NaN                                    | NaN   | 8.7                                               | $4.6 \cdot 10^{-3}$   | 8.7                                                | $4.6 \cdot 10^{-3}$  |
| $z^8$    | 0                                        | 0     | 0                                      | 0     | 0.14                                              | 0.07                  | 0.14                                               | 0.069                |
| $z^7$    | 78.0                                     | 0     | NaN                                    | NaN   | 78.0                                              | 0.13                  | 78.0                                               | 0.13                 |
| $z^6$    | 0                                        | 0     | 0                                      | 0     | 0.036                                             | 0.12                  | 0.035                                              | 0.12                 |
| $z^5$    | 120.0                                    | 0     | NaN                                    | NaN   | 120.0                                             | 0.34                  | 120.0                                              | 0.34                 |
| $z^4$    | 0                                        | 0     | 0                                      | 0     | -0.31                                             | $-3.8 \cdot 10^{-3}$  | -0.31                                              | $-7.4 \cdot 10^{-3}$ |
| $z^3$    | 44.0                                     | 0     | NaN                                    | NaN   | 44.0                                              | 0.17                  | 44.0                                               | 0.17                 |
| $z^2$    | 0                                        | 0     | 0                                      | 0     | -0.1                                              | -0.016                | -0.11                                              | -0.017               |
| $z$      | 2.7                                      | 0     | NaN                                    | NaN   | 2.8                                               | 0.014                 | 2.8                                                | 0.014                |
| 1.0      | 0                                        | 0     | 0                                      | 0     | $-2.7 \cdot 10^{-3}$                              | $-5.5 \cdot 10^{-4}$  | $-2.7 \cdot 10^{-3}$                               | $-5.6 \cdot 10^{-4}$ |
| $z^{10}$ | 1.0                                      | 0     | NaN                                    | NaN   | 1.0                                               | $-3.4 \cdot 10^{-17}$ | 1.0                                                | 0                    |
| $z^9$    | 0                                        | 0     | 0                                      | 0     | 0.051                                             | 0.023                 | 0.051                                              | 0.022                |
| $z^8$    | 34.0                                     | 0     | NaN                                    | NaN   | 34.0                                              | 0.036                 | 34.0                                               | 0.036                |
| $z^7$    | 0                                        | 0     | 0                                      | 0     | 0.17                                              | 0.12                  | 0.17                                               | 0.12                 |
| $z^6$    | 120.0                                    | 0     | NaN                                    | NaN   | 120.0                                             | 0.26                  | 120.0                                              | 0.26                 |
| $z^5$    | 0                                        | 0     | 0                                      | 0     | -0.2                                              | 0.062                 | -0.2                                               | 0.057                |
| $z^4$    | 89.0                                     | 0     | NaN                                    | NaN   | 90.0                                              | 0.3                   | 90.0                                               | 0.3                  |
| $z^3$    | 0                                        | 0     | 0                                      | 0     | -0.24                                             | -0.026                | -0.24                                              | -0.028               |
| $z^2$    | 14.0                                     | 0     | NaN                                    | NaN   | 14.0                                              | 0.064                 | 14.0                                               | 0.064                |
| $z$      | 0                                        | 0     | 0                                      | 0     | -0.026                                            | $-4.7 \cdot 10^{-3}$  | -0.026                                             | $-4.9 \cdot 10^{-3}$ |
| 1.0      | 0.24                                     | 0     | NaN                                    | NaN   | 0.24                                              | $1.4 \cdot 10^{-3}$   | 0.24                                               | $1.4 \cdot 10^{-3}$  |

Table 85: Case 1a (cpv MACA64 MR2023b),  $r = 10$ , Z4: numerator (first lines) and denominator (last lines) coefficients.

|          | Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$ |       | LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$ |                       | AAA-sign-L200, $\sigma_{10} = 3.6 \cdot 10^{-12}$ |                      | AAA-sign-L1000, $\sigma_{10} = 3.6 \cdot 10^{-12}$ |                     |
|----------|------------------------------------------|-------|----------------------------------------|-----------------------|---------------------------------------------------|----------------------|----------------------------------------------------|---------------------|
| Basis    | real                                     | imag. | real                                   | imag.                 | real                                              | imag.                | real                                               | imag.               |
| $z^{10}$ | 0                                        | 0     | 0                                      | 0                     | $4.2 \cdot 10^{-8}$                               | $-3.6 \cdot 10^{-5}$ | $7.5 \cdot 10^{-8}$                                | $4.1 \cdot 10^{-6}$ |
| $z^9$    | 8.7                                      | 0     | 8.9                                    | $-1.1 \cdot 10^{-13}$ | 8.7                                               | $-6.5 \cdot 10^{-7}$ | 8.7                                                | $1.3 \cdot 10^{-7}$ |
| $z^8$    | 0                                        | 0     | $-1.6 \cdot 10^{-14}$                  | $-3.8 \cdot 10^{-12}$ | $1.3 \cdot 10^{-6}$                               | $-1.2 \cdot 10^{-3}$ | $2.6 \cdot 10^{-6}$                                | $1.4 \cdot 10^{-4}$ |
| $z^7$    | 78.0                                     | 0     | 85.0                                   | $-3.7 \cdot 10^{-12}$ | 78.0                                              | $-1.8 \cdot 10^{-5}$ | 78.0                                               | $3.5 \cdot 10^{-6}$ |
| $z^6$    | 0                                        | 0     | $1.9 \cdot 10^{-12}$                   | $-2.6 \cdot 10^{-11}$ | $4.0 \cdot 10^{-6}$                               | $-4.3 \cdot 10^{-3}$ | $9.1 \cdot 10^{-6}$                                | $4.9 \cdot 10^{-4}$ |
| $z^5$    | 120.0                                    | 0     | 140.0                                  | $-7.9 \cdot 10^{-12}$ | 120.0                                             | $-4.8 \cdot 10^{-5}$ | 120.0                                              | $9.3 \cdot 10^{-6}$ |
| $z^4$    | 0                                        | 0     | $4.0 \cdot 10^{-12}$                   | $-3.0 \cdot 10^{-11}$ | $2.6 \cdot 10^{-6}$                               | $-3.2 \cdot 10^{-3}$ | $6.9 \cdot 10^{-6}$                                | $3.7 \cdot 10^{-4}$ |
| $z^3$    | 44.0                                     | 0     | 54.0                                   | $-2.9 \cdot 10^{-12}$ | 44.0                                              | $-2.4 \cdot 10^{-5}$ | 44.0                                               | $4.7 \cdot 10^{-6}$ |
| $z^2$    | 0                                        | 0     | $1.2 \cdot 10^{-12}$                   | $-6.9 \cdot 10^{-12}$ | $3.7 \cdot 10^{-7}$                               | $-5.2 \cdot 10^{-4}$ | $1.1 \cdot 10^{-6}$                                | $6.0 \cdot 10^{-5}$ |
| $z$      | 2.7                                      | 0     | 3.5                                    | $-1.3 \cdot 10^{-13}$ | 2.7                                               | $-2.0 \cdot 10^{-6}$ | 2.7                                                | $3.9 \cdot 10^{-7}$ |
| 1.0      | 0                                        | 0     | $3.2 \cdot 10^{-14}$                   | $-1.5 \cdot 10^{-13}$ | $5.1 \cdot 10^{-9}$                               | $-8.7 \cdot 10^{-6}$ | $1.9 \cdot 10^{-8}$                                | $1.0 \cdot 10^{-6}$ |
| $z^{10}$ | 1.0                                      | 0     | 1.0                                    | 0                     | 1.0                                               | 0                    | 1.0                                                | 0                   |
| $z^9$    | 0                                        | 0     | $-4.1 \cdot 10^{-14}$                  | $-5.2 \cdot 10^{-13}$ | $3.5 \cdot 10^{-7}$                               | $-3.1 \cdot 10^{-4}$ | $6.6 \cdot 10^{-7}$                                | $3.5 \cdot 10^{-5}$ |
| $z^8$    | 34.0                                     | 0     | 36.0                                   | $-1.1 \cdot 10^{-12}$ | 34.0                                              | $-5.1 \cdot 10^{-6}$ | 34.0                                               | $1.0 \cdot 10^{-6}$ |
| $z^7$    | 0                                        | 0     | $5.3 \cdot 10^{-13}$                   | $-1.3 \cdot 10^{-11}$ | $2.8 \cdot 10^{-6}$                               | $-2.8 \cdot 10^{-3}$ | $6.0 \cdot 10^{-6}$                                | $3.2 \cdot 10^{-4}$ |
| $z^6$    | 120.0                                    | 0     | 130.0                                  | $-6.8 \cdot 10^{-12}$ | 120.0                                             | $-3.7 \cdot 10^{-5}$ | 120.0                                              | $7.1 \cdot 10^{-6}$ |
| $z^5$    | 0                                        | 0     | $3.6 \cdot 10^{-12}$                   | $-3.4 \cdot 10^{-11}$ | $3.9 \cdot 10^{-6}$                               | $-4.5 \cdot 10^{-3}$ | $9.5 \cdot 10^{-6}$                                | $5.1 \cdot 10^{-4}$ |
| $z^4$    | 89.0                                     | 0     | 110.0                                  | $-6.0 \cdot 10^{-12}$ | 89.0                                              | $-4.2 \cdot 10^{-5}$ | 89.0                                               | $8.1 \cdot 10^{-6}$ |
| $z^3$    | 0                                        | 0     | $2.8 \cdot 10^{-12}$                   | $-1.8 \cdot 10^{-11}$ | $1.2 \cdot 10^{-6}$                               | $-1.6 \cdot 10^{-3}$ | $3.4 \cdot 10^{-6}$                                | $1.8 \cdot 10^{-4}$ |
| $z^2$    | 14.0                                     | 0     | 18.0                                   | $-8.5 \cdot 10^{-13}$ | 14.0                                              | $-9.1 \cdot 10^{-6}$ | 14.0                                               | $1.8 \cdot 10^{-6}$ |
| $z$      | 0                                        | 0     | $2.9 \cdot 10^{-13}$                   | $-1.5 \cdot 10^{-12}$ | $6.5 \cdot 10^{-8}$                               | $-1.0 \cdot 10^{-4}$ | $2.2 \cdot 10^{-7}$                                | $1.2 \cdot 10^{-5}$ |
| 1.0      | 0.24                                     | 0     | 0.31                                   | $-8.1 \cdot 10^{-15}$ | 0.24                                              | $-1.9 \cdot 10^{-7}$ | 0.24                                               | $3.7 \cdot 10^{-8}$ |

Table 86: Case 1a (cpv MACI64 MR2022b),  $r = 10$ , Z4: numerator (first lines) and denominator (last lines) coefficients.

|          | Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$ |       | LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$ |       | AAA-sign-L200, $\sigma_{10} = 3.6 \cdot 10^{-12}$ |                      | AAA-sign-L1000, $\sigma_{10} = 3.6 \cdot 10^{-12}$ |                      |
|----------|------------------------------------------|-------|----------------------------------------|-------|---------------------------------------------------|----------------------|----------------------------------------------------|----------------------|
| Basis    | real                                     | imag. | real                                   | imag. | real                                              | imag.                | real                                               | imag.                |
| $z^{10}$ | 0                                        | 0     | 0                                      | 0     | $2.4 \cdot 10^{-9}$                               | $4.3 \cdot 10^{-6}$  | $2.2 \cdot 10^{-9}$                                | $-3.9 \cdot 10^{-8}$ |
| $z^9$    | 8.7                                      | 0     | NaN                                    | NaN   | 8.7                                               | $-1.0 \cdot 10^{-7}$ | 8.7                                                | $-1.3 \cdot 10^{-8}$ |
| $z^8$    | 0                                        | 0     | 0                                      | 0     | $5.9 \cdot 10^{-8}$                               | $1.5 \cdot 10^{-4}$  | $7.2 \cdot 10^{-8}$                                | $-1.4 \cdot 10^{-6}$ |
| $z^7$    | 78.0                                     | 0     | NaN                                    | NaN   | 78.0                                              | $-2.8 \cdot 10^{-6}$ | 78.0                                               | $-3.7 \cdot 10^{-7}$ |
| $z^6$    | 0                                        | 0     | 0                                      | 0     | $1.4 \cdot 10^{-7}$                               | $5.0 \cdot 10^{-4}$  | $2.4 \cdot 10^{-7}$                                | $-5.2 \cdot 10^{-6}$ |
| $z^5$    | 120.0                                    | 0     | NaN                                    | NaN   | 120.0                                             | $-7.5 \cdot 10^{-6}$ | 120.0                                              | $-9.7 \cdot 10^{-7}$ |
| $z^4$    | 0                                        | 0     | 0                                      | 0     | $7.2 \cdot 10^{-8}$                               | $3.8 \cdot 10^{-4}$  | $1.8 \cdot 10^{-7}$                                | $-4.2 \cdot 10^{-6}$ |
| $z^3$    | 44.0                                     | 0     | NaN                                    | NaN   | 44.0                                              | $-3.7 \cdot 10^{-6}$ | 44.0                                               | $-4.9 \cdot 10^{-7}$ |
| $z^2$    | 0                                        | 0     | 0                                      | 0     | $7.8 \cdot 10^{-9}$                               | $6.0 \cdot 10^{-5}$  | $2.8 \cdot 10^{-8}$                                | $-7.1 \cdot 10^{-7}$ |
| $z$      | 2.7                                      | 0     | NaN                                    | NaN   | 2.7                                               | $-3.0 \cdot 10^{-7}$ | 2.7                                                | $-3.9 \cdot 10^{-8}$ |
| 1.0      | 0                                        | 0     | 0                                      | 0     | $9.9 \cdot 10^{-11}$                              | $1.0 \cdot 10^{-6}$  | $4.6 \cdot 10^{-10}$                               | $-1.3 \cdot 10^{-8}$ |
| $z^{10}$ | 1.0                                      | 0     | NaN                                    | NaN   | 1.0                                               | 0                    | 1.0                                                | 0                    |
| $z^9$    | 0                                        | 0     | 0                                      | 0     | $1.8 \cdot 10^{-8}$                               | $3.7 \cdot 10^{-5}$  | $1.9 \cdot 10^{-8}$                                | $-3.5 \cdot 10^{-7}$ |
| $z^8$    | 34.0                                     | 0     | NaN                                    | NaN   | 34.0                                              | $-8.1 \cdot 10^{-7}$ | 34.0                                               | $-1.1 \cdot 10^{-7}$ |
| $z^7$    | 0                                        | 0     | 0                                      | 0     | $1.1 \cdot 10^{-7}$                               | $3.3 \cdot 10^{-4}$  | $1.6 \cdot 10^{-7}$                                | $-3.3 \cdot 10^{-6}$ |
| $z^6$    | 120.0                                    | 0     | NaN                                    | NaN   | 120.0                                             | $-5.7 \cdot 10^{-6}$ | 120.0                                              | $-7.4 \cdot 10^{-7}$ |
| $z^5$    | 0                                        | 0     | 0                                      | 0     | $1.2 \cdot 10^{-7}$                               | $5.2 \cdot 10^{-4}$  | $2.5 \cdot 10^{-7}$                                | $-5.6 \cdot 10^{-6}$ |
| $z^4$    | 89.0                                     | 0     | NaN                                    | NaN   | 89.0                                              | $-6.5 \cdot 10^{-6}$ | 89.0                                               | $-8.4 \cdot 10^{-7}$ |
| $z^3$    | 0                                        | 0     | 0                                      | 0     | $2.9 \cdot 10^{-8}$                               | $1.9 \cdot 10^{-4}$  | $8.7 \cdot 10^{-8}$                                | $-2.1 \cdot 10^{-6}$ |
| $z^2$    | 14.0                                     | 0     | NaN                                    | NaN   | 14.0                                              | $-1.4 \cdot 10^{-6}$ | 14.0                                               | $-1.8 \cdot 10^{-7}$ |
| $z$      | 0                                        | 0     | 0                                      | 0     | $1.3 \cdot 10^{-9}$                               | $1.2 \cdot 10^{-5}$  | $5.4 \cdot 10^{-9}$                                | $-1.4 \cdot 10^{-7}$ |
| 1.0      | 0.24                                     | 0     | NaN                                    | NaN   | 0.24                                              | $-2.9 \cdot 10^{-8}$ | 0.24                                               | $-3.8 \cdot 10^{-9}$ |

Table 87: Case 1a (cpv PCWIN64 MR2025b),  $r = 10$ , Z4: numerator (first lines) and denominator (last lines) coefficients.

|          | Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$ |       | LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$ |       | AAA-sign-L200, $\sigma_{10} = 3.6 \cdot 10^{-12}$ |                      | AAA-sign-L1000, $\sigma_{10} = 3.6 \cdot 10^{-12}$ |                      |
|----------|------------------------------------------|-------|----------------------------------------|-------|---------------------------------------------------|----------------------|----------------------------------------------------|----------------------|
| Basis    | real                                     | imag. | real                                   | imag. | real                                              | imag.                | real                                               | imag.                |
| $z^{10}$ | 0                                        | 0     | 0                                      | 0     | $2.4 \cdot 10^{-9}$                               | $4.3 \cdot 10^{-6}$  | $2.2 \cdot 10^{-9}$                                | $-3.9 \cdot 10^{-8}$ |
| $z^9$    | 8.7                                      | 0     | NaN                                    | NaN   | 8.7                                               | $-1.0 \cdot 10^{-7}$ | 8.7                                                | $-1.3 \cdot 10^{-8}$ |
| $z^8$    | 0                                        | 0     | 0                                      | 0     | $5.9 \cdot 10^{-8}$                               | $1.5 \cdot 10^{-4}$  | $7.2 \cdot 10^{-8}$                                | $-1.4 \cdot 10^{-6}$ |
| $z^7$    | 78.0                                     | 0     | NaN                                    | NaN   | 78.0                                              | $-2.8 \cdot 10^{-6}$ | 78.0                                               | $-3.7 \cdot 10^{-7}$ |
| $z^6$    | 0                                        | 0     | 0                                      | 0     | $1.4 \cdot 10^{-7}$                               | $5.0 \cdot 10^{-4}$  | $2.4 \cdot 10^{-7}$                                | $-5.2 \cdot 10^{-6}$ |
| $z^5$    | 120.0                                    | 0     | NaN                                    | NaN   | 120.0                                             | $-7.5 \cdot 10^{-6}$ | 120.0                                              | $-9.7 \cdot 10^{-7}$ |
| $z^4$    | 0                                        | 0     | 0                                      | 0     | $7.2 \cdot 10^{-8}$                               | $3.8 \cdot 10^{-4}$  | $1.8 \cdot 10^{-7}$                                | $-4.2 \cdot 10^{-6}$ |
| $z^3$    | 44.0                                     | 0     | NaN                                    | NaN   | 44.0                                              | $-3.7 \cdot 10^{-6}$ | 44.0                                               | $-4.9 \cdot 10^{-7}$ |
| $z^2$    | 0                                        | 0     | 0                                      | 0     | $7.8 \cdot 10^{-9}$                               | $6.0 \cdot 10^{-5}$  | $2.8 \cdot 10^{-8}$                                | $-7.1 \cdot 10^{-7}$ |
| $z$      | 2.7                                      | 0     | NaN                                    | NaN   | 2.7                                               | $-3.0 \cdot 10^{-7}$ | 2.7                                                | $-3.9 \cdot 10^{-8}$ |
| 1.0      | 0                                        | 0     | 0                                      | 0     | $9.9 \cdot 10^{-11}$                              | $1.0 \cdot 10^{-6}$  | $4.6 \cdot 10^{-10}$                               | $-1.3 \cdot 10^{-8}$ |
| $z^{10}$ | 1.0                                      | 0     | NaN                                    | NaN   | 1.0                                               | 0                    | 1.0                                                | 0                    |
| $z^9$    | 0                                        | 0     | 0                                      | 0     | $1.8 \cdot 10^{-8}$                               | $3.7 \cdot 10^{-5}$  | $1.9 \cdot 10^{-8}$                                | $-3.5 \cdot 10^{-7}$ |
| $z^8$    | 34.0                                     | 0     | NaN                                    | NaN   | 34.0                                              | $-8.1 \cdot 10^{-7}$ | 34.0                                               | $-1.1 \cdot 10^{-7}$ |
| $z^7$    | 0                                        | 0     | 0                                      | 0     | $1.1 \cdot 10^{-7}$                               | $3.3 \cdot 10^{-4}$  | $1.6 \cdot 10^{-7}$                                | $-3.3 \cdot 10^{-6}$ |
| $z^6$    | 120.0                                    | 0     | NaN                                    | NaN   | 120.0                                             | $-5.7 \cdot 10^{-6}$ | 120.0                                              | $-7.4 \cdot 10^{-7}$ |
| $z^5$    | 0                                        | 0     | 0                                      | 0     | $1.2 \cdot 10^{-7}$                               | $5.2 \cdot 10^{-4}$  | $2.5 \cdot 10^{-7}$                                | $-5.6 \cdot 10^{-6}$ |
| $z^4$    | 89.0                                     | 0     | NaN                                    | NaN   | 89.0                                              | $-6.5 \cdot 10^{-6}$ | 89.0                                               | $-8.4 \cdot 10^{-7}$ |
| $z^3$    | 0                                        | 0     | 0                                      | 0     | $2.9 \cdot 10^{-8}$                               | $1.9 \cdot 10^{-4}$  | $8.7 \cdot 10^{-8}$                                | $-2.1 \cdot 10^{-6}$ |
| $z^2$    | 14.0                                     | 0     | NaN                                    | NaN   | 14.0                                              | $-1.4 \cdot 10^{-6}$ | 14.0                                               | $-1.8 \cdot 10^{-7}$ |
| $z$      | 0                                        | 0     | 0                                      | 0     | $1.3 \cdot 10^{-9}$                               | $1.2 \cdot 10^{-5}$  | $5.4 \cdot 10^{-9}$                                | $-1.4 \cdot 10^{-7}$ |
| 1.0      | 0.24                                     | 0     | NaN                                    | NaN   | 0.24                                              | $-2.9 \cdot 10^{-8}$ | 0.24                                               | $-3.8 \cdot 10^{-9}$ |

Table 88: Case 1a (AlexReis PCWIN64 MR2025b),  $r = 10$ , Z4: numerator (first lines) and denominator (last lines) coefficients.

## 9.2 Poles

The poles are given in the following tables. Each table gives the results obtained on different platform.

|      | Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$ | LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$ | AAA-sign-L200, $\sigma_{10} = 4.3 \cdot 10^{-12}$ | AAA-sign-L1000, $\sigma_{10} = 4.3 \cdot 10^{-12}$ |
|------|------------------------------------------|----------------------------------------|---------------------------------------------------|----------------------------------------------------|
| 1.0  | $2.22 \cdot 10^{-16} + 5.47i$            | $1.27 \cdot 10^{-14} + 0.141i$         | $0.00103 - 0.137i$                                | $0.00103 - 0.137i$                                 |
| 2.0  | $2.22 \cdot 10^{-16} - 5.47i$            | $1.93 \cdot 10^{-14} - 0.141i$         | $8.67 \cdot 10^{-4} + 0.138i$                     | $8.68 \cdot 10^{-4} + 0.138i$                      |
| 3.0  | $2.78 \cdot 10^{-16} + 1.7i$             | $1.97 \cdot 10^{-14} + 0.453i$         | $0.00103 - 0.442i$                                | $0.00103 - 0.442i$                                 |
| 4.0  | $2.78 \cdot 10^{-16} - 1.7i$             | $3.14 \cdot 10^{-14} - 0.453i$         | $5.19 \cdot 10^{-4} + 0.442i$                     | $5.2 \cdot 10^{-4} + 0.442i$                       |
| 5.0  | $-1.25 \cdot 10^{-16} + 0.866i$          | $9.09 \cdot 10^{-16} + 0.891i$         | $-2.36 \cdot 10^{-4} + 0.867i$                    | $-2.35 \cdot 10^{-4} + 0.867i$                     |
| 6.0  | $-1.25 \cdot 10^{-16} - 0.866i$          | $2.9 \cdot 10^{-14} - 0.891i$          | $7.39 \cdot 10^{-4} - 0.868i$                     | $7.4 \cdot 10^{-4} - 0.868i$                       |
| 7.0  | $-6.94 \cdot 10^{-18} + 0.441i$          | $-7.64 \cdot 10^{-15} + 1.75i$         | $-0.00261 + 1.7i$                                 | $-0.00261 + 1.7i$                                  |
| 8.0  | $-6.94 \cdot 10^{-18} - 0.441i$          | $-1.98 \cdot 10^{-14} - 1.75i$         | $-7.39 \cdot 10^{-4} - 1.7i$                      | $-7.38 \cdot 10^{-4} - 1.7i$                       |
| 9.0  | $-1.34 \cdot 10^{-17} + 0.137i$          | $-2.66 \cdot 10^{-14} - 5.64i$         | $-0.0286 + 5.47i$                                 | $-0.0285 + 5.47i$                                  |
| 10.0 | $-1.34 \cdot 10^{-17} - 0.137i$          | $-5.92 \cdot 10^{-14} + 5.64i$         | $-0.0228 - 5.49i$                                 | $-0.0228 - 5.49i$                                  |

Table 89: Case 1a (cpv MAC64 MR2023b),  $r = 10$ , Z4: poles.

|      | Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$ | LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$ | AAA-sign-L200, $\sigma_{10} = 3.6 \cdot 10^{-12}$ | AAA-sign-L1000, $\sigma_{10} = 3.6 \cdot 10^{-12}$ |
|------|------------------------------------------|----------------------------------------|---------------------------------------------------|----------------------------------------------------|
| 1.0  | $6.66 \cdot 10^{-16} + 5.47i$            | $-6.77 \cdot 10^{-15} - 0.141i$        | $-1.38 \cdot 10^{-8} - 0.137i$                    | $-9.33 \cdot 10^{-9} + 0.137i$                     |
| 2.0  | $6.66 \cdot 10^{-16} - 5.47i$            | $-1.0 \cdot 10^{-14} + 0.141i$         | $9.86 \cdot 10^{-9} + 0.137i$                     | $-4.69 \cdot 10^{-9} - 0.137i$                     |
| 3.0  | $2.22 \cdot 10^{-16} + 1.7i$             | $-3.47 \cdot 10^{-15} - 0.453i$        | $-3.99 \cdot 10^{-8} - 0.441i$                    | $-1.58 \cdot 10^{-8} + 0.441i$                     |
| 4.0  | $2.22 \cdot 10^{-16} - 1.7i$             | $-1.04 \cdot 10^{-14} + 0.453i$        | $3.43 \cdot 10^{-8} + 0.441i$                     | $-1.28 \cdot 10^{-9} - 0.441i$                     |
| 5.0  | $-5.2 \cdot 10^{-17} + 0.866i$           | $-3.18 \cdot 10^{-15} - 0.891i$        | $-7.56 \cdot 10^{-8} - 0.866i$                    | $-2.7 \cdot 10^{-8} + 0.866i$                      |
| 6.0  | $-5.2 \cdot 10^{-17} - 0.866i$           | $1.9 \cdot 10^{-15} + 0.891i$          | $6.45 \cdot 10^{-8} + 0.866i$                     | $2.74 \cdot 10^{-10} - 0.866i$                     |
| 7.0  | $3.47 \cdot 10^{-18} + 0.441i$           | $-1.86 \cdot 10^{-14} - 1.75i$         | $-1.48 \cdot 10^{-7} - 1.7i$                      | $-5.77 \cdot 10^{-8} + 1.7i$                       |
| 8.0  | $3.47 \cdot 10^{-18} - 0.441i$           | $3.05 \cdot 10^{-14} + 1.75i$          | $1.16 \cdot 10^{-7} + 1.7i$                       | $-6.33 \cdot 10^{-9} - 1.7i$                       |
| 9.0  | $-1.52 \cdot 10^{-18} + 0.137i$          | $-5.53 \cdot 10^{-14} - 5.64i$         | $-5.63 \cdot 10^{-7} - 5.47i$                     | $-3.48 \cdot 10^{-7} + 5.47i$                      |
| 10.0 | $-1.52 \cdot 10^{-18} - 0.137i$          | $1.18 \cdot 10^{-13} + 5.64i$          | $2.67 \cdot 10^{-7} + 5.47i$                      | $-1.87 \cdot 10^{-7} - 5.47i$                      |

Table 90: Case 1a (cpv MACI64 MR2022b),  $r = 10$ , Z4: poles.

|      | Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$ | LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$ | AAA-sign-L200, $\sigma_{10} = 3.6 \cdot 10^{-12}$ | AAA-sign-L1000, $\sigma_{10} = 3.6 \cdot 10^{-12}$ |
|------|------------------------------------------|----------------------------------------|---------------------------------------------------|----------------------------------------------------|
| 1.0  | $2.22 \cdot 10^{-16} + 5.47i$            | $-1.13 \cdot 10^{-14} + 0.141i$        | $1.68 \cdot 10^{-9} + 0.137i$                     | $-3.96 \cdot 10^{-10} - 0.137i$                    |
| 2.0  | $2.22 \cdot 10^{-16} - 5.47i$            | $-3.06 \cdot 10^{-14} - 0.141i$        | $-1.75 \cdot 10^{-9} - 0.137i$                    | $4.93 \cdot 10^{-11} + 0.137i$                     |
| 3.0  | $2.78 \cdot 10^{-16} + 1.7i$             | $6.86 \cdot 10^{-15} + 0.453i$         | $5.46 \cdot 10^{-9} + 0.441i$                     | $-9.24 \cdot 10^{-10} - 0.441i$                    |
| 4.0  | $2.78 \cdot 10^{-16} - 1.7i$             | $-3.6 \cdot 10^{-14} - 0.453i$         | $-5.53 \cdot 10^{-9} - 0.441i$                    | $5.0 \cdot 10^{-10} + 0.441i$                      |
| 5.0  | $-1.25 \cdot 10^{-16} + 0.866i$          | $1.56 \cdot 10^{-14} + 0.891i$         | $1.05 \cdot 10^{-8} + 0.866i$                     | $-1.72 \cdot 10^{-9} - 0.866i$                     |
| 6.0  | $-1.25 \cdot 10^{-16} - 0.866i$          | $-4.39 \cdot 10^{-14} - 0.891i$        | $-1.08 \cdot 10^{-8} - 0.866i$                    | $1.04 \cdot 10^{-9} + 0.866i$                      |
| 7.0  | $-6.94 \cdot 10^{-18} + 0.441i$          | $8.07 \cdot 10^{-15} + 1.75i$          | $2.0 \cdot 10^{-8} + 1.7i$                        | $-3.56 \cdot 10^{-9} - 1.7i$                       |
| 8.0  | $-6.94 \cdot 10^{-18} - 0.441i$          | $-5.94 \cdot 10^{-14} - 1.75i$         | $-2.13 \cdot 10^{-8} - 1.7i$                      | $1.79 \cdot 10^{-9} + 1.7i$                        |
| 9.0  | $-1.34 \cdot 10^{-17} + 0.137i$          | $-1.74 \cdot 10^{-13} - 5.64i$         | $5.76 \cdot 10^{-8} + 5.47i$                      | $-1.63 \cdot 10^{-8} - 5.47i$                      |
| 10.0 | $-1.34 \cdot 10^{-17} - 0.137i$          | $1.08 \cdot 10^{-13} + 5.64i$          | $-7.4 \cdot 10^{-8} - 5.47i$                      | $7.27 \cdot 10^{-10} + 5.47i$                      |

Table 91: Case 1a (cpv PCWIN64 MR2025b),  $r = 10$ , Z4: poles.

|      | Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$ | LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$ | AAA-sign-L200, $\sigma_{10} = 3.6 \cdot 10^{-12}$ | AAA-sign-L1000, $\sigma_{10} = 3.6 \cdot 10^{-12}$ |
|------|------------------------------------------|----------------------------------------|---------------------------------------------------|----------------------------------------------------|
| 1.0  | $2.22 \cdot 10^{-16} + 5.47i$            | $-1.13 \cdot 10^{-14} + 0.141i$        | $1.68 \cdot 10^{-9} + 0.137i$                     | $-3.96 \cdot 10^{-10} - 0.137i$                    |
| 2.0  | $2.22 \cdot 10^{-16} - 5.47i$            | $-3.06 \cdot 10^{-14} - 0.141i$        | $-1.75 \cdot 10^{-9} - 0.137i$                    | $4.93 \cdot 10^{-11} + 0.137i$                     |
| 3.0  | $2.78 \cdot 10^{-16} + 1.7i$             | $6.86 \cdot 10^{-15} + 0.453i$         | $5.46 \cdot 10^{-9} + 0.441i$                     | $-9.24 \cdot 10^{-10} - 0.441i$                    |
| 4.0  | $2.78 \cdot 10^{-16} - 1.7i$             | $-3.6 \cdot 10^{-14} - 0.453i$         | $-5.53 \cdot 10^{-9} - 0.441i$                    | $5.0 \cdot 10^{-10} + 0.441i$                      |
| 5.0  | $-1.25 \cdot 10^{-16} + 0.866i$          | $1.56 \cdot 10^{-14} + 0.891i$         | $1.05 \cdot 10^{-8} + 0.866i$                     | $-1.72 \cdot 10^{-9} - 0.866i$                     |
| 6.0  | $-1.25 \cdot 10^{-16} - 0.866i$          | $-4.39 \cdot 10^{-14} - 0.891i$        | $-1.08 \cdot 10^{-8} - 0.866i$                    | $1.04 \cdot 10^{-9} + 0.866i$                      |
| 7.0  | $-6.94 \cdot 10^{-18} + 0.441i$          | $8.07 \cdot 10^{-15} + 1.75i$          | $2.0 \cdot 10^{-8} + 1.7i$                        | $-3.56 \cdot 10^{-9} - 1.7i$                       |
| 8.0  | $-6.94 \cdot 10^{-18} - 0.441i$          | $-5.94 \cdot 10^{-14} - 1.75i$         | $-2.13 \cdot 10^{-8} - 1.7i$                      | $1.79 \cdot 10^{-9} + 1.7i$                        |
| 9.0  | $-1.34 \cdot 10^{-17} + 0.137i$          | $-1.74 \cdot 10^{-13} - 5.64i$         | $5.76 \cdot 10^{-8} + 5.47i$                      | $-1.63 \cdot 10^{-8} - 5.47i$                      |
| 10.0 | $-1.34 \cdot 10^{-17} - 0.137i$          | $1.08 \cdot 10^{-13} + 5.64i$          | $-7.4 \cdot 10^{-8} - 5.47i$                      | $7.27 \cdot 10^{-10} + 5.47i$                      |

Table 92: Case 1a (AlexReis PCWIN64 MR2025b),  $r = 10$ , Z4: poles.

### 9.3 Zeros

The zeros are given in the following tables. Each table gives the results obtained on different platform.

|      | Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$ | LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$        | AAA-sign-L200, $\sigma_{10} = 4.3 \cdot 10^{-12}$ | AAA-sign-L1000, $\sigma_{10} = 4.3 \cdot 10^{-12}$ |
|------|------------------------------------------|-----------------------------------------------|---------------------------------------------------|----------------------------------------------------|
| 1.0  | NaN                                      | NaN                                           | $9.66 \cdot 10^{-4} + 1.92 \cdot 10^{-4}_z$       | $9.67 \cdot 10^{-4} + 1.96 \cdot 10^{-4}_z$        |
| 2.0  | 0                                        | $1.35 \cdot 10^{-14} - 8.67 \cdot 10^{-15}_z$ | $0.00105 - 0.282_z$                               | $0.00105 - 0.282_z$                                |
| 3.0  | $3.33 \cdot 10^{-16} + 2.67_z$           | $1.74 \cdot 10^{-14} + 0.289_z$               | $7.24 \cdot 10^{-4} + 0.282_z$                    | $7.24 \cdot 10^{-4} + 0.282_z$                     |
| 4.0  | $3.33 \cdot 10^{-16} - 2.67_z$           | $2.58 \cdot 10^{-14} - 0.289_z$               | $9.39 \cdot 10^{-4} - 0.63_z$                     | $9.39 \cdot 10^{-4} - 0.63_z$                      |
| 5.0  | $2.43 \cdot 10^{-16} + 1.19_z$           | $1.34 \cdot 10^{-14} + 0.647_z$               | $2.23 \cdot 10^{-4} + 0.63_z$                     | $2.23 \cdot 10^{-4} + 0.63_z$                      |
| 6.0  | $2.43 \cdot 10^{-16} - 1.19_z$           | $3.58 \cdot 10^{-14} - 0.647_z$               | $-0.00102 + 1.19_z$                               | $-0.00102 + 1.19_z$                                |
| 7.0  | $4.16 \cdot 10^{-17} + 0.629_z$          | $-5.1 \cdot 10^{-15} + 1.23_z$                | $3.05 \cdot 10^{-4} - 1.19_z$                     | $3.06 \cdot 10^{-4} - 1.19_z$                      |
| 8.0  | $4.16 \cdot 10^{-17} - 0.629_z$          | $6.08 \cdot 10^{-15} - 1.23_z$                | $-0.00688 + 2.67_z$                               | $-0.00687 + 2.67_z$                                |
| 9.0  | $-3.47 \cdot 10^{-18} + 0.281_z$         | $-2.24 \cdot 10^{-14} + 2.75_z$               | $-0.00398 - 2.67_z$                               | $-0.00398 - 2.67_z$                                |
| 10.0 | $-3.47 \cdot 10^{-18} - 0.281_z$         | $-3.62 \cdot 10^{-14} - 2.75_z$               | $-971.0 + 392.0_z$                                | $-975.0 + 389.0_z$                                 |

Table 93: Case 1a (cpv MAC64 MR2023b),  $r = 10$ , Z4: zeros.

|      | Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$ | LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$         | AAA-sign-L200, $\sigma_{10} = 3.6 \cdot 10^{-12}$ | AAA-sign-L1000, $\sigma_{10} = 3.6 \cdot 10^{-12}$ |
|------|------------------------------------------|------------------------------------------------|---------------------------------------------------|----------------------------------------------------|
| 1.0  | NaN                                      | NaN                                            | $-1.87 \cdot 10^{-9} + 3.18 \cdot 10^{-6}_z$      | $-6.85 \cdot 10^{-9} - 3.69 \cdot 10^{-7}_z$       |
| 2.0  | 0                                        | $-9.07 \cdot 10^{-15} + 4.27 \cdot 10^{-14}_z$ | $-2.63 \cdot 10^{-8} - 0.281_z$                   | $-1.22 \cdot 10^{-8} + 0.281_z$                    |
| 3.0  | $3.33 \cdot 10^{-16} + 2.67_z$           | $-4.41 \cdot 10^{-15} - 0.289_z$               | $2.18 \cdot 10^{-8} + 0.281_z$                    | $-2.83 \cdot 10^{-9} - 0.281_z$                    |
| 4.0  | $3.33 \cdot 10^{-16} - 2.67_z$           | $-1.09 \cdot 10^{-14} + 0.289_z$               | $-5.58 \cdot 10^{-8} - 0.629_z$                   | $-2.05 \cdot 10^{-8} + 0.629_z$                    |
| 5.0  | $-2.22 \cdot 10^{-16} + 1.19_z$          | $-4.14 \cdot 10^{-15} - 0.647_z$               | $4.82 \cdot 10^{-8} + 0.629_z$                    | $-1.51 \cdot 10^{-10} - 0.629_z$                   |
| 6.0  | $-2.22 \cdot 10^{-16} - 1.19_z$          | $-5.14 \cdot 10^{-15} + 0.647_z$               | $-1.03 \cdot 10^{-7} - 1.19_z$                    | $-3.76 \cdot 10^{-8} + 1.19_z$                     |
| 7.0  | $-4.16 \cdot 10^{-17} + 0.629_z$         | $-4.28 \cdot 10^{-15} - 1.23_z$                | $8.56 \cdot 10^{-8} + 1.19_z$                     | $-8.47 \cdot 10^{-10} - 1.19_z$                    |
| 8.0  | $-4.16 \cdot 10^{-17} - 0.629_z$         | $9.49 \cdot 10^{-15} + 1.23_z$                 | $-2.41 \cdot 10^{-7} - 2.67_z$                    | $-1.08 \cdot 10^{-7} + 2.67_z$                     |
| 9.0  | $1.73 \cdot 10^{-18} + 0.281_z$          | $-3.66 \cdot 10^{-14} - 2.75_z$                | $1.68 \cdot 10^{-7} + 2.67_z$                     | $-2.91 \cdot 10^{-8} - 2.67_z$                     |
| 10.0 | $1.73 \cdot 10^{-18} - 0.281_z$          | $6.7 \cdot 10^{-14} + 2.75_z$                  | $-284.0 - 2.41 \cdot 10^5_z$                      | $-3.94 \cdot 10^4 + 2.13 \cdot 10^6_z$             |

Table 94: Case 1a (cpv MACI64 MR2022b),  $r = 10$ , Z4: zeros.

|      | Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$ | LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$         | AAA-sign-L200, $\sigma_{10} = 3.6 \cdot 10^{-12}$ | AAA-sign-L1000, $\sigma_{10} = 3.6 \cdot 10^{-12}$ |
|------|------------------------------------------|------------------------------------------------|---------------------------------------------------|----------------------------------------------------|
| 1.0  | NaN                                      | NaN                                            | $-3.62 \cdot 10^{-11} - 3.64 \cdot 10^{-7}_z$     | $-1.7 \cdot 10^{-10} + 4.59 \cdot 10^{-9}_z$       |
| 2.0  | 0                                        | $-2.24 \cdot 10^{-14} - 5.81 \cdot 10^{-15}_z$ | $3.48 \cdot 10^{-9} + 0.281_z$                    | $-6.42 \cdot 10^{-10} - 0.281_z$                   |
| 3.0  | $3.33 \cdot 10^{-16} + 2.67_z$           | $-4.04 \cdot 10^{-16} + 0.289_z$               | $-3.55 \cdot 10^{-9} - 0.281_z$                   | $2.69 \cdot 10^{-10} + 0.281_z$                    |
| 4.0  | $3.33 \cdot 10^{-16} - 2.67_z$           | $-3.45 \cdot 10^{-14} - 0.289_z$               | $7.74 \cdot 10^{-9} + 0.629_z$                    | $-1.27 \cdot 10^{-9} - 0.629_z$                    |
| 5.0  | $2.43 \cdot 10^{-16} + 1.19_z$           | $-3.9 \cdot 10^{-14} - 0.647_z$                | $-7.85 \cdot 10^{-9} - 0.629_z$                   | $7.51 \cdot 10^{-10} + 0.629_z$                    |
| 6.0  | $2.43 \cdot 10^{-16} - 1.19_z$           | $1.22 \cdot 10^{-14} + 0.647_z$                | $1.43 \cdot 10^{-8} + 1.19_z$                     | $-2.4 \cdot 10^{-9} - 1.19_z$                      |
| 7.0  | $4.16 \cdot 10^{-17} + 0.629_z$          | $1.25 \cdot 10^{-14} + 1.23_z$                 | $-1.48 \cdot 10^{-8} - 1.19_z$                    | $1.38 \cdot 10^{-9} + 1.19_z$                      |
| 8.0  | $4.16 \cdot 10^{-17} - 0.629_z$          | $-5.01 \cdot 10^{-14} - 1.23_z$                | $3.04 \cdot 10^{-8} + 2.67_z$                     | $-6.13 \cdot 10^{-9} - 2.67_z$                     |
| 9.0  | $-3.47 \cdot 10^{-18} + 0.281_z$         | $-8.32 \cdot 10^{-14} - 2.75_z$                | $-3.4 \cdot 10^{-8} - 2.67_z$                     | $2.21 \cdot 10^{-9} + 2.67_z$                      |
| 10.0 | $-3.47 \cdot 10^{-18} - 0.281_z$         | $2.31 \cdot 10^{-14} + 2.75_z$                 | $-1130.0 + 2.0 \cdot 10^6_z$                      | $-1.28 \cdot 10^7 - 2.24 \cdot 10^8_z$             |

Table 95: Case 1a (cpv PCWIN64 MR2025b),  $r = 10$ , Z4: zeros.

|      | Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$ | LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$         | AAA-sign-L200, $\sigma_{10} = 3.6 \cdot 10^{-12}$ | AAA-sign-L1000, $\sigma_{10} = 3.6 \cdot 10^{-12}$ |
|------|------------------------------------------|------------------------------------------------|---------------------------------------------------|----------------------------------------------------|
| 1.0  | NaN                                      | NaN                                            | $-3.62 \cdot 10^{-11} - 3.64 \cdot 10^{-7}_z$     | $-1.7 \cdot 10^{-10} + 4.59 \cdot 10^{-9}_z$       |
| 2.0  | 0                                        | $-2.24 \cdot 10^{-14} - 5.81 \cdot 10^{-15}_z$ | $3.48 \cdot 10^{-9} + 0.281_z$                    | $-6.42 \cdot 10^{-10} - 0.281_z$                   |
| 3.0  | $3.33 \cdot 10^{-16} + 2.67_z$           | $-4.04 \cdot 10^{-16} + 0.289_z$               | $-3.55 \cdot 10^{-9} - 0.281_z$                   | $2.69 \cdot 10^{-10} + 0.281_z$                    |
| 4.0  | $3.33 \cdot 10^{-16} - 2.67_z$           | $-3.45 \cdot 10^{-14} - 0.289_z$               | $7.74 \cdot 10^{-9} + 0.629_z$                    | $-1.27 \cdot 10^{-9} - 0.629_z$                    |
| 5.0  | $2.43 \cdot 10^{-16} + 1.19_z$           | $-3.9 \cdot 10^{-14} - 0.647_z$                | $-7.85 \cdot 10^{-9} - 0.629_z$                   | $7.51 \cdot 10^{-10} + 0.629_z$                    |
| 6.0  | $2.43 \cdot 10^{-16} - 1.19_z$           | $1.22 \cdot 10^{-14} + 0.647_z$                | $1.43 \cdot 10^{-8} + 1.19_z$                     | $-2.4 \cdot 10^{-9} - 1.19_z$                      |
| 7.0  | $4.16 \cdot 10^{-17} + 0.629_z$          | $1.25 \cdot 10^{-14} + 1.23_z$                 | $-1.48 \cdot 10^{-8} - 1.19_z$                    | $1.38 \cdot 10^{-9} + 1.19_z$                      |
| 8.0  | $4.16 \cdot 10^{-17} - 0.629_z$          | $-5.01 \cdot 10^{-14} - 1.23_z$                | $3.04 \cdot 10^{-8} + 2.67_z$                     | $-6.13 \cdot 10^{-9} - 2.67_z$                     |
| 9.0  | $-3.47 \cdot 10^{-18} + 0.281_z$         | $-8.32 \cdot 10^{-14} - 2.75_z$                | $-3.4 \cdot 10^{-8} - 2.67_z$                     | $2.21 \cdot 10^{-9} + 2.67_z$                      |
| 10.0 | $-3.47 \cdot 10^{-18} - 0.281_z$         | $2.31 \cdot 10^{-14} + 2.75_z$                 | $-1130.0 + 2.0 \cdot 10^6_z$                      | $-1.28 \cdot 10^7 - 2.24 \cdot 10^8_z$             |

Table 96: Case 1a (AlexReis PCWIN64 MR2025b),  $r = 10$ , Z4: zeros.

## 9.4 Poles and zeros visualisation

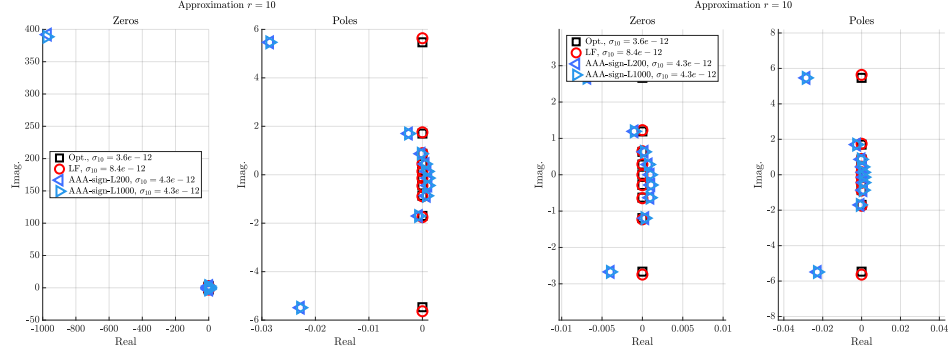


Figure 15: Case 1a (cpv MACA64 MR2023b),  $r = 10$ , Z4. Left frame: zeros (left) and poles (right). Right frame: same but discarding the largest real entry.

## 9.5 Approximation visualization

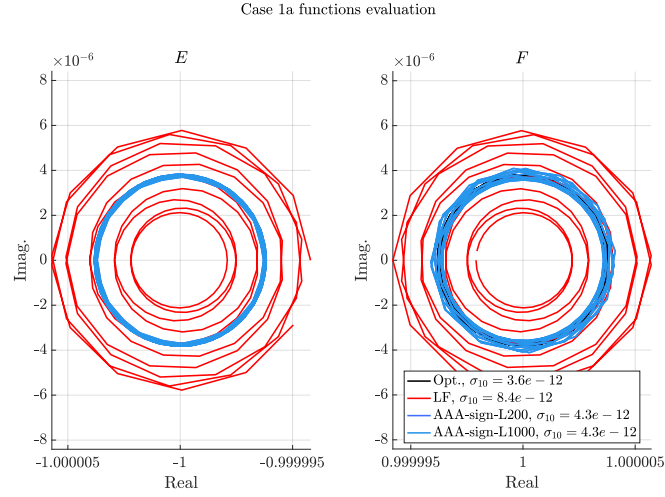


Figure 16: Case 1a (cpv MACA64 MR2023b),  $r = 10$ , Z4.  $E$  and  $F$  evaluations.